

TimeMorph: Dual-view LLM-based Few-Shot Time Series Classification

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Abstract—Few-shot time-series classification (TSC) is challenging due to scarce labeled sequences, where each class provides only a few examples. Reliable prediction requires capturing both temporal dynamics and waveform morphology, reflecting how humans interpret sequential signals by tracking numeric trends and recognizing overall shape patterns. Recent pretrained large language models exhibit strong potential for few-shot learning due to their pretrained knowledge and reasoning capabilities. However, directly applying them remains nontrivial: continuous time-series features are not naturally aligned with language embeddings, and predictions must be restricted to a fixed set of target classes. Therefore, we propose TimeMorph, a dual-view framework that extracts temporal dynamics via a temporal branch and waveform morphology via a visual pseudo-image branch. To bridge time-series features and language embeddings, we employ a stage-wise transfer strategy to project the fused tokens into a pretrained causal LLM. During inference, TimeMorph replaces open-ended text generation with closed-set class-token scoring over the valid label set, and further incorporates semantic label priors when meaningful class names are available. Extensive few-shot experiments on the UCR archive and additional external datasets show that TimeMorph consistently outperforms strong task-specific, Transformer-based, and LLM/foundation-model baselines. Further analyses validate the effectiveness of dual-view modeling, transfer pretraining, and the class-token interface.

Index Terms—Few-shot learning, time-series classification, large language models, multimodal learning, closed-set classification.

1 INTRODUCTION

Time series classification (TSC) is challenging when labeled sequences are scarce in healthcare monitoring, industrial diagnosis, and related domains [1]. Reliable TSC requires capturing both temporal dynamics and waveform morphology. This reflects how humans interpret sequential data: they track numerical evolution over time while recognizing characteristic shape patterns [2], [3]. Some classes are mainly distinguished by temporal dynamics, such as local trends and long-range evolution, whereas others are more naturally characterized by local subsequence shapes, raw waveform patterns, or recurring morphological structures [4], [5]. Inspired by this dual-view observation of human, it believes that effective few-shot TSC models would benefit from extracting and combining these temporal and morphological features. However, existing approaches often focus primarily on numerical trends while overlooking the intuitive visual structure of time-series waveforms [6]–[8].

Traditional TSC methods, including convolutional networks [9], Transformers [10], and prototype-based models [7], can achieve strong performance when sufficient labeled data are available. In few-shot settings, however, they often overfit to the limited support set and generalize poorly [11]. Pretrained large language models (LLMs) offer a promising direction for data-scarce classification. Recent studies adapt LLMs to time-series forecasting, classification, and anomaly detection through prompting, sequence reprogramming, or cross-modal alignment, showing encouraging generalization potential [12]–[15]. Meanwhile, the in-context learning and prompt-based adaptation abilities of LLMs have demonstrated strong task adaptability in few-shot and zero-shot

settings [16], [17]. Nevertheless, directly applying LLMs to TSC remains non-trivial, since time series are continuous, domain-specific, and not naturally expressed in language. [18].

To make pretrained LLMs effective for few-shot TSC with dual-view time-series representations, three key challenges must be addressed: (i) the model should preserve both temporal dynamics and morphology-oriented cues. (ii) the fused continuous time-series features must be aligned with the input space of a pretrained LLM. (iii) predictions should be restricted to valid candidate classes rather than produced by unconstrained text generation. The latter may generate invalid labels and introduce undesirable biases from tokenization or label length. This issue is closely related to the broader problem of enforcing valid outputs in language-model decoding, which has motivated constrained decoding methods for sequence generation and structured prediction [19], [20].

We propose **TimeMorph**, a dual-view generative classification framework for few-shot TSC. For each input sequence, TimeMorph constructs two complementary views: a temporal branch that models ordered local relations and long-range dynamics, and a morphological branch that converts the waveform into a morphological map, enabling a frozen visual encoder to extract shape-level structures. Through a stage-wise cross-modal alignment strategy, the resulting temporal and morphological tokens are projected into the embedding space of a frozen causal language model, mitigating the mismatch between time-series representations and language embeddings. For downstream prediction, TimeMorph replaces unconstrained text generation with supervised closed-set class-token scoring over valid target labels. When semantically meaningful class names are available, label verbalizers can further initialize the class tokens,

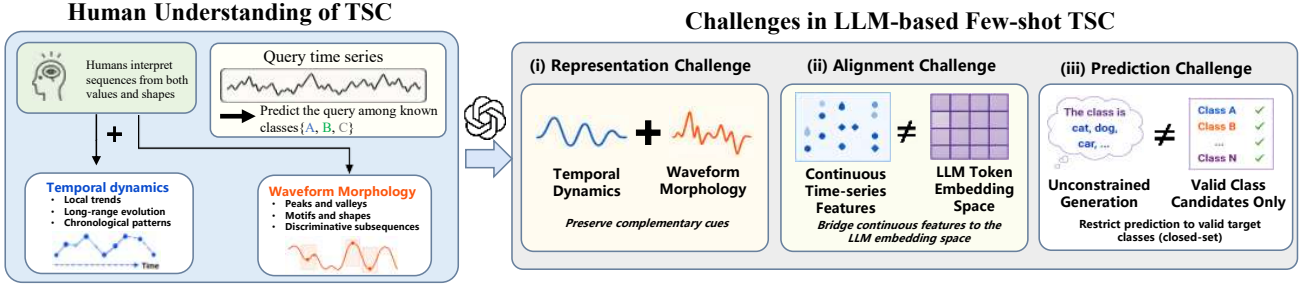


Fig. 1. Motivation of TimeMorph. Effective few-shot TSC requires both temporal dynamics and waveform morphology, while adapting LLMs to this setting introduces representation, alignment, and closed-set prediction challenges.

injecting useful semantic priors while avoiding the instability of multi-token generation.

Our main contributions are summarized as follows:

- We propose a **dual-view time-series modeling paradigm** for few-shot TSC that jointly models temporal dynamics and waveform morphology through complementary views.
- We introduce a **closed-set class-token scoring** interface for LLM-based TSC. It replaces autoregressive label generation with supervised scoring over valid classes, ensuring legal predictions, stabilizes optimization, and effectively leverages semantic label priors.
- We design a **stage-wise cross-modal pretraining strategy** that aligns continuous time-series tokens with the embedding space of a pretrained causal LLM, enabling the model to leverage foundation-model priors without numeric-to-text tokenization.

2 RELATED WORK

2.0.0.1 Few-shot time-series classification. Few-shot time-series classification (TSC) aims to recognize unseen classes or tasks from only a few labeled sequences. Existing methods typically inject data-efficient inductive biases through meta-learning, metric learning, or prototype-based decision rules. Narwariya et al. [21] meta-train residual networks across heterogeneous UCR tasks for rapid adaptation, while TapNet [7] and Dual Prototypical Shapelet Networks [8] combine metric prototypes with temporal or shapelet-based representations. Spectral and graph-based propagation has also been used to exploit relations among scarce examples [22], and COSCO improves few-shot multivariate TSC by coupling prototypical learning with sharpness-aware optimization [23]. These methods provide strong task-specific priors, but they are mainly discriminative models over numerical sequences. They usually do not leverage pretrained foundation-model priors, nor do they explicitly couple temporal dynamics with morphology-oriented waveform evidence.

2.0.0.2 LLMs and foundation models for time series. Recent work has explored adapting large language models and foundation models to time-series tasks through prompting, reprogramming, and cross-modal alignment [24]–[26]. GPT4TS/One-Fits-All adapts pretrained language or vision Transformers to multiple time-series tasks [13], and zero-shot forecasting studies serialize numerical values as text for next-token prediction [12]. TEST aligns time-series

embeddings with the LLM embedding space using text-prototype-aligned objectives [15], whereas Time-LLM reprograms time-series patches into a frozen LLM through text prototypes and prompt-as-prefix conditioning [14]. LLM4TS further studies staged alignment and fine-tuning [27], and OpenTSLM integrates time series as a native modality for multimodal reasoning through soft-prompt or cross-attention interfaces [28]. However, recent analyses show that LLM backbones do not automatically improve time-series models when the modality interface is weak or the task head is mismatched [18], [29], [30]. The closest few-shot classification work is LLMFew [11], which uses a patch-wise temporal encoder and LoRA-tuned decoder for few-shot multivariate TSC. TimeMorph differs by aligning dual-view time-series tokens to a causal LLM and replacing open-ended label generation with closed-set class-token scoring.

2.0.0.3 Temporal and morphology-aware representations. Time-series labels can depend on both temporal dynamics and waveform morphology. Sequence-domain methods model this evidence through shapelets, convolutional kernels, or local patches: shapelet methods extract class-discriminative subsequences [4], [8], convolutional models capture multi-scale local patterns [6], [9], and patch-based Transformers preserve local-to-global temporal dependencies [31]. A complementary direction maps signals to two-dimensional structures, such as Gramian angular fields and Markov transition fields, enabling image models to capture global shape and transition patterns [32]. This view is consistent with classical work on graphical perception and visualization, where suitable visual encodings make quantitative structure easier to decode [2], [3]. Recent studies further show that visual representations and pretrained vision models can benefit time-series reasoning and classification [33], [34]. TimeMorph adopts this dual view: its temporal branch preserves chronological dynamics, while its morphology branch exposes waveform shape and cross-window structural transitions as visual pseudo-images.

2.0.0.4 Generative classification and label-space constraints. Text-to-text models cast classification as conditional generation, where labels are produced as target tokens [35]. This interface is flexible, but it is poorly matched to closed-set classification under scarce supervision. Free-form decoding can generate invalid labels, and likelihood-based label selection is sensitive to answer priors, surface-form competition, tokenization, and verbalizer choices [36], [37]. Constrained decoding restricts the output space to valid lexical or structural forms [19], [20], while prompt-based

classification uses label words to connect semantic classes with vocabulary tokens [38], [39]. TimeMorph takes a more direct closed-set route. It introduces dataset-specific class tokens and scores only valid classes, avoiding autoregressive label generation. When class names are informative, verbalizers initialize and regularize the corresponding class-token embeddings, injecting semantic priors without relying on multi-token free-form outputs.

3 METHOD

We propose **TimeMorph**, a dual-view LLM-based framework for few-shot time-series classification (TSC). TimeMorph adapts a pretrained causal language model to continuous time-series inputs without serializing raw observations into text. Given an input sequence, it extracts complementary temporal-dynamic and waveform-morphology patterns, projects the resulting time-series tokens into the LLM embedding space, and performs prediction by closed-set scoring over dataset-specific class tokens.

3.1 Problem Setup

3.1.0.1 Univariate time-series classification. Let D be a target classification dataset with label set $\mathcal{Y}_D = \{0, \dots, C-1\}$. Each example consists of a univariate time series $x \in \mathbb{R}^T$ and a class label $y \in \mathcal{Y}_D$. Given a query sequence x , the goal of TSC is to infer its label from the temporal signal. We consider closed-set classification over the full label space: all C classes are available during adaptation and evaluation, and predictions are restricted to the valid label set $\mathcal{Y}_D$.

3.1.0.2 Few-shot learning setup. In the K -shot setting, the support set S_D^K contains at most K labeled examples per class:

$$S_D^K = \{(x_i, y_i)\}_{i=1}^{n_D}, \quad |\{i : y_i = c\}| \leq K, \quad c \in \mathcal{Y}_D. \quad (1)$$

The model is adapted on S_D^K and evaluated on query examples from the official test split.

3.2 Framework

The framework consists of three components: a dual-view encoder, an embedding-space projection interface, and a closed-set class-token scoring rule. The dual-view encoder G_θ extracts complementary representations using a temporal-view encoder and a morphology-view encoder. The projection interface P_ψ maps the fused time-series tokens to the hidden dimension of a pretrained causal LLM. The class-token scoring rule converts the LLM output logits into predictions over the valid labels of the target dataset. All components are optimized through the staged training schedule described in Section 3.2.4.

3.2.1 Dual-View Time-Series Encoding

Class-discriminative information in time series can arise from heterogeneous cues. Some classes are primarily distinguished by temporal dynamics, such as local trends, phase shifts, amplitude variations, or long-range evolution. Others are more naturally characterized by waveform morphology, including peaks, valleys, cycles, motifs, and discriminative subsequences. TimeMorph therefore encodes each input

sequence with two complementary views, processed by a temporal branch and a morphology branch, respectively. Figure 3 illustrates this design by comparing the raw input, ordered temporal patches, and the corresponding morphology-map view.

3.2.1.1 Temporal branch. The temporal view preserves the chronological structure of the original sequence. Let x^{pad} denote the sequence after right-side replicate padding. Given patch length ℓ and stride s , we extract overlapping temporal patches as

$$R_i = x_{(i-1)s+1:(i-1)s+\ell}^{\text{pad}}, \quad i = 1, \dots, N_t, \quad (2)$$

$$N_t = \left\lfloor \frac{T + p_t - \ell}{s} \right\rfloor + 1,$$

where p_t is the padding length required to form complete patches. Each patch is linearly embedded, added to a learnable positional embedding, and processed by a temporal Transformer:

$$Z^t(x) = G_t([W_t R_1 + \pi_1; \dots; W_t R_{N_t} + \pi_{N_t}]). \quad (3)$$

The output $Z^t(x)$ is a sequence of temporal tokens encoding ordered patch-level dynamics of the waveform.

3.2.1.2 Morphology branch. The morphology view exposes waveform morphology by converting the same one-dimensional sequence into a two-dimensional sliding-window morphology map. We first normalize x with robust per-sequence statistics:

$$\bar{x}_t = \frac{x_t - \text{median}(x)}{\text{IQR}(x) + \epsilon}. \quad (4)$$

The normalized sequence is then unfolded into overlapping local windows. Let $q = \max(1, \lfloor \sqrt{T} \rfloor)$ be the window length and $\delta = \max(1, \lfloor \rho q \rfloor)$ be the stride. We construct the morphology map by normalizing the unfolded matrix, resizing it to the input resolution of the vision backbone, and replicating it across RGB channels:

$$I(x) = \text{RGB}(\text{Resize}(\Gamma_{\text{img}}(\text{Unfold}_{q,\delta}(\bar{x})))), \quad (5)$$

where $\Gamma_{\text{img}}(\cdot)$ denotes per-map normalization and contrast adjustment. Each row of $I(x)$ corresponds to a local waveform window, so the map preserves local subsequence shapes while making cross-window structural transitions visible to a vision encoder. This construction exposes repeated local shapes and cross-window transitions rather than treating the time series as an arbitrary natural image. A frozen vision backbone G_m extracts morphology tokens:

$$Z^m(x) = G_m(I(x)). \quad (6)$$

We discard the vision class token and use only the patch tokens. The vision backbone remains frozen; trainable parameters are introduced in the temporal branch, projection modules, class-token rows, and LLM adapters.

3.2.1.3 Dual-view fusion. The temporal and morphology tokens are projected to a common dimension and concatenated:

$$\tilde{Z}^t(x) = Q_t(Z^t(x)), \quad \tilde{Z}^m(x) = Q_m(Z^m(x)), \quad (7)$$

$$Z_x = [\tilde{Z}^t(x); \tilde{Z}^m(x)].$$

The fused sequence $Z_x \in \mathbb{R}^{N_x \times d_e}$ serves as the dual-view time-series token sequence passed to the LLM interface.

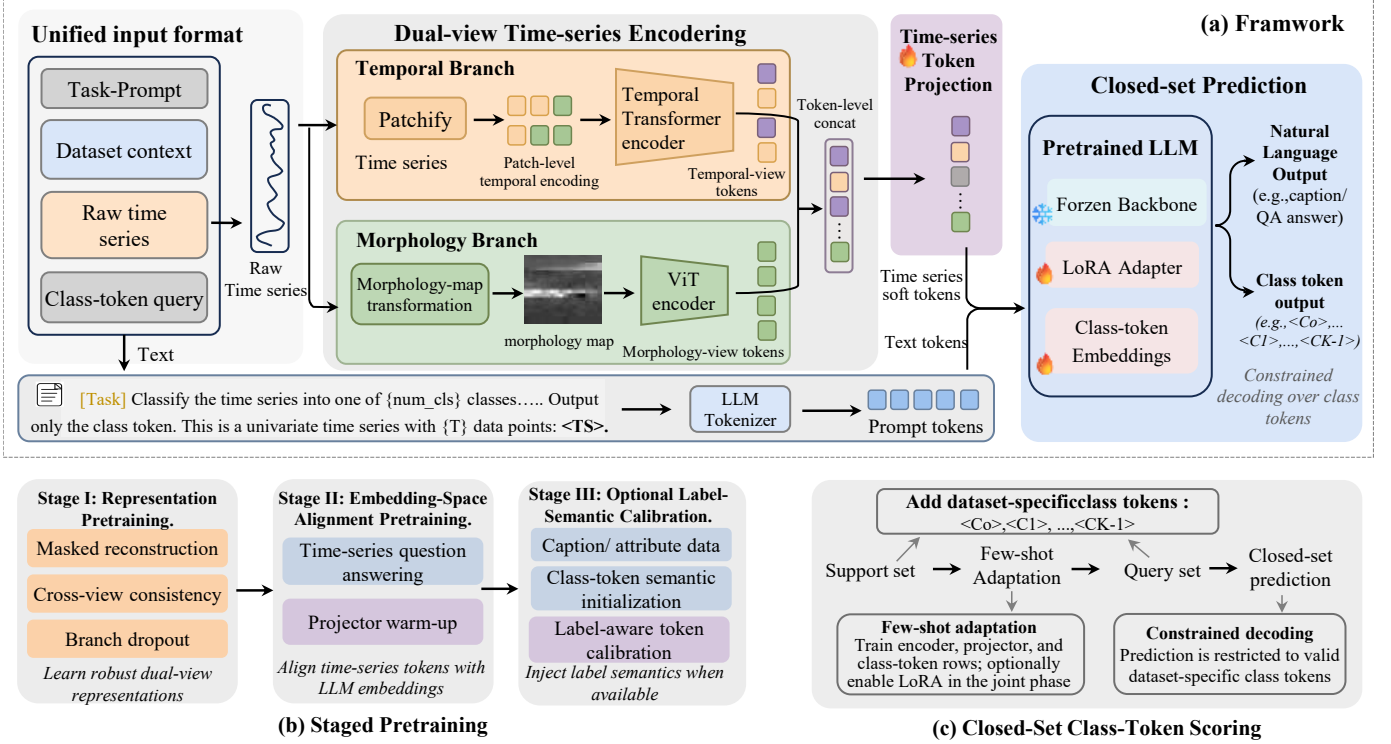


Fig. 2. Overview of TimeMorph. (a) TimeMorph encodes a raw univariate time series with temporal and morphology branches, fuses the resulting tokens, and projects them into the embedding space of a pretrained causal LLM for closed-set prediction. (b) Staged pretraining learns robust dual-view representations, aligns time-series tokens with LLM embeddings, and optionally injects label semantics. (c) During few-shot adaptation, dataset-specific class tokens are added and predictions are made by scoring only valid class tokens.

3.2.2 Time-Series Token Projection

The dual-view encoder outputs continuous time-series tokens, whereas a causal LLM consumes embeddings associated with discrete text tokens. TimeMorph bridges this modality gap through an input-side projection interface, which maps the fused time-series tokens Z_x into the LLM embedding space without serializing numerical observations as text.

Let P_ψ be a trainable projection module from encoder dimension d_e to the LLM hidden dimension, $\tau(\cdot)$ be the tokenizer, and E_{lm} be the LLM input embedding matrix. For a prompt $p = (p^{\text{pre}}, p^{\text{post}})$, the conditioned input embeddings are

$$H(x, p) = [E_{lm}(\tau(p^{\text{pre}})); P_\psi(Z_x); E_{lm}(\tau(p^{\text{post}}))] . \quad (8)$$

The projected time-series tokens are therefore inserted into the same embedding sequence as text tokens and can condition the causal LLM through its standard forward pass. When auxiliary textual descriptions are available, they are appended to the prompt and encoded by the same tokenizer. The objectives used to learn this interface are introduced together with the staged training schedule in Section 3.2.4.

3.2.3 Closed-Set Class-Token Scoring

Few-shot TSC requires prediction over a fixed set of valid labels rather than open-ended text generation. For each target dataset D , TimeMorph assigns a compact set of single-token class identifiers:

$$\mathcal{C}_D = \{\langle \text{cls}_0 \rangle, \dots, \langle \text{cls}_{C-1} \rangle\}, \quad \mathcal{V}_D = \{v_0, \dots, v_{C-1}\}, \quad (9)$$

where v_c is the vocabulary index of class token $\langle \text{cls}_c \rangle$. We implement these identifiers as newly added special tokens, so each class corresponds to exactly one vocabulary index. Let p_D denote the classification prompt for dataset D . After the LLM conditions on $H(x, p_D)$, let $h_\Theta(x, p_D)$ be the final hidden state at the last prompt position. With output embedding matrix W_{out} , the score for class c is the next-token logit assigned to its class token:

$$s_c(x) = [W_{\text{out}} h_\Theta(x, p_D)]_{v_c} . \quad (10)$$

Prediction is restricted to the valid class-token set:

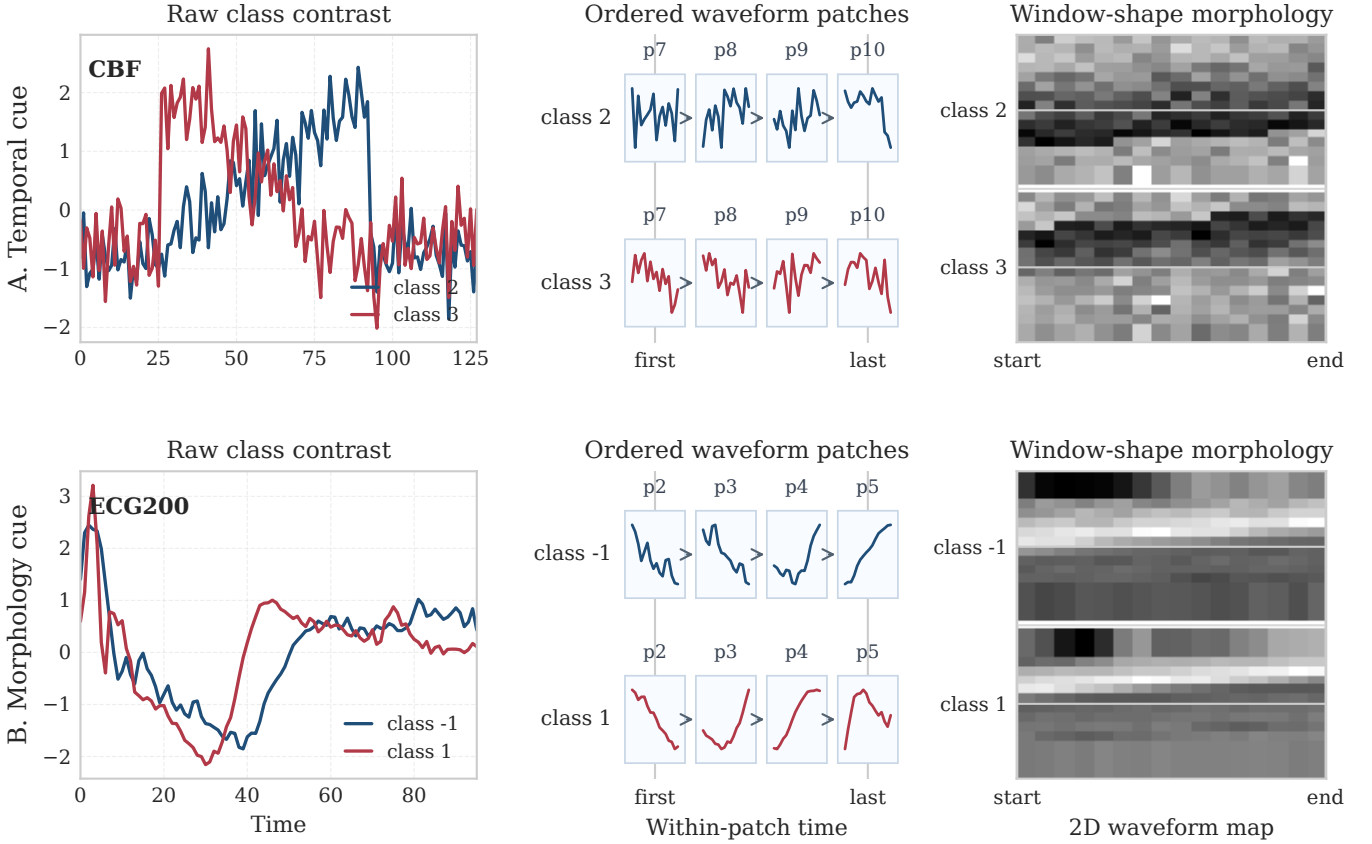
$$\hat{y} = \arg \max_{c \in \{0, \dots, C-1\}} s_c(x). \quad (11)$$

The supervised objective is the cross entropy over only the valid class tokens:

$$\mathcal{L}_{\text{ct}} = -\log \frac{\exp s_y(x)}{\sum_{c=0}^{C-1} \exp s_c(x)}. \quad (12)$$

This design keeps the autoregressive vocabulary head but converts downstream classification into a single-forward-pass closed-set comparison over C legal labels. Among vocabulary parameters, only the input embeddings and output rows of the newly introduced class tokens are trainable; all other vocabulary rows remain fixed.

3.2.3.1 Label-semantic priors. For anonymous-label benchmarks, the class tokens are learned as dataset-specific identifiers. When class names are semantically meaningful, TimeMorph uses label verbalizers to initialize and regularize the same class-token rows. For class c , let $\mathcal{R}_c$ denote a set



Real UCR TRAIN medoids; each row contrasts two representative class examples under temporal and morphology views.

Fig. 3. Qualitative motivation for the dual-view encoder. For representative UCR tasks, we visualize real training examples nearest to their class centroids, together with their raw sequences, ordered temporal patches, and morphology-map representations. The temporal view preserves ordered patch-level evolution, whereas the morphology view exposes local subsequence shapes and cross-window structural transitions through a two-dimensional waveform map.

of template-consistent verbalizers derived from its canonical label name. We form a semantic prototype by averaging the LLM input embeddings of all verbalizer tokens:

$$p_c = \frac{1}{|\mathcal{R}_c|} \sum_{r \in \mathcal{R}_c} \frac{1}{|\tau(r)|} \sum_{u \in \tau(r)} E_{lm}(u). \quad (13)$$

The prototype is rescaled to match the average norm of vocabulary rows and is used to initialize the input and output rows of the corresponding class token. During adaptation, we regularize these rows toward a detached prototype $\tilde{p}_c$:

$$\mathcal{L}_{sem} = \frac{1}{C} \sum_{c=0}^{C-1} [1 - \cos(e_{v_c}, \tilde{p}_c) + 1 - \cos(w_{v_c}, \tilde{p}_c)], \quad (14)$$

where e_{v_c} and w_{v_c} are the input and output rows of class token v_c . The adaptation loss is

$$\mathcal{L}_{adapt} = \mathcal{L}_{ct} + \lambda_{sem} \mathcal{L}_{sem}. \quad (15)$$

For datasets without meaningful label names, we set $\lambda_{sem} = 0$.

3.2.4 Staged Pretraining and Target Adaptation

TimeMorph learns the time-series-to-language interface through three pretraining stages followed by target-dataset

adaptation. Stage I learns reusable dual-view time-series representations. Stage II aligns projected time-series tokens with the embedding space of a pretrained causal LLM. Stage III optionally calibrates the interface with label-related language when semantic supervision is available. The default UCR benchmark uses the Stage-II checkpoint, while semantic-label external evaluations use the Stage-III calibrated checkpoint.

3.2.4.1 Stage I: Representation Pretraining. Stage I pretrains the dual-view encoder without class labels. In the default setting, we use unlabeled raw sequences from TSQA-train [40], M4-train [41], and the official TRAIN splits of a fixed 98-dataset UCR pretraining pool [42]. Only raw sequences from these TRAIN splits are used, with no access to UCR TEST sequences or class labels during pretraining.

The objective combines masked temporal patch reconstruction with an invariance–variance–covariance regularizer over augmented temporal, morphology, and fused representations:

$$\mathcal{L}_{rep} = \lambda_{rec} \mathcal{L}_{rec} + \sum_{b \in \{t, m, f\}} \lambda_b \mathcal{L}_{ivc}^b, \quad (16)$$

where t , m , and f denote the temporal, morphological, and fused branches, respectively. The reconstruction loss $\mathcal{L}_{rec}$ requires the temporal branch to recover original temporal patches from a contiguously masked sequence. The

regularizer $\mathcal{L}_{\text{ivc}}^b$ follows the invariance–variance–covariance principle: it aligns augmented views of the same sequence, prevents dimension-wise collapse, and penalizes redundant covariance [43]. We further apply branch dropout to prevent the fused representation from relying on a single view.

3.2.4.2 Stage II: Embedding-Space Alignment Pre-training.: Stage II attaches the projector P_ψ and the pre-trained causal LLM to the Stage-I dual-view encoder. Let $\mathcal{D}_{\text{TSQA}}$ denote the TSQA training split from ChatTime [40]. Each example is represented as (x, p, a) , where x is a raw time series, p is a natural-language or template-based question about the series, and a is the textual answer. The projected tokens $P_\psi(Z_x)$ are inserted into the prompt embedding sequence as in Eq. (8). The alignment objective is the answer-token language-modeling loss:

$$\mathcal{L}_{\text{align}} = \mathbb{E}_{(x,p,a) \sim \mathcal{D}_{\text{TSQA}}} [-\log p_\Theta(a \mid H(x, p))], \quad (17)$$

where the loss is applied only to answer tokens. This stage does not train TimeMorph to classify any target dataset. Instead, it teaches the projection interface to produce continuous tokens that the frozen LLM can condition on under its standard next-token objective. Following the implementation, we first optimize the projector with the dual-view encoder fixed, and then jointly optimize the encoder and projector. The LLM backbone remains frozen throughout Stage II. The resulting checkpoint initializes the default UCR target adaptation.

3.2.4.3 Stage III: Optional Label-Semantic Calibration.: Stage III is used when target evaluations provide meaningful class names or textual label descriptions. Starting from the Stage-II checkpoint, it further calibrates the interface toward label-related language using a mixture of TSQA retention examples, M4 captioning examples, and synthetic attribute supervision derived from TSQA and M4 raw series [40], [41]. The objective follows the same answer-token language-modeling form as Eq. (17), with auxiliary caption or attribute targets describing trends, periodicity, spikes, abrupt changes, and other interpretable temporal properties. For semantic-label external datasets such as MIT-BIH and CinC2017 [44], [45], target adaptation is initialized from the Stage-III checkpoint and uses the label-semantic class-token prior in Eq. (14).

3.2.4.4 Target-Dataset Adaptation.: For each target dataset D , adaptation proceeds in two phases. In the warm-up phase, TimeMorph updates the dual-view encoder, the projection module, and the dataset-specific class-token rows using the adaptation loss in Eq. (15). This phase calibrates the projected time-series tokens and the closed-set decision space before updating any LLM adapters.

In the joint adaptation phase, TimeMorph additionally updates low-rank adapters on the LLM side, while the pretrained LLM weights remain frozen. At inference time, TimeMorph performs a single forward pass for each query sequence and ranks the valid class tokens using Eq. (10). Implementation details, expanded training objectives, and staged-training data sources are provided in Appendix B.

4 EXPERIMENTS

4.0.0.1 Few-shot Protocol.: We evaluate on all 128 datasets in the UCR Archive 2018 with a fixed full-label-space

few-shot protocol. For each dataset D and $K \in \{1, 2, 5, 10\}$, all classes are included; we sample up to K labeled examples per class from the official training split and evaluate on all examples in the official test split. All methods use the same sampled support sets within each of five independent runs. We also evaluate semantic-label transfer on MIT-BIH and CinC2017 with $K \in \{10, 20, 30\}$, using the same full-label-space protocol and enabling label-name priors when available.

4.0.0.2 Pretraining and adaptation.: The main UCR benchmark uses the Stage-II checkpoint. Stage I is trained on raw TSQA-train series, raw M4-train series, and unlabeled official TRAIN-split sequences from a fixed 98-dataset UCR pretraining pool. Stage II is trained on the TSQA training split. Stage III is not used for the main UCR benchmark. External semantic-label evaluations use the Stage-III calibrated checkpoint.

4.0.0.3 Baselines.: We compare TimeMorph with representative few-shot and time-series classification baselines, including RESNET, TAPNET, COSCO-RESNET, PATCHTST, DLINEAR, TIMESNET, AUTOFORMER, CROSSFORMER, FEDFORMER, INFORMER, and GPT4TS. Accuracy is averaged over five runs and then macro-averaged over datasets. Avg averages the four shot settings, and AvgRank averages dataset-level ranks across shots. Standard deviations and full per-dataset tables are in Appendix A. Appendix B.8 lists the training recipes and hyperparameters.

4.0.0.4 Implementation.: TimeMorph uses Llama-3.2-1B as the causal language backbone and dinov2-base as the frozen morphology encoder. The temporal branch uses patch length 16, stride 8, and hidden size 128. For each target dataset, we perform 5 warm-up epochs followed by 55 joint-training epochs. LLM adaptation uses LoRA with rank 16 and alpha 32. Further training details are given in Appendix B.

4.1 Main Results

4.1.0.1 UCR few-shot classification.: Table 1 reports the main results on 128 UCR datasets. TimeMorph achieves the best mean accuracy in all four shot settings, obtaining 48.31, 55.55, 64.14, and 69.16 in the 1-, 2-, 5-, and 10-shot settings. It also obtains the best overall average accuracy and the best average rank. Compared with the strongest baseline, COSCO-RESNET, TimeMorph improves average accuracy from 58.39 to 59.29 and average rank from 4.70 to 4.24. The gains are larger over generic sequence models and LLM-style adaptation: TimeMorph improves the average accuracy by 4.53 points over INFORMER and 18.36 points over GPT4TS. These results show that TimeMorph consistently benefits from combining temporal dynamics, morphological structure, and LLM-based class-token scoring.

4.1.0.2 Dataset-level comparison.: We next examine whether the aggregate gains are distributed across datasets. Figure 4 shows win–tie–loss statistics against representative baselines. TimeMorph wins on substantially more datasets than it loses against PATCHTST, TIMESNET, and GPT4TS. The comparison with specialized CNN/prototype methods is closer, but TimeMorph still achieves the strongest overall average and rank.

Figure 5 reports the per-dataset accuracy distributions of representative baselines and TimeMorph. TimeMorph has

TABLE 1

Few-shot classification accuracy (%) on 128 UCR datasets, grouped by model family. We report the mean over five runs. Avg and AvgRank average the four shot-specific mean accuracies and average ranks, respectively. Best is bold and second-best is underlined.

Family	Model	1-shot	2-shot	5-shot	10-shot	Avg	AvgRank
CNN/prototype	ResNet	47.79	53.98	61.27	66.24	57.32	4.89
CNN/prototype	TapNet	45.40	50.79	59.47	64.93	55.15	5.45
CNN/prototype	COSCO-ResNet	<u>47.88</u>	<u>54.95</u>	<u>63.55</u>	<u>67.18</u>	<u>58.39</u>	<u>4.70</u>
Transformer/TS	DLinear	44.21	48.01	52.30	55.43	49.99	7.24
Transformer/TS	PatchTST	42.39	48.43	56.94	62.72	52.62	6.54
Transformer/TS	TimesNet	45.16	49.95	57.86	62.79	53.94	5.90
Transformer/TS	Autoformer	36.78	39.45	43.20	46.04	41.37	9.47
Transformer/TS	Crossformer	44.31	49.45	57.20	61.94	53.23	6.04
Transformer/TS	FEDformer	37.71	41.71	47.30	51.50	44.56	8.68
Transformer/TS	Informer	46.42	51.14	58.35	63.13	54.76	5.59
LLM/foundation	GPT4TS	35.61	38.30	42.94	46.88	40.93	9.26
Ours	TimeMorph	48.31	55.55	64.14	69.16	59.29	4.24

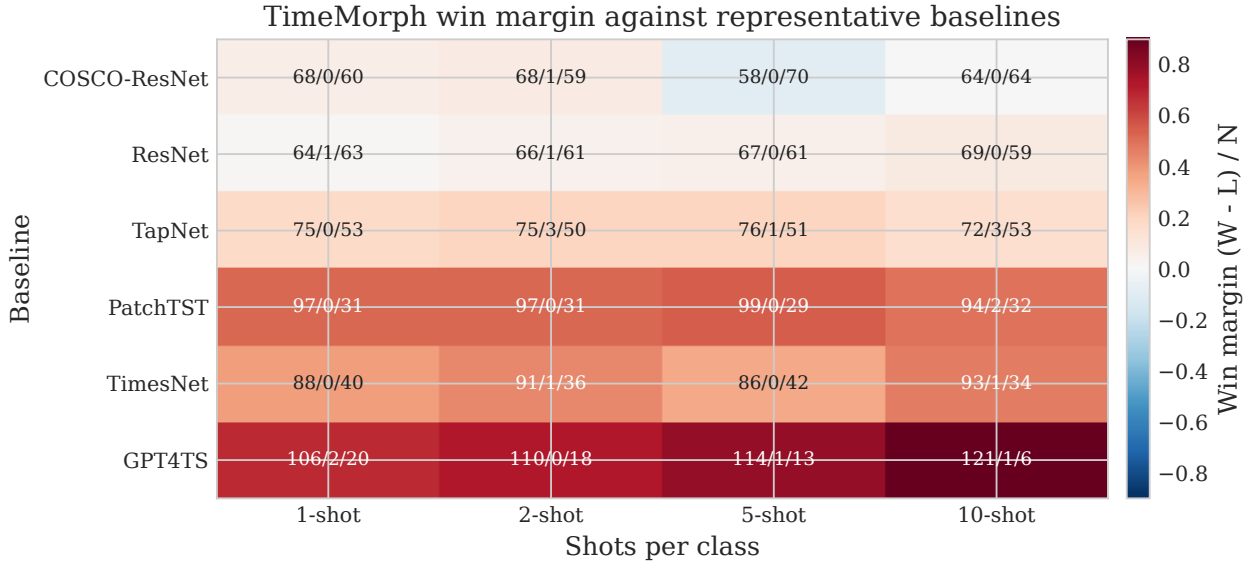


Fig. 4. Win–tie–loss heatmap against representative UCR baselines. Each cell reports wins/ties/losses over 128 datasets, with color indicating the normalized win margin $(W - L) / N$.

the highest mean accuracy in all four shot settings, with a distribution that is especially separated from the Transformer-style and LLM-style baselines. The comparison with specialized CNN/prototype baselines is closer, matching the aggregate results in Table 1: TimeMorph provides broad gains over generic sequence models and remains competitive with specialized few-shot classifiers.

4.1.0.3 External semantic-label datasets.: We further evaluate TimeMorph on MIT-BIH and CinC2017 under 10-, 20-, and 30-shot settings. These datasets provide meaningful class names, allowing TimeMorph to use Stage-III semantic calibration and label-verbalizer priors. Table 2 reports the average accuracy over the three shot settings. TimeMorph obtains the best average accuracy on both datasets. The result

confirms that the semantic class-token interface transfers beyond UCR and can exploit informative label names when they are available.

4.2 Efficiency and Adaptation Cost

Table 3 reports the parameter and adaptation cost of TimeMorph. Although TimeMorph uses a large frozen LLM backbone, only 12.3M parameters are updated for each target dataset through LoRA and task-specific components. This gives TimeMorph a favorable adaptation-cost profile while preserving the transfer capacity of the pretrained language model. Full efficiency statistics, including throughput and memory usage, are reported in Appendix B.7.

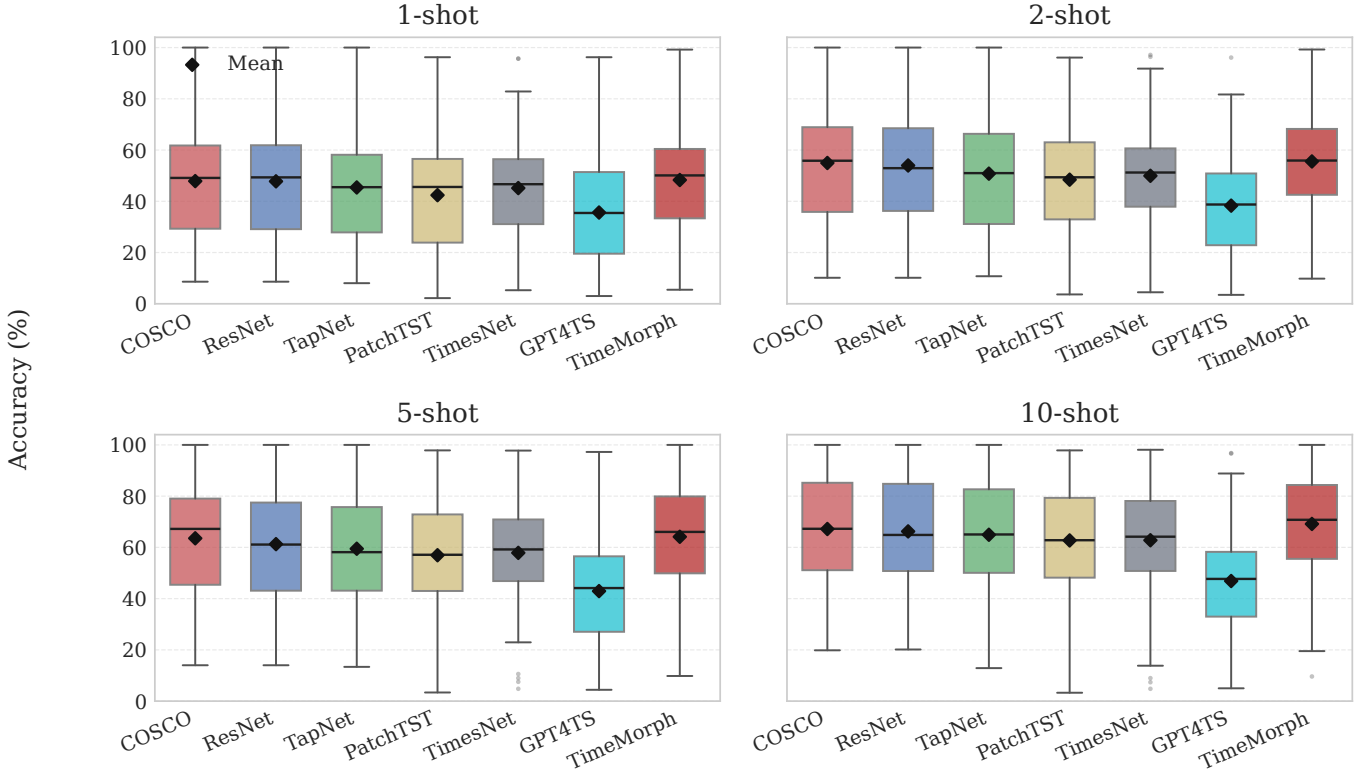


Fig. 5. Per-dataset accuracy distribution on the 128 UCR datasets. Each box summarizes one method's accuracy across datasets for a fixed shot setting; diamonds mark mean accuracies.

TABLE 2

Average few-shot accuracy (%) on two external semantic-label datasets, computed over 10-, 20-, and 30-shot settings with 10 runs per setting. Best is bold and second-best is underlined.

Model	MIT-BIH	CinC2017	Avg.
TimeMorph	61.04	34.63	47.84
ResNet	50.23	<u>33.63</u>	41.93
TapNet	38.73	31.29	35.01
COSCO-ResNet	36.19	32.48	34.34
PatchTST	50.46	29.21	39.84
OneFitsAll	43.42	29.19	36.31
TimesNet	53.79	28.36	41.08
DLinear	<u>57.29</u>	24.87	41.08
Autoformer	41.64	27.45	34.55
Crossformer	56.76	25.62	<u>41.19</u>
FEDformer	56.93	25.05	40.99
Informer	53.25	28.39	40.82

4.3 Ablation Studies

Table 4 summarizes the main ablations on all 128 UCR datasets. Each variant follows the same few-shot protocol and changes only one component of TimeMorph. Figure 6 visualizes the average accuracy drop caused by removing each component.

4.3.0.1 Effect of staged pretraining: Removing staged pretraining drops Avg from 59.29 to 55.94, with shot-level accuracies decreasing from 48.31/55.55/64.14/69.16 to 44.70/51.81/60.65/66.60 for 1-/2-/5-/10-shot, respectively. The focused view in Appendix Table 20 reports a 3.35-point average drop, suggesting that the aggregate Stage I+II initialization helps when target supervision is too limited

TABLE 3

Compact efficiency comparison under the 10-shot UCR protocol. Runtime and memory are measured on the representative UCR runtime subset. Total parameters count the deployed model; updated parameters count downstream-adapted parameters only.

Model	Total params	Updated params	Peak GB	Infer ms/ex.
TimeMorph	1.28B	12.3M	13.75	156.85
COSCO-ResNet	662.1K	662.1K	0.58	0.25
ResNet	629.1K	629.1K	0.33	0.26
PatchTST	665.4K	598.3K	1.54	0.75
GPT4TS	82.2M	1.1M	0.56	0.86

TABLE 4

Summary of top-level ablations on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs. Best is bold, and the best ablated variant in each column is underlined. Since UCR labels are anonymous, the label-space ablation isolates closed-set class-token scoring rather than semantic label priors.

Variant	1-shot	2-shot	5-shot	10-shot	Avg
TimeMorph	48.31	55.55	64.14	69.16	59.29
w/o Curriculum Pretraining	44.70	51.81	60.65	66.60	55.94
Temporal only	43.51	50.55	58.40	64.39	54.21
Visual only	<u>45.88</u>	50.02	<u>60.86</u>	63.58	55.09
w/o Class-Token Label-Space Scoring	44.46	50.44	60.08	65.59	55.14

to learn the representation and LLM interface from scratch. This ablation removes the full pretraining pipeline at once and should not be read as isolating each substage.

4.3.0.2 Effect of dual-view time-series-to-language modeling: Temporal-only and morphology-only variants reach Avg scores of 54.21 and 55.09, below the full model's

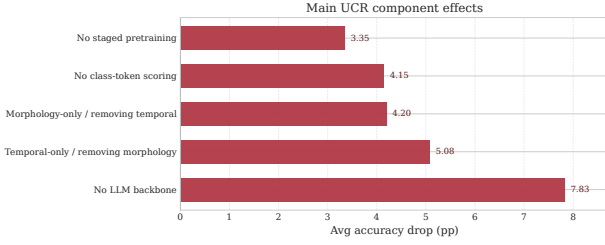


Fig. 6. Summary of the main component effects. Each bar reports the average UCR accuracy drop when removing one TimeMorph component under the same 128-dataset few-shot protocol.

TABLE 5

Semantic-prior class-token ablation on two external semantic-label datasets. Values are average accuracy (%) over 10-, 20-, and 30-shot settings with 10 runs per setting. Δ denotes Semantic prior minus Anonymous labels.

Variant	MIT-BIH	CinC2017	Avg.
Anonymous class tokens	61.04	34.63	47.84
Semantic-prior class tokens	64.18	36.27	50.23
Δ	+3.14	+1.64	+2.39

59.29. Appendix Table 21 shows that removing the morphology branch yields a 5.08-point average drop, while removing the temporal branch yields a 4.20-point drop. The temporal-only variant is weaker at 1-, 2-, and 5-shot, whereas the morphology-only variant is weaker at 2- and 10-shot. Neither single view dominates across all shots, which supports the intended complementarity between chronological and morphological evidence.

4.3.0.3 Effect of class-token scoring.: Disabling class-token scoring reduces Avg from 59.29 to 55.14. As detailed in Appendix Table 22, the corresponding shot-level accuracies decrease from 48.31/55.55/64.14/69.16 to 44.46/50.44/60.08/65.59, giving a 4.15-point average drop. This indicates that, under the same adaptation budget, unconstrained generative adaptation is less stable for closed-set few-shot classification than directly scoring the dataset-specific label tokens.

4.3.0.4 Effect of semantic label priors.: For external datasets with meaningful label names, Table 5 compares anonymous class-token rows with rows initialized and regularized from label-verbalizer prototypes, using the Stage III calibrated checkpoint for both variants. Semantic priors improve the 10/20/30-shot average on MIT-BIH and CinC2017, increasing the overall average from 47.84 to 50.23. The full appendix table shows that the benefit is not uniform at every shot, with a decrease on CinC2017 20-shot. The averaged result supports the semantic-prior class-token space as a useful extension of supervised closed-set decoding.

4.3.0.5 Effect of the LLM backbone.: The no-LLM variant replaces the language pathway with a lightweight linear classifier on the same dual-view features. As shown in Table 23, this ablation is evaluated on all 128 UCR datasets under the same five-run protocol. Removing the LLM backbone reduces Avg from 59.29 to 51.46, a 7.83-point drop. The decrease grows with the number of support examples: accuracy changes from 48.31 to 44.89 at 1-shot, 55.55 to

48.29 at 2-shot, 64.14 to 54.22 at 5-shot, and 69.16 to 58.43 at 10-shot. This suggests that, under our implementation and adaptation schedule, the language pathway contributes more than a lightweight classifier on the same features; however, this ablation does not isolate whether the gain comes from pretrained knowledge, additional capacity, or the token-scoring interface.

Overall, the ablations support the central design: staged pretraining improves transferability, dual-view modeling preserves more complete evidence, closed-set class-token scoring stabilizes the final decision, and semantic-prior row initialization and regularization further improve semantic-label transfer.

APPENDIX A**FULL UCR BENCHMARK RESULTS**

This appendix reports the complete per-dataset few-shot benchmark results for all 128 UCR datasets. Each cell is shown as mean $\pm$ standard deviation over five runs, and the summary rows at the end of each table report average accuracy, average rank, and the number of datasets attaining the best mean accuracy (counting ties).

TABLE 6: Per-dataset 1-shot classification results on 128 UCR datasets. Each value is reported as mean $\pm$ standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
ACSF1	53.80 $\pm$ 4.97	42.00 $\pm$ 4.95	48.40 $\pm$ 7.30	24.00 $\pm$ 4.24	35.60 $\pm$ 3.21	27.67 $\pm$ 7.02	24.33 $\pm$ 9.71	28.33 $\pm$ 2.52	27.80 $\pm$ 7.09	31.67 $\pm$ 9.81	18.60 $\pm$ 4.51	44.00 $\pm$ 5.20
Adiac	33.81 $\pm$ 1.23	12.84 $\pm$ 3.09	33.81 $\pm$ 1.23	8.85 $\pm$ 2.36	11.97 $\pm$ 1.69	21.40 $\pm$ 4.24	34.53 $\pm$ 1.77	8.10 $\pm$ 1.92	40.82 $\pm$ 3.61	24.21 $\pm$ 3.24	3.02 $\pm$ 1.05	45.17 $\pm$ 5.28
AllGestureWiimoteX	25.71 $\pm$ 4.01	26.86 $\pm$ 2.52	24.43 $\pm$ 2.55	15.57 $\pm$ 2.95	21.86 $\pm$ 4.59	20.71 $\pm$ 2.87	11.33 $\pm$ 1.19	16.76 $\pm$ 2.52	13.43 $\pm$ 1.74	18.90 $\pm$ 1.08	21.86 $\pm$ 4.59	25.11 $\pm$ 4.65
AllGestureWiimoteY	26.37 $\pm$ 2.89	26.17 $\pm$ 2.88	22.71 $\pm$ 4.62	17.49 $\pm$ 2.02	25.17 $\pm$ 5.50	21.81 $\pm$ 2.78	12.29 $\pm$ 1.27	20.29 $\pm$ 1.98	13.06 $\pm$ 1.34	21.33 $\pm$ 2.38	25.17 $\pm$ 5.50	25.54 $\pm$ 3.42
AllGestureWiimoteZ	27.11 $\pm$ 1.95	22.66 $\pm$ 4.13	25.49 $\pm$ 2.57	18.26 $\pm$ 2.01	19.71 $\pm$ 2.87	19.24 $\pm$ 1.57	13.29 $\pm$ 0.43	18.76 $\pm$ 2.79	14.14 $\pm$ 1.72	19.71 $\pm$ 1.79	19.71 $\pm$ 2.87	22.26 $\pm$ 4.96
ArrowHead	47.66 $\pm$ 8.94	46.63 $\pm$ 9.45	48.00 $\pm$ 7.74	52.00 $\pm$ 6.38	49.71 $\pm$ 1.40	47.62 $\pm$ 9.46	38.29 $\pm$ 4.00	50.86 $\pm$ 1.71	35.89 $\pm$ 2.60	50.86 $\pm$ 6.05	41.49 $\pm$ 12.34	52.57 $\pm$ 3.43
BME	49.60 $\pm$ 2.85	50.80 $\pm$ 2.13	49.73 $\pm$ 3.42	56.67 $\pm$ 4.43	62.93 $\pm$ 5.35	55.33 $\pm$ 7.86	51.56 $\pm$ 3.15	61.33 $\pm$ 4.06	55.87 $\pm$ 7.71	58.89 $\pm$ 6.19	35.73 $\pm$ 3.22	66.40 $\pm$ 3.88
Beef	36.00 $\pm$ 6.41	31.33 $\pm$ 5.06	27.33 $\pm$ 7.96	38.67 $\pm$ 9.89	32.00 $\pm$ 7.67	40.00 $\pm$ 8.82	34.44 $\pm$ 6.94	41.11 $\pm$ 1.92	30.67 $\pm$ 6.83	40.00 $\pm$ 6.67	31.33 $\pm$ 9.31	34.00 $\pm$ 2.79
BeetleFly	55.00 $\pm$ 24.24	60.00 $\pm$ 25.00	61.00 $\pm$ 24.85	67.00 $\pm$ 4.47	61.00 $\pm$ 9.62	60.00 $\pm$ 13.23	55.00 $\pm$ 5.00	61.67 $\pm$ 11.55	48.00 $\pm$ 2.74	60.00 $\pm$ 13.23	70.00 $\pm$ 11.18	70.00 $\pm$ 11.73
BirdChicken	62.00 $\pm$ 8.37	71.00 $\pm$ 9.62	70.00 $\pm$ 17.68	56.00 $\pm$ 4.18	56.00 $\pm$ 4.18	51.67 $\pm$ 2.89	51.67 $\pm$ 12.58	53.33 $\pm$ 5.77	55.00 $\pm$ 5.00	51.67 $\pm$ 7.64	59.00 $\pm$ 9.62	68.00 $\pm$ 14.83
CBF	81.67 $\pm$ 6.47	87.62 $\pm$ 2.73	71.71 $\pm$ 7.91	54.02 $\pm$ 3.44	66.38 $\pm$ 7.40	55.85 $\pm$ 2.16	33.63 $\pm$ 1.20	58.00 $\pm$ 5.61	36.89 $\pm$ 2.37	54.78 $\pm$ 1.68	35.20 $\pm$ 3.75	72.48 $\pm$ 4.85
Car	37.00 $\pm$ 4.62	39.00 $\pm$ 4.80	41.67 $\pm$ 9.28	38.67 $\pm$ 11.63	27.33 $\pm$ 9.90	42.22 $\pm$ 5.09	24.44 $\pm$ 3.47	43.89 $\pm$ 5.85	27.00 $\pm$ 4.15	40.00 $\pm$ 7.26	30.00 $\pm$ 9.65	55.67 $\pm$ 8.94
Chinatown	66.53 $\pm$ 19.14	63.73 $\pm$ 26.98	62.51 $\pm$ 27.20	67.41 $\pm$ 23.86	67.35 $\pm$ 11.32	77.07 $\pm$ 16.93	81.83 $\pm$ 2.19	77.94 $\pm$ 9.20	77.61 $\pm$ 8.88	75.41 $\pm$ 7.92	56.68 $\pm$ 20.52	93.59 $\pm$ 3.92
ChlorineConcentration	34.61 $\pm$ 2.26	35.41 $\pm$ 4.36	34.04 $\pm$ 2.01	33.26 $\pm$ 2.96	35.54 $\pm$ 3.20	33.64 $\pm$ 3.96	32.92 $\pm$ 4.26	33.42 $\pm$ 5.09	33.78 $\pm$ 2.74	34.11 $\pm$ 3.50	39.93 $\pm$ 15.64	32.61 $\pm$ 3.30
CinCECGTorso	37.20 $\pm$ 9.73	36.03 $\pm$ 6.56	36.51 $\pm$ 8.82	40.06 $\pm$ 3.86	35.72 $\pm$ 7.09	44.86 $\pm$ 6.03	35.36 $\pm$ 5.77	50.53 $\pm$ 4.66	34.97 $\pm$ 4.68	45.68 $\pm$ 5.50	42.49 $\pm$ 6.76	48.74 $\pm$ 6.47
Coffee	78.57 $\pm$ 9.78	92.86 $\pm$ 8.38	84.29 $\pm$ 13.02	72.86 $\pm$ 14.42	74.29 $\pm$ 18.80	63.10 $\pm$ 10.91	65.48 $\pm$ 7.43	71.43 $\pm$ 16.37	76.43 $\pm$ 4.07	64.29 $\pm$ 7.14	51.43 $\pm$ 4.07	75.71 $\pm$ 13.69
Computers	57.52 $\pm$ 7.64	60.40 $\pm$ 6.13	59.12 $\pm$ 4.47	47.12 $\pm$ 5.87	52.64 $\pm$ 10.26	50.93 $\pm$ 1.29	48.80 $\pm$ 0.69	51.87 $\pm$ 2.57	48.96 $\pm$ 2.79	52.27 $\pm$ 3.23	45.20 $\pm$ 5.79	56.40 $\pm$ 5.77
CricketX	26.15 $\pm$ 3.29	26.10 $\pm$ 3.67	20.56 $\pm$ 4.67	17.28 $\pm$ 1.96	18.15 $\pm$ 4.50	20.34 $\pm$ 0.39	8.89 $\pm$ 0.82	17.52 $\pm$ 1.26	11.38 $\pm$ 2.35	20.68 $\pm$ 0.74	7.28 $\pm$ 1.05	26.26 $\pm$ 5.29
CricketY	26.97 $\pm$ 3.03	24.46 $\pm$ 4.25	22.72 $\pm$ 1.74	18.56 $\pm$ 4.13	17.90 $\pm$ 3.61	22.82 $\pm$ 1.56	10.34 $\pm$ 1.89	20.94 $\pm$ 3.72	11.23 $\pm$ 0.80	21.20 $\pm$ 3.98	9.54 $\pm$ 2.33	23.33 $\pm$ 5.19
CricketZ	32.51 $\pm$ 2.32	28.72 $\pm$ 2.15	26.56 $\pm$ 4.19	19.64 $\pm$ 2.73	22.26 $\pm$ 3.91	20.34 $\pm$ 1.11	11.03 $\pm$ 1.54	20.00 $\pm$ 2.31	12.26 $\pm$ 1.25	20.34 $\pm$ 2.57	7.64 $\pm$ 1.05	29.28 $\pm$ 4.82
Crop	25.10 $\pm$ 4.94	28.14 $\pm$ 2.90	25.10 $\pm$ 4.94	10.23 $\pm$ 2.66	17.27 $\pm$ 1.43	27.73 $\pm$ 2.89	24.10 $\pm$ 4.30	26.19 $\pm$ 2.96	27.27 $\pm$ 3.58	30.31 $\pm$ 5.00	3.80 $\pm$ 0.58	36.48 $\pm$ 3.38
DiatomSizeReduction	79.67 $\pm$ 5.51	55.36 $\pm$ 19.14	84.12 $\pm$ 4.82	89.22 $\pm$ 8.91	49.61 $\pm$ 24.86	75.71 $\pm$ 7.74	55.01 $\pm$ 11.77	56.43 $\pm$ 20.40	50.07 $\pm$ 12.66	76.47 $\pm$ 11.70	41.44 $\pm$ 13.25	85.82 $\pm$ 8.55
DistalPhalanxOutlineAgeGroup	43.02 $\pm$ 17.16	43.02 $\pm$ 18.34	43.60 $\pm$ 14.92	42.73 $\pm$ 19.01	46.76 $\pm$ 16.49	45.80 $\pm$ 19.81	46.28 $\pm$ 21.27	56.59 $\pm$ 9.36	50.94 $\pm$ 19.66	46.28 $\pm$ 20.78	35.68 $\pm$ 14.61	42.01 $\pm$ 16.44
DistalPhalanxOutlineCorrect	62.10 $\pm$ 1.32	57.32 $\pm$ 9.22	63.04 $\pm$ 2.44	57.54 $\pm$ 11.13	53.55 $\pm$ 11.63	53.99 $\pm$ 14.78	53.26 $\pm$ 12.27	53.62 $\pm$ 11.97	55.14 $\pm$ 10.52	53.26 $\pm$ 12.97	56.74 $\pm$ 7.30	60.58 $\pm$ 2.98
DistalPhalanxTW	51.22 $\pm$ 5.15	44.03 $\pm$ 11.70	53.38 $\pm$ 4.02	43.74 $\pm$ 5.17	51.22 $\pm$ 6.58	45.32 $\pm$ 9.89	42.69 $\pm$ 4.40	38.13 $\pm$ 7.51	48.20 $\pm$ 5.84	47.48 $\pm$ 5.62	18.56 $\pm$ 14.51	54.53 $\pm$ 1.72
DodgerLoopDay	32.50 $\pm$ 7.34	25.00 $\pm$ 4.05	29.75 $\pm$ 7.68	41.50 $\pm$ 2.05	26.00 $\pm$ 6.58	40.42 $\pm$ 4.73	20.42 $\pm$ 1.91	32.92 $\pm$ 3.15	21.50 $\pm$ 4.45	40.42 $\pm$ 7.94	26.00 $\pm$ 6.58	34.50 $\pm$ 4.78
DodgerLoopGame	52.17 $\pm$ 6.30	55.07 $\pm$ 17.38	49.13 $\pm$ 8.90	60.87 $\pm$ 12.35	59.86 $\pm$ 22.05	51.45 $\pm$ 9.96	51.21 $\pm$ 5.14	56.52 $\pm$ 10.82	50.43 $\pm$ 2.44	51.45 $\pm$ 9.59	59.86 $\pm$ 22.05	56.96 $\pm$ 11.69
DodgerLoopWeekend	84.49 $\pm$ 8.37	74.78 $\pm$ 12.27	83.19 $\pm$ 10.24	93.62 $\pm$ 3.81	96.23 $\pm$ 1.19	95.65 $\pm$ 1.26	63.77 $\pm$ 4.41	94.44 $\pm$ 0.42	68.41 $\pm$ 4.57	93.72 $\pm$ 5.86	96.23 $\pm$ 1.19	85.36 $\pm$ 4.30
ECG200	63.00 $\pm$ 12.61	72.00 $\pm$ 6.75	66.20 $\pm$ 8.32	71.80 $\pm$ 4.44	61.60 $\pm$ 12.82	61.33 $\pm$ 10.97	65.00 $\pm$ 11.79	63.00 $\pm$ 20.66	64.20 $\pm$ 14.29	68.67 $\pm$ 7.09	51.40 $\pm$ 12.93	60.60 $\pm$ 6.91
ECG5000	54.22 $\pm$ 21.00	50.05 $\pm$ 20.59	61.53 $\pm$ 13.66	62.80 $\pm$ 19.19	53.35 $\pm$ 20.14	69.07 $\pm$ 12.52	54.44 $\pm$ 9.79	57.20 $\pm$ 16.09	62.35 $\pm$ 12.99	68.84 $\pm$ 13.99	19.70 $\pm$ 25.30	67.18 $\pm$ 10.57
ECGFiveDays	58.65 $\pm$ 7.69	57.24 $\pm$ 6.51	54.80 $\pm$ 13.00	57.63 $\pm$ 2.63	62.21 $\pm$ 8.35	57.99 $\pm$ 4.84	54.86 $\pm$ 3.17	55.87 $\pm$ 1.56	54.08 $\pm$ 6.73	58.46 $\pm$ 5.08	51.54 $\pm$ 6.38	61.77 $\pm$ 6.61
EOGHorizontalSignal	20.11 $\pm$ 2.01	22.38 $\pm$ 1.86	22.60 $\pm$ 2.77	22.43 $\pm$ 4.01	21.22 $\pm$ 3.15	20.63 $\pm$ 4.63	15.29 $\pm$ 2.63	23.57 $\pm$ 5.45	10.39 $\pm$ 2.26	22.01 $\pm$ 4.37	27.24 $\pm$ 2.99	29.06 $\pm$ 4.16
EOGVerticalSignal	11.27 $\pm$ 3.67	14.09 $\pm$ 3.23	13.26 $\pm$ 3.09	24.20 $\pm$ 3.27	23.76 $\pm$ 3.73	23.39 $\pm$ 2.79	11.23 $\pm$ 0.70	24.86 $\pm$ 4.17	11.27 $\pm$ 3.83	23.11 $\pm$ 4.07	19.06 $\pm$ 6.08	27.35 $\pm$ 1.90
Earthquakes	50.07 $\pm$ 8.55	51.08 $\pm$ 6.33	50.94 $\pm$ 6.46	49.35 $\pm$ 5.05	45.61 $\pm$ 20.94	48.68 $\pm$ 19.16	49.40 $\pm$ 1.10	49.40 $\pm$ 14.03	47.05 $\pm$ 5.90	47.00 $\pm$ 24.65	46.04 $\pm$ 13.71	55.54 $\pm$ 11.57
ElectricDevices	28.83 $\pm$ 9.01	32.82 $\pm$ 12.31	28.35 $\pm$ 12.24	17.59 $\pm$ 1.67	27.21 $\pm$ 7.38	21.93 $\pm$ 4.54	17.11 $\pm$ 1.87	15.12 $\pm$ 2.81	17.04 $\pm$ 2.05	15.61 $\pm$ 7.10	17.06 $\pm$ 3.10	33.14 $\pm$ 10.96
EthanolLevel	24.44 $\pm$ 1.30	25.40 $\pm$ 0.85	25.28 $\pm$ 1.57	24.92 $\pm$ 1.50	23.80 $\pm$ 1.65	26.13 $\pm$ 1.58	26.53 $\pm$ 1.55	26.67 $\pm$ 1.29	25.32 $\pm$ 1.50	25.80 $\pm$ 1.39	24.76 $\pm$ 1.40	26.68 $\pm$ 1.86
FaceAll	42.17 $\pm$ 4.79	32.36 $\pm$ 9.47	42.17 $\pm$ 4.79	28.66 $\pm$ 9.09	19.36 $\pm$ 3.07	29.11 $\pm$ 4.19	33.12 $\pm$ 5.46	25.94 $\pm$ 1.75	34.05 $\pm$ 3.40	32.92 $\pm$ 2.10	8.64 $\pm$ 3.46	32.84 $\pm$ 6.86
FaceFour	68.64 $\pm$ 10.56	47.50 $\pm$ 11.12	56.36 $\pm$ 4.30	64.32 $\pm$ 4.73	54.09 $\pm$ 1.93	70.08 $\pm$ 4.30	65.65 $\pm$ 3.99	67.05 $\pm$ 7.10	51.36 $\pm$ 4.28	73.11 $\pm$ 3.99	42.95 $\pm$ 6.60	54.77 $\pm$ 8.25
FacesUCR	54.83 $\pm$ 5.21	37.63 $\pm$ 5.89	43.65 $\pm$ 3.88	35.69 $\pm$ 7.47	20.81 $\pm$ 1.95	36.24 $\pm$ 3.26	41.98 $\pm$ 2.95	40.05 $\pm$ 2.80	41.87 $\pm$ 3.09	42.63 $\pm$ 2.58	16.92 $\pm$ 7.54	50.37 $\pm$ 2.98
FiftyWords	13.27 $\pm$ 1.89	12.79 $\pm$ 3.45	13.27 $\pm$ 1.89	32.70 $\pm$ 4.33	19.60 $\pm$ 4.90	35.90 $\pm$ 3.31	3.44 $\pm$ 1.59	38.97 $\pm$ 6.44	11.30 $\pm$ 2.86	36.70 $\pm$ 4.41	3.74 $\pm$ 2.40	36.48 $\pm$ 4.96
Fish	49.03 $\pm$ 5.96	21.26 $\pm$ 3.60	45.71 $\pm$ 1.71	36.00 $\pm$ 3.98	23.20 $\pm$ 3.80	32.57 $\pm$ 4.57	16.95 $\pm$ 2.93	30.29 $\pm$ 2.62	21.71 $\pm$ 2.77	31.81 $\pm$ 4.76	20.80 $\pm$ 5.10	34.74 $\pm$ 9.41
FordA	59.23 $\pm$ 9.68	57.68 $\pm$ 10.00	59.83 $\pm$ 8.76	50.73 $\pm$ 1.01	56.70 $\pm$ 8.04	50.30 $\pm$ 0.33	49.57 $\pm$ 1.42	49.72 $\pm$ 0.63	49.6			

TABLE 6: Per-dataset 1-shot classification results on 128 UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
MixedShapesSmallTrain	52.78 $\pm$ 8.02	39.55 $\pm$ 6.53	48.84 $\pm$ 9.40	53.90 $\pm$ 7.66	45.57 $\pm$ 7.91	52.81 $\pm$ 6.78	17.32 $\pm$ 0.61	53.84 $\pm$ 4.25	24.17 $\pm$ 3.24	53.31 $\pm$ 5.69	48.73 $\pm$ 8.44	63.30 $\pm$ 3.34
MoteStrain	62.99 $\pm$ 13.96	60.67 $\pm$ 12.58	64.35 $\pm$ 11.70	63.40 $\pm$ 15.50	63.27 $\pm$ 13.76	68.45 $\pm$ 11.94	56.26 $\pm$ 12.55	63.82 $\pm$ 16.98	61.65 $\pm$ 9.57	68.66 $\pm$ 11.89	57.99 $\pm$ 9.98	54.36 $\pm$ 12.77
NonInvasiveFetalECGThorax1	19.05 $\pm$ 4.42	8.04 $\pm$ 2.52	19.05 $\pm$ 4.42	38.23 $\pm$ 1.81	12.07 $\pm$ 7.41	36.76 $\pm$ 3.30	12.57 $\pm$ 2.38	28.62 $\pm$ 3.97	14.44 $\pm$ 2.07	42.58 $\pm$ 4.62	5.58 $\pm$ 2.48	39.16 $\pm$ 2.28
NonInvasiveFetalECGThorax2	20.69 $\pm$ 5.30	9.71 $\pm$ 2.65	20.69 $\pm$ 5.30	47.48 $\pm$ 2.72	14.81 $\pm$ 5.33	46.67 $\pm$ 0.18	20.97 $\pm$ 3.22	34.59 $\pm$ 7.78	23.69 $\pm$ 1.73	49.89 $\pm$ 4.96	8.30 $\pm$ 1.72	47.48 $\pm$ 2.06
OSULeaf	45.62 $\pm$ 7.45	30.41 $\pm$ 8.35	38.35 $\pm$ 12.00	24.05 $\pm$ 2.14	22.15 $\pm$ 5.63	27.82 $\pm$ 3.13	17.77 $\pm$ 1.80	26.03 $\pm$ 1.89	20.00 $\pm$ 5.09	26.17 $\pm$ 1.56	17.60 $\pm$ 2.44	29.83 $\pm$ 5.82
OliveOil	28.00 $\pm$ 23.64	26.67 $\pm$ 11.06	47.33 $\pm$ 13.21	65.33 $\pm$ 15.20	36.00 $\pm$ 22.66	32.22 $\pm$ 16.44	44.44 $\pm$ 10.72	28.89 $\pm$ 11.71	32.00 $\pm$ 9.60	61.11 $\pm$ 15.40	27.33 $\pm$ 11.88	28.00 $\pm$ 14.26
PLAID	15.46 $\pm$ 6.16	13.97 $\pm$ 4.45	16.95 $\pm$ 4.17	17.77 $\pm$ 3.34	21.79 $\pm$ 0.97	21.54 $\pm$ 5.77	13.78 $\pm$ 3.08	19.93 $\pm$ 5.41	14.41 $\pm$ 2.63	20.30 $\pm$ 4.75	21.79 $\pm$ 0.97	32.10 $\pm$ 5.72
PhalangesOutlinesCorrect	54.90 $\pm$ 6.98	51.28 $\pm$ 10.93	54.41 $\pm$ 7.38	53.40 $\pm$ 8.33	52.54 $\pm$ 10.38	48.87 $\pm$ 4.48	51.01 $\pm$ 10.34	53.38 $\pm$ 8.94	52.73 $\pm$ 9.70	52.49 $\pm$ 9.30	<u>54.83 $\pm$ 6.32</u>	51.93 $\pm$ 9.54
Phoneme	10.43 $\pm$ 1.42	8.46 $\pm$ 1.00	10.43 $\pm$ 1.42	4.21 $\pm$ 0.54	5.61 $\pm$ 2.38	6.14 $\pm$ 1.81	2.39 $\pm$ 1.01	5.87 $\pm$ 1.54	3.20 $\pm$ 2.06	4.98 $\pm$ 3.20	3.42 $\pm$ 0.41	6.87 $\pm$ 1.15
PickupGestureWiimoteZ	35.60 $\pm$ 7.80	44.80 $\pm$ 4.82	42.00 $\pm$ 5.10	40.80 $\pm$ 5.40	36.00 $\pm$ 3.16	46.67 $\pm$ 9.87	17.33 $\pm$ 5.03	39.33 $\pm$ 6.43	20.40 $\pm$ 4.98	48.67 $\pm$ 8.33	36.00 $\pm$ 3.16	<u>47.20 $\pm$ 7.29</u>
PigAirwayPressure	23.94 $\pm$ 3.76	9.13 $\pm$ 2.66	38.65 $\pm$ 2.92	3.65 $\pm$ 1.47	3.75 $\pm$ 1.33	5.29 $\pm$ 0.83	4.01 $\pm$ 0.73	4.01 $\pm$ 1.11	3.27 $\pm$ 0.92	4.81 $\pm$ 0.96	6.15 $\pm$ 1.64	5.48 $\pm$ 2.19
PigArtPressure	<u>40.58 $\pm$ 5.14</u>	10.10 $\pm$ 2.36	60.10 $\pm$ 2.28	9.90 $\pm$ 2.42	2.21 $\pm$ 0.80	10.58 $\pm$ 2.10	8.01 $\pm$ 2.00	10.58 $\pm$ 1.73	4.81 $\pm$ 0.76	9.94 $\pm$ 2.27	15.58 $\pm$ 3.05	15.38 $\pm$ 4.07
PigCVP	<u>22.50 $\pm$ 3.47</u>	14.33 $\pm$ 2.79	50.29 $\pm$ 2.90	5.00 $\pm$ 1.25	3.27 $\pm$ 0.71	5.93 $\pm$ 0.73	3.53 $\pm$ 1.39	6.89 $\pm$ 0.56	2.02 $\pm$ 0.22	4.33 $\pm$ 0.83	8.94 $\pm$ 1.94	11.44 $\pm$ 2.08
Plane	98.86 $\pm$ 2.56	91.62 $\pm$ 6.26	95.43 $\pm$ 3.65	87.24 $\pm$ 8.67	85.14 $\pm$ 9.49	82.86 $\pm$ 4.15	78.73 $\pm$ 5.25	88.57 $\pm$ 9.09	86.48 $\pm$ 4.39	88.57 $\pm$ 6.67	26.10 $\pm$ 13.02	93.90 $\pm$ 6.92
PowerCons	76.89 $\pm$ 9.68	80.22 $\pm$ 9.41	78.11 $\pm$ 13.88	60.22 $\pm$ 11.82	61.00 $\pm$ 7.74	62.96 $\pm$ 3.70	49.44 $\pm$ 7.47	58.33 $\pm$ 2.78	51.33 $\pm$ 4.56	62.78 $\pm$ 6.41	55.22 $\pm$ 21.33	60.89 $\pm$ 11.09
ProximalPhalanxOutlineAgeGroup	48.49 $\pm$ 34.51	48.20 $\pm$ 31.13	48.68 $\pm$ 30.82	39.61 $\pm$ 19.82	39.12 $\pm$ 18.60	43.74 $\pm$ 32.89	49.59 $\pm$ 25.02	60.81 $\pm$ 40.85	38.15 $\pm$ 22.87	46.50 $\pm$ 27.14	32.68 $\pm$ 16.57	48.98 $\pm$ 12.07
ProximalPhalanxOutlineCorrect	54.43 $\pm$ 13.05	53.40 $\pm$ 15.85	51.20 $\pm$ 16.57	56.22 $\pm$ 5.47	53.26 $\pm$ 10.68	44.67 $\pm$ 13.09	46.96 $\pm$ 12.63	58.76 $\pm$ 6.63	52.78 $\pm$ 12.27	43.99 $\pm$ 6.30	57.59 $\pm$ 13.02	52.23 $\pm$ 12.03
ProximalPhalanxTW	54.63 $\pm$ 10.31	57.56 $\pm$ 9.33	58.15 $\pm$ 6.64	54.34 $\pm$ 11.16	49.76 $\pm$ 11.67	60.81 $\pm$ 11.71	55.93 $\pm$ 9.23	38.37 $\pm$ 0.75	55.80 $\pm$ 7.24	57.07 $\pm$ 10.70	17.37 $\pm$ 16.23	55.22 $\pm$ 6.63
RefrigerationDevices	36.05 $\pm$ 6.49	35.68 $\pm$ 5.16	37.44 $\pm$ 5.37	34.99 $\pm$ 2.77	33.97 $\pm$ 4.76	41.16 $\pm$ 5.10	34.58 $\pm$ 1.47	34.93 $\pm$ 0.92	34.72 $\pm$ 0.99	38.84 $\pm$ 5.04	32.75 $\pm$ 2.24	29.76 $\pm$ 11.02
Rock	29.20 $\pm$ 8.07	34.40 $\pm$ 7.92	28.40 $\pm$ 6.54	41.60 $\pm$ 16.52	26.40 $\pm$ 8.53	38.67 $\pm$ 25.32	57.33 $\pm$ 5.03	<u>44.67 $\pm$ 18.90</u>	36.80 $\pm$ 8.07	40.00 $\pm$ 24.25	36.40 $\pm$ 14.93	39.20 $\pm$ 13.75
ScreenType	33.92 $\pm$ 4.73	34.19 $\pm$ 7.39	34.29 $\pm$ 6.61	33.87 $\pm$ 2.72	33.87 $\pm$ 4.61	32.44 $\pm$ 5.27	32.18 $\pm$ 3.99	28.00 $\pm$ 3.40	30.99 $\pm$ 3.01	33.60 $\pm$ 4.19	30.61 $\pm$ 2.66	32.53 $\pm$ 4.84
SemgHandGenderCh2	50.87 $\pm$ 2.35	55.37 $\pm$ 5.38	53.17 $\pm$ 4.35	50.17 $\pm$ 11.45	61.00 $\pm$ 15.57	42.78 $\pm$ 13.47	52.78 $\pm$ 1.84	39.89 $\pm$ 8.47	44.67 $\pm$ 12.14	35.61 $\pm$ 1.06	55.73 $\pm$ 4.74	51.90 $\pm$ 14.92
SemgHandMovementCh2	28.89 $\pm$ 4.22	29.87 $\pm$ 4.86	29.60 $\pm$ 5.24	25.29 $\pm$ 1.13	24.89 $\pm$ 3.57	20.37 $\pm$ 0.13	21.70 $\pm$ 2.31	22.52 $\pm$ 2.76	18.71 $\pm$ 1.91	20.07 $\pm$ 3.46	27.02 $\pm$ 5.10	28.13 $\pm$ 4.95
SemgHandSubjectCh2	28.89 $\pm$ 5.77	29.07 $\pm$ 4.60	28.00 $\pm$ 2.65	35.60 $\pm$ 4.19	29.91 $\pm$ 10.77	26.37 $\pm$ 1.89	23.11 $\pm$ 2.89	26.67 $\pm$ 4.87	21.56 $\pm$ 1.94	26.96 $\pm$ 3.68	27.91 $\pm$ 3.11	37.42 $\pm$ 8.82
ShakeGestureWiimoteZ	63.20 $\pm$ 7.69	58.00 $\pm$ 11.83	67.60 $\pm$ 11.61	40.40 $\pm$ 11.26	43.20 $\pm$ 9.55	45.33 $\pm$ 3.06	16.00 $\pm$ 7.21	40.00 $\pm$ 3.46	24.40 $\pm$ 4.98	46.00 $\pm$ 6.00	43.20 $\pm$ 9.55	60.40 $\pm$ 7.90
ShapeletSim	66.33 $\pm$ 12.25	87.00 $\pm$ 16.91	73.89 $\pm$ 23.85	50.11 $\pm$ 2.27	75.22 $\pm$ 8.30	53.89 $\pm$ 1.67	52.96 $\pm$ 1.40	48.89 $\pm$ 5.64	50.44 $\pm$ 1.82	51.11 $\pm$ 0.96	47.78 $\pm$ 3.07	62.44 $\pm$ 7.44
ShapesAll	23.63 $\pm$ 4.60	15.60 $\pm$ 1.98	23.63 $\pm$ 4.60	45.70 $\pm$ 2.59	19.27 $\pm$ 4.58	44.72 $\pm$ 2.58	6.00 $\pm$ 0.58	43.94 $\pm$ 2.14	8.07 $\pm$ 3.90	44.06 $\pm$ 1.11	8.13 $\pm$ 2.75	50.20 $\pm$ 3.03
SmallKitchenAppliances	40.37 $\pm$ 15.51	46.24 $\pm$ 15.42	44.32 $\pm$ 15.50	42.19 $\pm$ 3.16	38.13 $\pm$ 22.42	32.98 $\pm$ 5.23	32.80 $\pm$ 3.67	35.82 $\pm$ 4.27	35.41 $\pm$ 3.13	36.89 $\pm$ 7.74	38.88 $\pm$ 7.80	42.88 $\pm$ 15.27
SmoothSubspace	74.93 $\pm$ 5.59	74.40 $\pm$ 5.13	74.80 $\pm$ 8.48	31.73 $\pm$ 3.96	50.00 $\pm$ 10.52	46.67 $\pm$ 1.76	53.33 $\pm$ 9.45	50.22 $\pm$ 10.96	62.27 $\pm$ 6.56	60.67 $\pm$ 10.35	56.67 $\pm$ 5.19	68.53 $\pm$ 8.95
SonyAIBORobotSurface1	77.57 $\pm$ 10.30	80.80 $\pm$ 8.99	80.37 $\pm$ 5.89	59.80 $\pm$ 6.73	64.53 $\pm$ 18.83	52.86 $\pm$ 10.38	56.96 $\pm$ 5.82	55.35 $\pm$ 9.37	56.07 $\pm$ 4.43	57.68 $\pm$ 9.32	55.61 $\pm$ 9.09	57.20 $\pm$ 8.62
SonyAIBORobotSurface2	80.29 $\pm$ 6.43	84.09 $\pm$ 5.18	80.90 $\pm$ 6.99	67.39 $\pm$ 19.08	73.03 $\pm$ 13.78	66.18 $\pm$ 11.61	60.37 $\pm$ 27.05	61.35 $\pm$ 27.10	69.49 $\pm$ 15.13	68.07 $\pm$ 16.06	57.31 $\pm$ 15.03	66.09 $\pm$ 10.78
StarLightCurves	63.75 $\pm$ 3.45	62.30 $\pm$ 9.88	57.59 $\pm$ 9.08	57.39 $\pm$ 13.40	57.55 $\pm$ 15.28	67.54 $\pm$ 17.11	37.78 $\pm$ 3.53	57.92 $\pm$ 15.55	27.99 $\pm$ 0.00	67.36 $\pm$ 17.40	64.81 $\pm$ 14.61	63.03 $\pm$ 14.23
Strawberry	52.27 $\pm$ 4.26	51.84 $\pm$ 3.86	54.43 $\pm$ 4.33	54.86 $\pm$ 4.34	53.62 $\pm$ 3.45	58.29 $\pm$ 0.95	54.86 $\pm$ 13.21	60.18 $\pm$ 0.68	56.11 $\pm$ 9.26	57.93 $\pm$ 0.83	45.89 $\pm$ 11.15	51.46 $\pm$ 5.21
SwedishLeaf	65.95 $\pm$ 6.01	46.08 $\pm$ 6.36	65.95 $\pm$ 6.01	27.42 $\pm$ 3.44	40.83 $\pm$ 4.40	36.69 $\pm$ 1.99	27.20 $\pm$ 2.42	24.53 $\pm$ 4.08	31.65 $\pm$ 2.48	38.19 $\pm$ 0.81	8.10 $\pm$ 1.23	57.98 $\pm$ 2.72
Symbols	85.77 $\pm$ 4.83	70.17 $\pm$ 4.91	76.68 $\pm$ 7.53	79.60 $\pm$ 7.65	72.50 $\pm$ 6.12	81.54 $\pm$ 3.92	37.15 $\pm$ 0.97	<u>82.18 $\pm$ 0.42</u>	46.13 $\pm$ 9.06	80.57 $\pm$ 0.95	51.54 $\pm$ 6.72	80.94 $\pm$ 7.54
SyntheticControl	72.53 $\pm$ 6.61	76.40 $\pm$ 5.24	68.67 $\pm$ 10.59	38.47 $\pm$ 6.28	80.33 $\pm$ 9.12	47.78 $\pm$ 5.87	44.89 $\pm$ 3.20	48.22 $\pm$ 0.38	42.13 $\pm$ 2.87	50.00 $\pm$ 4.48	24.53 $\pm$ 7.68	84.20 $\pm$ 6.54
ToeSegmentation1	69.82 $\pm$ 7.39	<u>69.65 $\pm$ 4.27</u>	65.09 $\pm$ 8.05	51.67 $\pm$ 1.47	55.79 $\pm$ 6.85	49.27 $\pm$ 1.10	49.71 $\pm$ 3.29	46.64 $\pm$ 4.73	52.46 $\pm$ 2.82	48.39 $\pm$ 3.23	49.47 $\pm$ 2.47	58.16 $\pm$ 3.70
ToeSegmentation2	68.92 $\pm$ 8.63	70.31 $\pm$ 3.38	70.92 $\pm$ 10.76	50.31 $\pm$ 5.37	65.54 $\pm$ 9.84	50.26 $\pm$ 2.91	35.90 $\pm$ 10.81	50.51 $\pm$ 4.24	33.23 $\pm$ 16.47	52.05 $\pm$ 6.54	41.54 $\pm$ 14.95	75.23 $\pm$ 12.13
Trace	61.00 $\pm$ 4.64	65.20 $\pm$ 9.09	65.80 $\pm$ 6.61	49.60 $\pm$ 3.65	74.40 $\pm$ 6.15	51.33 $\pm$ 5.51	42.33 $\pm$ 4.51	47.67 $\pm$ 0.58	44.80 $\pm$ 7.69	51.33 $\pm$ 4.93	37.80 $\pm$ 6.42	99.20 $\pm$ 1.79
TwoLeadECG	69.48 $\pm$ 12.74	64.97 $\pm$ 10.26	66.51 $\pm$ 13.23	54.03 $\pm$ 2.33	57.82 $\pm$ 5.04	59.32 $\pm$ 3.05	53.32 $\pm$ 7.30	52.62 $\pm$ 3.32	54.47 $\pm$ 5.41	60.32 $\pm$ 1.06	50.73 $\pm$ 1.02	64.88 $\pm$ 11.28
TwoPatterns	48.77 $\pm$ 2.41	61.51 $\pm$ 6.85	46.17 $\pm$ 11.65	31.86 $\pm$ 3.24	36.86 $\pm$ 6.10	29.76 $\pm$ 2.56	25.52 $\pm$ 1.93	28.23 $\pm$ 0.74	26.06 $\pm$ 1.27	29.87 $\pm$ 1.68	26.32 $\pm$ 2.53	45.74 $\pm$ 2.64
UMD	43.06 $\pm$ 4.02	54.58 $\pm$ 9.53	45.56 $\pm$ 6.14	46.81 $\pm$ 7.70	50.14 $\pm$ 6.28	46.30 $\pm$ 5.02	45.60 $\pm$ 3.21	47.69 $\pm$ 4.19	48.61 $\pm$ 2.55	47.45 $\pm$ 6.56	31.39 $\pm$ 4.35	45.47 $\pm$ 8.73
UWaveGestureLibraryAll	29.36 $\pm$ 4.20	24.46 $\pm$ 5.19	28.34 $\pm$ 3.82	66.77 $\pm$ 5.89	41.63 $\pm$ 5.40							

TABLE 7: Per-dataset 2-shot classification results on 128 UCR datasets. Each value is reported as mean $\pm$ standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
ACSF1	61.60 $\pm$ 7.86	47.20 $\pm$ 6.30	59.20 $\pm$ 5.26	25.80 $\pm$ 4.38	42.00 $\pm$ 3.16	40.33 $\pm$ 2.31	30.67 $\pm$ 2.52	44.00 $\pm$ 6.24	35.80 $\pm$ 3.19	40.67 $\pm$ 5.69	24.80 $\pm$ 4.66	54.33 $\pm$ 4.16
Adiac	35.45 $\pm$ 2.60	14.27 $\pm$ 3.74	35.45 $\pm$ 2.60	18.87 $\pm$ 2.86	17.80 $\pm$ 5.08	27.79 $\pm$ 2.14	44.67 $\pm$ 0.39	16.62 $\pm$ 1.77	55.24 $\pm$ 4.67	32.91 $\pm$ 3.49	3.48 $\pm$ 1.27	56.01 $\pm$ 1.26
AllGestureWiimoteX	32.43 $\pm$ 2.94	27.69 $\pm$ 3.48	32.09 $\pm$ 3.58	19.46 $\pm$ 2.47	23.77 $\pm$ 3.63	24.43 $\pm$ 4.71	11.86 $\pm$ 1.00	22.67 $\pm$ 2.29	14.09 $\pm$ 1.28	21.10 $\pm$ 5.46	23.77 $\pm$ 3.63	26.51 $\pm$ 4.35
AllGestureWiimoteY	35.11 $\pm$ 5.43	28.57 $\pm$ 4.31	28.91 $\pm$ 5.81	20.54 $\pm$ 2.69	33.31 $\pm$ 2.65	24.33 $\pm$ 2.50	12.00 $\pm$ 1.38	21.90 $\pm$ 1.33	16.14 $\pm$ 1.67	22.57 $\pm$ 1.00	33.31 $\pm$ 2.65	29.43 $\pm$ 5.61
AllGestureWiimoteZ	29.80 $\pm$ 2.53	26.94 $\pm$ 2.84	29.31 $\pm$ 4.56	16.77 $\pm$ 2.02	22.97 $\pm$ 5.35	19.86 $\pm$ 4.12	12.76 $\pm$ 0.36	19.57 $\pm$ 2.48	14.97 $\pm$ 0.98	20.29 $\pm$ 3.48	22.97 $\pm$ 5.35	27.91 $\pm$ 1.29
ArrowHead	50.17 $\pm$ 5.13	44.80 $\pm$ 3.82	45.14 $\pm$ 5.25	57.03 $\pm$ 4.92	49.14 $\pm$ 13.02	53.14 $\pm$ 6.02	34.10 $\pm$ 6.27	50.67 $\pm$ 2.01	42.29 $\pm$ 4.92	55.43 $\pm$ 6.44	43.77 $\pm$ 8.76	57.03 $\pm$ 2.81
BME	48.40 $\pm$ 4.91	55.33 $\pm$ 7.02	54.40 $\pm$ 8.12	50.80 $\pm$ 5.11	67.33 $\pm$ 7.90	48.89 $\pm$ 2.12	64.00 $\pm$ 5.21	58.89 $\pm$ 3.91	67.33 $\pm$ 7.44	54.89 $\pm$ 9.20	50.80 $\pm$ 8.05	79.47 $\pm$ 11.40
Beef	42.00 $\pm$ 2.98	36.67 $\pm$ 8.50	36.70 $\pm$ 9.60	49.33 $\pm$ 7.23	39.33 $\pm$ 6.41	48.89 $\pm$ 5.09	40.00 $\pm$ 3.33	34.44 $\pm$ 5.09	42.67 $\pm$ 4.94	51.11 $\pm$ 7.70	32.67 $\pm$ 7.23	43.33 $\pm$ 5.27
BeetleFly	50.00 $\pm$ 9.35	67.00 $\pm$ 18.23	55.00 $\pm$ 12.75	52.00 $\pm$ 11.51	57.00 $\pm$ 16.43	45.00 $\pm$ 10.00	48.33 $\pm$ 12.58	38.33 $\pm$ 7.64	45.00 $\pm$ 9.35	56.67 $\pm$ 7.64	57.00 $\pm$ 4.47	75.00 $\pm$ 18.03
BirdChicken	71.00 $\pm$ 6.52	72.00 $\pm$ 4.47	76.00 $\pm$ 7.42	52.00 $\pm$ 14.40	65.00 $\pm$ 10.61	45.00 $\pm$ 8.66	51.67 $\pm$ 7.64	46.67 $\pm$ 5.77	53.00 $\pm$ 2.74	50.00 $\pm$ 13.23	49.00 $\pm$ 2.24	69.00 $\pm$ 22.19
CBF	85.76 $\pm$ 5.83	90.62 $\pm$ 7.16	82.18 $\pm$ 10.67	64.78 $\pm$ 6.67	82.69 $\pm$ 10.89	68.07 $\pm$ 6.00	34.89 $\pm$ 2.41	73.78 $\pm$ 6.88	36.78 $\pm$ 1.75	70.00 $\pm$ 6.65	42.84 $\pm$ 7.09	85.91 $\pm$ 4.23
Car	42.00 $\pm$ 15.43	41.67 $\pm$ 13.49	52.67 $\pm$ 7.87	55.33 $\pm$ 10.63	43.00 $\pm$ 11.45	54.44 $\pm$ 6.74	32.78 $\pm$ 0.96	52.78 $\pm$ 10.84	37.00 $\pm$ 7.76	55.56 $\pm$ 4.19	31.00 $\pm$ 10.71	63.33 $\pm$ 7.45
Chinatown	95.74 $\pm$ 1.26	96.73 $\pm$ 1.19	96.73 $\pm$ 0.81	54.52 $\pm$ 24.75	91.08 $\pm$ 5.39	85.33 $\pm$ 8.50	86.10 $\pm$ 2.71	84.55 $\pm$ 3.36	80.70 $\pm$ 5.41	82.80 $\pm$ 4.37	45.01 $\pm$ 24.65	91.84 $\pm$ 4.85
ChlorineConcentration	34.61 $\pm$ 4.95	35.24 $\pm$ 5.79	35.19 $\pm$ 5.79	34.54 $\pm$ 2.93	34.28 $\pm$ 2.22	31.22 $\pm$ 6.21	33.24 $\pm$ 2.95	33.84 $\pm$ 2.69	33.91 $\pm$ 2.07	33.69 $\pm$ 2.59	39.08 $\pm$ 9.47	32.47 $\pm$ 3.03
CinCECGTorso	46.52 $\pm$ 12.75	48.35 $\pm$ 9.58	51.06 $\pm$ 10.06	39.77 $\pm$ 4.46	54.91 $\pm$ 12.73	48.14 $\pm$ 2.63	36.18 $\pm$ 3.15	53.94 $\pm$ 2.51	50.30 $\pm$ 12.26	48.89 $\pm$ 2.81	49.00 $\pm$ 5.40	62.95 $\pm$ 6.66
Coffee	94.29 $\pm$ 3.19	88.57 $\pm$ 12.73	94.29 $\pm$ 3.19	92.14 $\pm$ 3.91	90.71 $\pm$ 10.89	73.81 $\pm$ 8.99	63.10 $\pm$ 7.43	85.71 $\pm$ 3.57	84.29 $\pm$ 10.29	77.38 $\pm$ 4.12	65.71 $\pm$ 15.07	88.57 $\pm$ 4.66
Computers	52.24 $\pm$ 10.15	54.40 $\pm$ 10.81	52.72 $\pm$ 12.14	48.08 $\pm$ 7.70	53.28 $\pm$ 11.25	48.00 $\pm$ 10.51	48.67 $\pm$ 4.60	48.00 $\pm$ 11.45	50.40 $\pm$ 1.72	46.67 $\pm$ 11.22	50.16 $\pm$ 7.53	87.73 $\pm$ 3.14
CricketX	36.51 $\pm$ 3.27	34.10 $\pm$ 3.98	33.03 $\pm$ 3.86	20.62 $\pm$ 2.07	30.56 $\pm$ 3.87	27.35 $\pm$ 2.06	8.12 $\pm$ 1.41	26.15 $\pm$ 1.85	12.15 $\pm$ 1.24	27.44 $\pm$ 2.96	11.54 $\pm$ 1.86	35.98 $\pm$ 13.43
CricketY	37.54 $\pm$ 5.48	30.62 $\pm$ 2.69	33.28 $\pm$ 3.86	22.56 $\pm$ 4.44	24.21 $\pm$ 2.69	26.50 $\pm$ 7.27	9.06 $\pm$ 0.39	25.90 $\pm$ 6.29	12.67 $\pm$ 0.94	27.52 $\pm$ 6.80	10.00 $\pm$ 3.24	30.43 $\pm$ 5.23
CricketZ	40.41 $\pm$ 4.15	35.49 $\pm$ 2.72	33.69 $\pm$ 5.21	20.36 $\pm$ 1.13	24.92 $\pm$ 3.09	26.75 $\pm$ 2.72	8.12 $\pm$ 1.50	24.44 $\pm$ 0.78	12.97 $\pm$ 2.47	26.50 $\pm$ 2.74	11.18 $\pm$ 2.42	33.33 $\pm$ 3.02
Crop	31.31 $\pm$ 3.87	37.82 $\pm$ 2.09	31.31 $\pm$ 3.87	14.67 $\pm$ 0.98	24.66 $\pm$ 2.54	36.33 $\pm$ 2.63	35.06 $\pm$ 3.15	37.89 $\pm$ 1.28	35.83 $\pm$ 2.39	38.66 $\pm$ 2.99	4.63 $\pm$ 0.75	43.94 $\pm$ 2.11
DiatomSizeReduction	78.43 $\pm$ 9.98	84.44 $\pm$ 4.38	89.74 $\pm$ 8.40	83.73 $\pm$ 5.41	63.01 $\pm$ 13.25	84.42 $\pm$ 6.89	59.91 $\pm$ 2.78	84.97 $\pm$ 5.38	59.28 $\pm$ 18.44	82.35 $\pm$ 7.47	35.10 $\pm$ 12.79	90.39 $\pm$ 5.62
DistalPhalanxOutlineAgeGroup	66.91 $\pm$ 6.08	67.77 $\pm$ 3.31	65.76 $\pm$ 6.53	55.54 $\pm$ 12.17	58.42 $\pm$ 11.15	57.07 $\pm$ 19.57	50.84 $\pm$ 14.56	56.12 $\pm$ 13.32	56.26 $\pm$ 12.92	57.07 $\pm$ 14.47	38.99 $\pm$ 8.80	67.77 $\pm$ 3.39
DistalPhalanxOutlineCorrect	52.32 $\pm$ 12.15	51.59 $\pm$ 11.49	52.54 $\pm$ 12.44	49.57 $\pm$ 11.83	51.45 $\pm$ 15.36	57.49 $\pm$ 13.09	56.76 $\pm$ 11.56	55.80 $\pm$ 11.93	51.38 $\pm$ 12.34	56.64 $\pm$ 12.97	50.65 $\pm$ 9.31	55.80 $\pm$ 14.95
DistalPhalanxTW	49.50 $\pm$ 8.83	55.54 $\pm$ 4.21	54.10 $\pm$ 4.92	39.28 $\pm$ 8.61	53.96 $\pm$ 7.56	41.97 $\pm$ 11.56	42.45 $\pm$ 10.74	39.81 $\pm$ 18.49	47.05 $\pm$ 8.79	45.80 $\pm$ 16.17	22.45 $\pm$ 8.85	51.37 $\pm$ 8.97
DodgerLoopDay	28.75 $\pm$ 2.34	23.75 $\pm$ 2.50	28.00 $\pm$ 5.70	44.00 $\pm$ 4.09	38.50 $\pm$ 7.42	39.58 $\pm$ 4.02	23.75 $\pm$ 2.50	42.50 $\pm$ 7.60	19.75 $\pm$ 4.09	42.50 $\pm$ 3.31	38.50 $\pm$ 7.42	35.50 $\pm$ 8.78
DodgerLoopGame	54.93 $\pm$ 6.45	63.04 $\pm$ 8.02	58.41 $\pm$ 11.09	53.62 $\pm$ 8.51	65.65 $\pm$ 18.08	67.87 $\pm$ 7.26	48.55 $\pm$ 4.53	71.50 $\pm$ 0.84	50.14 $\pm$ 3.81	68.36 $\pm$ 3.42	65.65 $\pm$ 18.08	61.01 $\pm$ 13.03
DodgerLoopWeekend	92.75 $\pm$ 2.40	87.68 $\pm$ 5.59	92.03 $\pm$ 3.69	94.64 $\pm$ 2.74	96.09 $\pm$ 1.41	96.38 $\pm$ 1.26	64.98 $\pm$ 4.71	97.34 $\pm$ 0.42	65.22 $\pm$ 8.45	96.14 $\pm$ 0.84	96.09 $\pm$ 1.41	84.64 $\pm$ 2.73
ECG200	69.40 $\pm$ 7.77	75.20 $\pm$ 7.05	71.80 $\pm$ 5.54	76.80 $\pm$ 3.56	71.80 $\pm$ 4.38	78.33 $\pm$ 1.53	74.33 $\pm$ 5.69	75.67 $\pm$ 3.21	73.60 $\pm$ 3.65	75.33 $\pm$ 0.58	52.20 $\pm$ 18.20	73.00 $\pm$ 3.46
ECG5000	61.90 $\pm$ 13.09	69.05 $\pm$ 11.38	61.86 $\pm$ 10.65	74.23 $\pm$ 10.96	58.61 $\pm$ 14.82	72.42 $\pm$ 4.03	65.61 $\pm$ 11.76	69.40 $\pm$ 4.84	67.93 $\pm$ 11.62	66.36 $\pm$ 9.81	30.14 $\pm$ 19.40	67.68 $\pm$ 9.76
ECGFiveDays	68.78 $\pm$ 9.46	56.79 $\pm$ 4.91	71.73 $\pm$ 8.76	57.12 $\pm$ 2.23	67.29 $\pm$ 3.58	66.55 $\pm$ 6.20	74.64 $\pm$ 13.99	75.18 $\pm$ 2.37	68.76 $\pm$ 10.24	68.52 $\pm$ 6.54	57.79 $\pm$ 7.24	69.12 $\pm$ 9.82
EOGHorizontalSignal	20.66 $\pm$ 1.70	23.43 $\pm$ 5.48	21.38 $\pm$ 4.60	25.86 $\pm$ 4.54	28.07 $\pm$ 2.90	30.66 $\pm$ 2.91	30.66 $\pm$ 2.91	31.68 $\pm$ 2.61	9.56 $\pm$ 1.50	29.01 $\pm$ 2.49	29.83 $\pm$ 3.23	37.35 $\pm$ 4.07
EOGVerticalSignal	13.37 $\pm$ 1.32	16.57 $\pm$ 3.25	14.86 $\pm$ 2.77	26.80 $\pm$ 1.98	29.56 $\pm$ 4.11	26.98 $\pm$ 4.28	11.88 $\pm$ 1.54	26.34 $\pm$ 4.16	10.83 $\pm$ 1.45	26.61 $\pm$ 6.09	20.88 $\pm$ 2.94	31.82 $\pm$ 4.47
Earthquakes	52.95 $\pm$ 7.17	57.84 $\pm$ 11.11	49.35 $\pm$ 10.38	53.24 $\pm$ 4.93	63.88 $\pm$ 6.42	63.31 $\pm$ 10.45	49.16 $\pm$ 1.10	54.92 $\pm$ 17.17	56.55 $\pm$ 8.34	60.43 $\pm$ 21.88	57.12 $\pm$ 15.34	55.25 $\pm$ 13.33
ElectricDevices	37.63 $\pm$ 9.44	41.80 $\pm$ 2.08	34.70 $\pm$ 5.95	21.27 $\pm$ 5.73	37.69 $\pm$ 10.00	28.63 $\pm$ 3.21	22.56 $\pm$ 2.12	20.63 $\pm$ 2.87	22.76 $\pm$ 4.51	29.88 $\pm$ 3.14	14.36 $\pm$ 4.97	47.14 $\pm$ 3.99
EthanolLevel	25.32 $\pm$ 0.64	25.52 $\pm$ 1.56	25.12 $\pm$ 2.09	25.67 $\pm$ 1.71	26.96 $\pm$ 0.82	26.87 $\pm$ 2.61	23.20 $\pm$ 1.91	25.67 $\pm$ 3.18	25.24 $\pm$ 0.98	26.47 $\pm$ 2.08	25.12 $\pm$ 0.81	28.00 $\pm$ 3.11
FaceAll	55.98 $\pm$ 6.19	45.22 $\pm$ 8.18	55.98 $\pm$ 6.19	40.27 $\pm$ 5.76	24.12 $\pm$ 4.85	38.56 $\pm$ 3.11	38.84 $\pm$ 3.81	36.71 $\pm$ 2.55	38.89 $\pm$ 3.64	41.40 $\pm$ 3.79	11.35 $\pm$ 5.01	39.09 $\pm$ 1.28
FaceFour	77.73 $\pm$ 4.99	52.05 $\pm$ 10.37	81.82 $\pm$ 7.14	74.09 $\pm$ 5.98	59.32 $\pm$ 5.76	79.55 $\pm$ 9.02	58.58 $\pm$ 7.31	80.68 $\pm$ 3.41	62.50 $\pm$ 7.14	84.85 $\pm$ 2.37	40.45 $\pm$ 14.16	68.55 $\pm$ 4.91
FacesUCR	67.20 $\pm$ 6.42	51.90 $\pm$ 6.85	62.04 $\pm$ 8.47	41.67 $\pm$ 3.36	25.42 $\pm$ 6.93	44.55 $\pm$ 2.52	45.95 $\pm$ 4.27	49.07 $\pm$ 5.45	48.56 $\pm$ 3.79	49.27 $\pm$ 6.47	10.96 $\pm$ 5.26	55.08 $\pm$ 2.32
FiftyWords	19.03 $\pm$ 1.51	13.49 $\pm$ 1.95	19.03 $\pm$ 1.51	40.19 $\pm$ 3.22	28.35 $\pm$ 3.05	44.69 $\pm$ 2.97	5.05 $\pm$ 0.44	46.74 $\pm$ 2.43	16.53 $\pm$ 1.75	44.18 $\pm$ 3.70	15.16 $\pm$ 3.75	47.30 $\pm$ 2.89
Fish	50.97 $\pm$ 10.77	26.97 $\pm$ 9.88	50.51 $\pm$ 16.65	50.07 $\pm$ 6.36	31.54 $\pm$ 5.18	45.90 $\pm$ 8.88	20.00 $\pm$ 4.12	39.05 $\pm$ 5.64	22.97 $\pm$ 3.62	45.71 $\pm$ 10.47	22.86 $\pm$ 2.83	49.91 $\pm$ 10.25
FordA	64.98 $\pm$ 8.59	62.27 $\pm$ 9.82	62.82 $\pm$ 9.33	49.92 $\pm$ 1.84	60.48 $\pm$ 7.29	50.30 $\pm$ 1.31	50.66 $\pm$ 0.90	50.15 $\pm$ 0.79	50.27 $\pm$ 1.01	50.58 $\pm$ 1.43	50.76 $\pm$ 2.11	53.35 $\pm$ 2.84
FordB	61.19 $\pm$ 3.67	58.10 $\pm$ 1.40	58.67 $\pm$ 2.63	49.09 $\pm$ 1.35	56.89 $\pm$ 2.99	49.84 $\pm$ 0.63	50.99 $\pm$ 2.05	50.25 $\pm$ 1.29	49.95 $\pm$ 1.77	50.58 $\pm$ 0		

TABLE 7: Per-dataset 2-shot classification results on 128 UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
MixedShapesSmallTrain	45.73 $\pm$ 9.51	37.76 $\pm$ 12.10	51.77 $\pm$ 7.21	61.69 $\pm$ 3.51	54.28 $\pm$ 10.55	63.00 $\pm$ 2.46	19.05 $\pm$ 0.77	64.04 $\pm$ 1.32	24.01 $\pm$ 6.26	62.64 $\pm$ 3.39	57.53 $\pm$ 4.77	68.35 $\pm$ 3.26
MoteTrain	81.95 $\pm$ 4.07	75.34 $\pm$ 9.74	78.95 $\pm$ 3.25	70.97 $\pm$ 10.57	75.59 $\pm$ 8.13	78.41 $\pm$ 9.87	68.53 $\pm$ 3.31	82.48 $\pm$ 1.96	68.63 $\pm$ 10.07	83.65 $\pm$ 4.68	55.75 $\pm$ 11.79	58.56 $\pm$ 15.12
NonInvasiveFetalECGThorax1	25.55 $\pm$ 2.02	14.44 $\pm$ 3.90	25.55 $\pm$ 2.02	55.96 $\pm$ 3.30	18.44 $\pm$ 9.38	50.77 $\pm$ 4.84	18.37 $\pm$ 2.16	51.65 $\pm$ 2.21	21.42 $\pm$ 2.01	53.89 $\pm$ 1.99	13.55 $\pm$ 0.51	50.58 $\pm$ 2.13
NonInvasiveFetalECGThorax2	27.07 $\pm$ 5.56	22.30 $\pm$ 6.32	27.07 $\pm$ 5.56	62.09 $\pm$ 2.37	28.27 $\pm$ 8.71	55.69 $\pm$ 4.28	27.23 $\pm$ 2.03	56.05 $\pm$ 0.94	30.73 $\pm$ 3.23	58.63 $\pm$ 2.42	18.29 $\pm$ 1.75	59.72 $\pm$ 2.76
OSULeaf	55.70 $\pm$ 9.28	38.35 $\pm$ 6.31	56.03 $\pm$ 5.89	25.79 $\pm$ 2.77	25.29 $\pm$ 5.19	29.89 $\pm$ 3.34	18.18 $\pm$ 1.43	28.10 $\pm$ 2.07	21.90 $\pm$ 3.33	28.51 $\pm$ 4.37	23.31 $\pm$ 3.85	36.28 $\pm$ 6.97
OliveOil	64.67 $\pm$ 10.70	32.67 $\pm$ 5.48	71.33 $\pm$ 10.70	80.67 $\pm$ 3.65	17.33 $\pm$ 10.90	32.22 $\pm$ 11.71	38.89 $\pm$ 7.70	25.56 $\pm$ 12.62	24.00 $\pm$ 10.90	58.89 $\pm$ 22.69	22.00 $\pm$ 13.04	23.33 $\pm$ 9.72
PLAID	19.29 $\pm$ 3.99	16.65 $\pm$ 2.54	20.89 $\pm$ 4.30	22.98 $\pm$ 3.80	35.53 $\pm$ 7.89	27.56 $\pm$ 4.82	17.44 $\pm$ 2.97	25.88 $\pm$ 4.40	19.81 $\pm$ 3.24	28.68 $\pm$ 3.91	35.53 $\pm$ 7.89	40.19 $\pm$ 5.06
PhalangesOutlinesCorrect	52.91 $\pm$ 9.24	53.03 $\pm$ 10.79	52.61 $\pm$ 9.16	56.64 $\pm$ 11.15	54.13 $\pm$ 11.04	60.64 $\pm$ 3.81	58.70 $\pm$ 2.16	60.37 $\pm$ 4.20	52.26 $\pm$ 8.88	60.06 $\pm$ 3.62	49.04 $\pm$ 14.25	50.84 $\pm$ 2.93
Phoneme	12.22 $\pm$ 1.88	10.75 $\pm$ 2.12	11.22 $\pm$ 1.88	5.41 $\pm$ 0.40	7.31 $\pm$ 0.69	7.77 $\pm$ 0.84	4.13 $\pm$ 1.74	7.00 $\pm$ 0.45	6.19 $\pm$ 2.34	5.29 $\pm$ 0.61	5.20 $\pm$ 0.75	10.02 $\pm$ 0.59
PickupGestureWiimoteZ	43.20 $\pm$ 11.37	53.20 $\pm$ 9.55	62.40 $\pm$ 1.67	43.20 $\pm$ 6.72	40.40 $\pm$ 8.29	52.67 $\pm$ 10.07	21.33 $\pm$ 1.15	53.33 $\pm$ 4.16	27.60 $\pm$ 3.58	51.33 $\pm$ 6.43	40.40 $\pm$ 8.29	58.80 $\pm$ 3.03
PigAirwayPressure	28.37 $\pm$ 3.10	13.08 $\pm$ 1.90	49.52 $\pm$ 0.76	5.00 $\pm$ 0.26	3.65 $\pm$ 0.87	4.49 $\pm$ 0.28	3.21 $\pm$ 1.21	7.05 $\pm$ 0.73	4.62 $\pm$ 1.05	6.09 $\pm$ 0.73	5.10 $\pm$ 0.80	9.81 $\pm$ 0.43
PigArtPressure	50.58 $\pm$ 2.62	23.65 $\pm$ 3.56	69.42 $\pm$ 3.86	7.88 $\pm$ 0.26	5.29 $\pm$ 1.63	10.10 $\pm$ 0.48	8.01 $\pm$ 1.39	9.78 $\pm$ 1.11	7.12 $\pm$ 1.24	8.33 $\pm$ 0.56	19.13 $\pm$ 1.04	30.67 $\pm$ 1.38
PigCVP	31.54 $\pm$ 0.94	18.08 $\pm$ 2.27	61.92 $\pm$ 1.78	7.02 $\pm$ 0.43	3.94 $\pm$ 1.10	6.73 $\pm$ 0.00	4.01 $\pm$ 0.56	7.85 $\pm$ 1.00	4.13 $\pm$ 0.80	5.29 $\pm$ 0.83	10.10 $\pm$ 1.40	19.33 $\pm$ 0.99
Plane	100.00 $\pm$ 0.00	94.29 $\pm$ 3.75	99.05 $\pm$ 1.17	86.10 $\pm$ 6.69	91.62 $\pm$ 3.39	86.98 $\pm$ 7.02	84.44 $\pm$ 7.02	81.90 $\pm$ 1.90	87.24 $\pm$ 4.98	86.98 $\pm$ 6.34	21.71 $\pm$ 8.89	99.24 $\pm$ 0.80
PowerCons	82.11 $\pm$ 6.61	83.78 $\pm$ 3.63	82.33 $\pm$ 7.91	72.11 $\pm$ 2.98	70.78 $\pm$ 14.65	69.81 $\pm$ 4.10	52.22 $\pm$ 1.47	70.93 $\pm$ 7.46	50.67 $\pm$ 4.38	70.00 $\pm$ 8.73	65.89 $\pm$ 16.73	66.78 $\pm$ 15.52
ProximalPhalanxOutlineAgeGroup	68.10 $\pm$ 13.07	64.29 $\pm$ 28.01	64.78 $\pm$ 22.07	58.44 $\pm$ 28.63	68.00 $\pm$ 16.32	53.17 $\pm$ 31.21	56.26 $\pm$ 31.37	56.91 $\pm$ 37.90	59.22 $\pm$ 23.54	51.38 $\pm$ 35.67	33.95 $\pm$ 22.63	66.24 $\pm$ 17.89
ProximalPhalanxOutlineCorrect	65.43 $\pm$ 5.55	60.07 $\pm$ 16.64	63.85 $\pm$ 6.55	63.37 $\pm$ 8.02	62.96 $\pm$ 6.50	54.75 $\pm$ 14.29	54.30 $\pm$ 16.96	60.94 $\pm$ 6.65	59.86 $\pm$ 17.09	55.21 $\pm$ 16.86	44.88 $\pm$ 18.17	59.93 $\pm$ 12.82
ProximalPhalanxTW	50.83 $\pm$ 4.70	56.00 $\pm$ 9.92	67.12 $\pm$ 8.12	49.76 $\pm$ 11.43	53.37 $\pm$ 7.43	58.70 $\pm$ 5.35	53.98 $\pm$ 16.76	42.28 $\pm$ 24.98	56.00 $\pm$ 8.06	48.29 $\pm$ 19.01	16.98 $\pm$ 12.37	57.76 $\pm$ 7.59
RefrigerationDevices	41.39 $\pm$ 9.49	40.37 $\pm$ 10.57	41.23 $\pm$ 9.97	33.92 $\pm$ 0.58	35.31 $\pm$ 3.72	36.71 $\pm$ 1.61	33.87 $\pm$ 4.03	35.64 $\pm$ 1.78	33.81 $\pm$ 1.50	33.60 $\pm$ 3.03	33.07 $\pm$ 2.39	33.39 $\pm$ 7.13
Rock	32.00 $\pm$ 5.48	38.40 $\pm$ 8.65	37.80 $\pm$ 13.31	67.60 $\pm$ 10.53	44.80 $\pm$ 18.25	58.67 $\pm$ 13.61	70.00 $\pm$ 11.14	66.67 $\pm$ 8.08	47.60 $\pm$ 12.28	60.00 $\pm$ 14.00	44.40 $\pm$ 14.24	64.00 $\pm$ 7.07
ScreenType	36.91 $\pm$ 4.95	40.21 $\pm$ 6.72	38.40 $\pm$ 6.31	35.25 $\pm$ 3.07	36.75 $\pm$ 2.52	39.56 $\pm$ 4.47	33.78 $\pm$ 3.74	38.67 $\pm$ 2.63	33.71 $\pm$ 2.21	39.11 $\pm$ 4.89	33.81 $\pm$ 3.79	35.63 $\pm$ 4.40
SemgHandGenderCh2	55.13 $\pm$ 5.29	58.23 $\pm$ 11.97	54.27 $\pm$ 6.20	53.27 $\pm$ 13.91	62.87 $\pm$ 8.65	49.22 $\pm$ 10.20	55.56 $\pm$ 1.55	62.44 $\pm$ 3.53	43.87 $\pm$ 10.45	56.22 $\pm$ 13.04	53.70 $\pm$ 4.90	59.73 $\pm$ 10.87
SemgHandMovementCh2	33.51 $\pm$ 7.12	31.29 $\pm$ 5.18	36.00 $\pm$ 6.56	29.78 $\pm$ 3.16	23.56 $\pm$ 5.26	24.44 $\pm$ 2.52	18.89 $\pm$ 0.59	24.44 $\pm$ 1.39	18.71 $\pm$ 1.13	24.07 $\pm$ 5.35	27.91 $\pm$ 3.59	31.16 $\pm$ 4.70
SemgHandSubjectCh2	32.00 $\pm$ 3.88	30.71 $\pm$ 4.13	33.56 $\pm$ 5.04	45.91 $\pm$ 5.66	40.67 $\pm$ 4.39	29.56 $\pm$ 4.02	23.56 $\pm$ 2.47	32.52 $\pm$ 7.62	24.09 $\pm$ 3.07	30.81 $\pm$ 0.84	29.47 $\pm$ 4.28	43.20 $\pm$ 2.62
ShakeGestureWiimoteZ	76.00 $\pm$ 6.63	76.80 $\pm$ 7.43	81.60 $\pm$ 0.89	47.20 $\pm$ 7.95	58.40 $\pm$ 5.18	54.00 $\pm$ 6.00	26.00 $\pm$ 4.00	54.67 $\pm$ 4.62	32.40 $\pm$ 2.61	56.00 $\pm$ 5.29	58.40 $\pm$ 5.18	81.60 $\pm$ 6.71
ShapeletSim	78.89 $\pm$ 14.16	98.33 $\pm$ 2.08	92.11 $\pm$ 5.26	50.11 $\pm$ 2.87	80.22 $\pm$ 16.23	47.59 $\pm$ 3.10	50.74 $\pm$ 3.06	48.15 $\pm$ 1.79	49.22 $\pm$ 2.93	46.48 $\pm$ 3.06	49.67 $\pm$ 1.50	67.44 $\pm$ 10.92
ShapesAll	36.80 $\pm$ 3.60	28.17 $\pm$ 3.75	36.80 $\pm$ 3.60	50.73 $\pm$ 1.71	32.23 $\pm$ 5.06	55.78 $\pm$ 1.56	6.11 $\pm$ 0.63	53.17 $\pm$ 1.00	18.33 $\pm$ 5.95	53.33 $\pm$ 2.62	27.63 $\pm$ 3.10	63.03 $\pm$ 2.22
SmallKitchenAppliances	48.85 $\pm$ 7.91	50.40 $\pm$ 8.87	50.19 $\pm$ 7.71	41.49 $\pm$ 3.33	47.15 $\pm$ 6.04	49.60 $\pm$ 6.54	35.20 $\pm$ 2.93	40.09 $\pm$ 2.14	42.03 $\pm$ 4.49	44.09 $\pm$ 8.36	34.61 $\pm$ 4.49	55.68 $\pm$ 8.74
SmoothSubspace	78.93 $\pm$ 10.17	77.87 $\pm$ 5.55	80.13 $\pm$ 5.80	33.33 $\pm$ 0.00	51.20 $\pm$ 5.65	58.22 $\pm$ 4.73	68.89 $\pm$ 7.31	59.11 $\pm$ 4.23	70.67 $\pm$ 4.88	81.33 $\pm$ 6.36	48.27 $\pm$ 9.05	85.07 $\pm$ 5.07
SonyAIBORobotSurface1	90.42 $\pm$ 4.34	94.08 $\pm$ 1.77	89.38 $\pm$ 5.35	64.83 $\pm$ 12.29	70.05 $\pm$ 3.77	62.90 $\pm$ 13.91	61.90 $\pm$ 7.89	66.61 $\pm$ 10.89	59.10 $\pm$ 5.18	62.73 $\pm$ 9.45	52.65 $\pm$ 8.42	71.85 $\pm$ 8.28
SonyAIBORobotSurface2	88.00 $\pm$ 2.82	84.87 $\pm$ 6.58	83.67 $\pm$ 9.41	74.59 $\pm$ 3.28	78.70 $\pm$ 2.68	69.60 $\pm$ 7.04	77.26 $\pm$ 1.79	78.49 $\pm$ 0.86	78.30 $\pm$ 1.49	79.29 $\pm$ 1.17	56.81 $\pm$ 17.39	68.21 $\pm$ 5.28
StarLightCurves	76.49 $\pm$ 11.97	66.10 $\pm$ 9.91	66.22 $\pm$ 10.70	67.92 $\pm$ 13.52	71.57 $\pm$ 12.97	63.90 $\pm$ 14.70	35.37 $\pm$ 11.93	65.31 $\pm$ 14.87	34.69 $\pm$ 14.98	63.78 $\pm$ 14.15	68.45 $\pm$ 12.08	74.47 $\pm$ 15.08
Strawberry	62.81 $\pm$ 6.83	57.14 $\pm$ 6.89	60.22 $\pm$ 9.39	51.84 $\pm$ 8.28	56.59 $\pm$ 2.41	53.33 $\pm$ 7.10	66.58 $\pm$ 2.89	52.97 $\pm$ 8.18	59.62 $\pm$ 11.05	54.59 $\pm$ 4.47	51.03 $\pm$ 11.13	59.41 $\pm$ 6.38
SwedishLeaf	72.45 $\pm$ 3.37	54.30 $\pm$ 6.89	72.45 $\pm$ 3.37	34.02 $\pm$ 3.00	43.26 $\pm$ 7.26	48.91 $\pm$ 2.97	33.28 $\pm$ 3.94	46.24 $\pm$ 1.25	38.85 $\pm$ 4.16	47.52 $\pm$ 2.78	7.94 $\pm$ 0.74	64.99 $\pm$ 3.83
Symbols	90.49 $\pm$ 1.42	75.88 $\pm$ 7.72	88.48 $\pm$ 4.41	81.23 $\pm$ 0.99	76.18 $\pm$ 6.87	81.37 $\pm$ 1.84	37.42 $\pm$ 3.12	82.04 $\pm$ 0.58	58.95 $\pm$ 6.86	80.00 $\pm$ 1.74	51.50 $\pm$ 7.93	82.69 $\pm$ 3.43
SyntheticControl	79.67 $\pm$ 4.05	85.07 $\pm$ 4.63	82.60 $\pm$ 6.01	41.40 $\pm$ 6.37	83.80 $\pm$ 5.24	55.00 $\pm$ 3.38	44.11 $\pm$ 1.39	59.89 $\pm$ 3.15	47.93 $\pm$ 4.58	62.89 $\pm$ 3.15	20.80 $\pm$ 4.95	87.93 $\pm$ 3.95
ToeSegmentation1	68.42 $\pm$ 7.02	60.79 $\pm$ 8.14	61.58 $\pm$ 7.54	53.95 $\pm$ 2.19	58.60 $\pm$ 10.97	51.90 $\pm$ 4.30	53.80 $\pm$ 2.21	53.95 $\pm$ 0.44	52.37 $\pm$ 4.44	51.61 $\pm$ 3.08	50.70 $\pm$ 2.53	60.26 $\pm$ 8.54
ToeSegmentation2	76.15 $\pm$ 9.44	79.69 $\pm$ 9.24	72.15 $\pm$ 7.68	48.92 $\pm$ 5.64	68.62 $\pm$ 12.68	66.41 $\pm$ 15.71	51.54 $\pm$ 5.38	63.08 $\pm$ 15.78	47.08 $\pm$ 8.33	63.59 $\pm$ 15.79	54.31 $\pm$ 14.96	76.15 $\pm$ 12.44
Trace	76.00 $\pm$ 3.00	76.40 $\pm$ 3.05	73.80 $\pm$ 2.95	52.40 $\pm$ 4.04	76.60 $\pm$ 4.39	47.33 $\pm$ 3.21	51.00 $\pm$ 4.00	49.00 $\pm$ 3.00	55.60 $\pm$ 4.62	49.00 $\pm$ 5.00	46.00 $\pm$ 6.78	98.40 $\pm$ 3.05
TwoLeadECG	82.53 $\pm$ 6.47	79.19 $\pm$ 7.45	70.63 $\pm$ 4.92	55.68 $\pm$ 3.84	59.95 $\pm$ 3.74	62.36 $\pm$ 5.86	56.60 $\pm$ 1.27	55.90 $\pm$ 1.72	55.49 $\pm$ 2.80	60.14 $\pm$ 7.03	48.48 $\pm$ 5.21	74.63 $\pm$ 8.09
TwoPatterns	52.71 $\pm$ 4.44	71.75 $\pm$ 5.17	65.30 $\pm$ 11.45	32.62 $\pm$ 1.84	44.61 $\pm$ 9.45	31.66 $\pm$ 1.59	24.88 $\pm$ 1.30	28.57 $\pm$ 1.14	25.14 $\pm$ 0.61	29.88 $\pm$ 2.34	27.30 $\pm$ 2.15	78.23 $\pm$ 0.41
UMD	54.86 $\pm$ 7.95	71.67 $\pm$ 9.36	68.75 $\pm$ 16.74	54.17 $\pm$ 4.55	62.50 $\pm$ 12.00	52.31 $\pm$ 6.22	47.69 $\pm$ 9.93	60.19 $\pm$ 1.45	53.06 $\pm$ 4.83	56.25 $\pm$ 6.05	47.50 $\pm$ 4.16	78.86 $\pm$ 14.11
UWaveGestureLibraryAll	32.50 $\pm$ 3.38	28.30 $\pm$ 5.01	31.51 $\pm$ 4.14	69.44 $\pm$ 6.55	43.65 $\pm$ 6.39	66.54 $\pm$						

TABLE 8: Per-dataset 5-shot classification results on 128 UCR datasets. Each value is reported as mean $\pm$ standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
ACSF1	76.00 $\pm$ 2.92	64.60 $\pm$ 5.18	72.80 $\pm$ 1.64	36.20 $\pm$ 3.49	50.40 $\pm$ 2.61	60.00 $\pm$ 2.00	40.00 $\pm$ 1.00	62.67 $\pm$ 2.52	46.80 $\pm$ 3.70	58.67 $\pm$ 1.53	30.60 $\pm$ 4.83	68.33 $\pm$ 2.89
Adiac	56.21 $\pm$ 4.18	42.81 $\pm$ 4.46	56.21 $\pm$ 4.18	35.29 $\pm$ 1.32	34.58 $\pm$ 2.33	43.82 $\pm$ 1.54	46.97 $\pm$ 3.07	38.62 $\pm$ 1.68	64.50 $\pm$ 5.47	43.99 $\pm$ 3.86	7.11 $\pm$ 1.15	69.51 $\pm$ 2.82
AllGestureWiimoteX	40.46 $\pm$ 4.50	34.14 $\pm$ 3.58	38.71 $\pm$ 5.08	23.63 $\pm$ 2.19	35.40 $\pm$ 4.31	32.24 $\pm$ 3.43	13.57 $\pm$ 1.57	29.95 $\pm$ 4.86	18.17 $\pm$ 3.28	29.38 $\pm$ 3.89	35.40 $\pm$ 4.31	39.40 $\pm$ 2.46
AllGestureWiimoteY	43.20 $\pm$ 1.70	40.60 $\pm$ 1.51	36.20 $\pm$ 3.42	27.89 $\pm$ 2.07	40.97 $\pm$ 3.06	36.76 $\pm$ 2.07	15.38 $\pm$ 0.81	34.43 $\pm$ 2.02	19.49 $\pm$ 2.37	35.43 $\pm$ 1.55	40.97 $\pm$ 3.06	48.11 $\pm$ 0.89
AllGestureWiimoteZ	41.46 $\pm$ 4.33	35.71 $\pm$ 5.12	36.46 $\pm$ 5.30	20.20 $\pm$ 1.22	27.11 $\pm$ 2.51	26.57 $\pm$ 0.25	13.86 $\pm$ 1.71	26.67 $\pm$ 2.65	17.37 $\pm$ 2.23	26.86 $\pm$ 0.99	27.11 $\pm$ 2.51	36.14 $\pm$ 2.51
ArrowHead	66.40 $\pm$ 5.44	61.03 $\pm$ 5.46	64.23 $\pm$ 7.62	59.43 $\pm$ 5.27	58.86 $\pm$ 12.66	60.19 $\pm$ 7.46	42.67 $\pm$ 0.87	60.76 $\pm$ 1.19	48.11 $\pm$ 2.87	63.43 $\pm$ 8.42	54.06 $\pm$ 9.79	73.70 $\pm$ 3.71
BME	55.60 $\pm$ 7.80	56.93 $\pm$ 3.90	68.00 $\pm$ 9.90	58.53 $\pm$ 7.20	89.60 $\pm$ 10.65	58.89 $\pm$ 5.55	73.33 $\pm$ 8.11	76.89 $\pm$ 7.34	80.00 $\pm$ 6.70	73.33 $\pm$ 5.93	53.07 $\pm$ 5.63	97.73 $\pm$ 1.33
Beef	54.67 $\pm$ 4.47	47.33 $\pm$ 2.79	55.33 $\pm$ 2.98	52.67 $\pm$ 7.60	43.33 $\pm$ 4.71	64.44 $\pm$ 3.85	50.00 $\pm$ 3.33	52.22 $\pm$ 8.39	62.00 $\pm$ 8.03	84.44 $\pm$ 1.92	35.33 $\pm$ 4.47	51.33 $\pm$ 6.91
BeetleFly	72.00 $\pm$ 7.58	60.00 $\pm$ 11.73	79.00 $\pm$ 10.84	71.00 $\pm$ 14.32	57.00 $\pm$ 8.37	70.00 $\pm$ 8.66	51.67 $\pm$ 2.89	66.67 $\pm$ 16.07	49.00 $\pm$ 6.52	70.00 $\pm$ 10.00	59.00 $\pm$ 18.51	91.67 $\pm$ 2.89
BirdChicken	69.00 $\pm$ 12.94	77.00 $\pm$ 9.75	90.00 $\pm$ 3.54	57.00 $\pm$ 5.70	61.00 $\pm$ 16.73	63.33 $\pm$ 7.64	36.67 $\pm$ 2.89	65.00 $\pm$ 10.00	54.00 $\pm$ 8.94	68.33 $\pm$ 15.28	54.00 $\pm$ 4.18	85.00 $\pm$ 11.73
CBF	95.87 $\pm$ 2.59	95.58 $\pm$ 3.67	95.89 $\pm$ 3.27	61.47 $\pm$ 7.78	95.67 $\pm$ 4.48	74.52 $\pm$ 5.66	39.04 $\pm$ 0.80	80.11 $\pm$ 4.97	42.29 $\pm$ 1.09	80.59 $\pm$ 4.09	41.20 $\pm$ 4.62	93.76 $\pm$ 3.90
Car	59.33 $\pm$ 9.02	51.00 $\pm$ 7.96	65.67 $\pm$ 5.73	52.33 $\pm$ 6.41	45.67 $\pm$ 2.79	68.89 $\pm$ 6.31	31.67 $\pm$ 5.00	48.89 $\pm$ 4.19	41.00 $\pm$ 12.56	57.78 $\pm$ 5.85	36.00 $\pm$ 4.50	69.67 $\pm$ 3.98
Chinatown	96.15 $\pm$ 2.16	96.97 $\pm$ 0.49	97.43 $\pm$ 0.38	50.96 $\pm$ 33.10	96.03 $\pm$ 2.24	84.94 $\pm$ 15.12	91.06 $\pm$ 1.94	93.00 $\pm$ 5.85	89.56 $\pm$ 2.00	87.95 $\pm$ 8.75	73.00 $\pm$ 0.76	93.46 $\pm$ 10.56
ChlorineConcentration	35.97 $\pm$ 2.09	35.63 $\pm$ 3.63	36.15 $\pm$ 2.53	35.57 $\pm$ 3.02	35.80 $\pm$ 3.16	35.02 $\pm$ 2.94	36.15 $\pm$ 1.24	35.49 $\pm$ 4.56	37.02 $\pm$ 1.05	35.96 $\pm$ 1.50	38.60 $\pm$ 5.65	39.58 $\pm$ 3.20
CinCECGTorso	49.41 $\pm$ 5.43	57.77 $\pm$ 3.88	56.17 $\pm$ 3.91	43.81 $\pm$ 1.84	69.26 $\pm$ 4.67	63.41 $\pm$ 2.73	37.80 $\pm$ 1.42	64.95 $\pm$ 2.40	64.83 $\pm$ 11.83	68.41 $\pm$ 5.18	54.58 $\pm$ 3.46	82.56 $\pm$ 7.22
Coffee	86.43 $\pm$ 14.81	97.86 $\pm$ 1.96	100.00 $\pm$ 0.00	92.86 $\pm$ 3.57	97.86 $\pm$ 1.96	86.90 $\pm$ 2.06	76.19 $\pm$ 2.06	82.14 $\pm$ 9.45	92.86 $\pm$ 10.41	85.71 $\pm$ 0.00	50.00 $\pm$ 6.19	94.29 $\pm$ 1.96
Computers	58.80 $\pm$ 7.81	61.60 $\pm$ 5.76	60.96 $\pm$ 6.58	46.64 $\pm$ 4.37	54.48 $\pm$ 7.54	49.87 $\pm$ 5.33	50.53 $\pm$ 1.29	48.00 $\pm$ 8.43	49.04 $\pm$ 4.09	48.67 $\pm$ 4.31	46.56 $\pm$ 5.23	62.67 $\pm$ 4.74
CricketX	45.90 $\pm$ 4.41	43.08 $\pm$ 3.97	43.08 $\pm$ 2.15	25.23 $\pm$ 1.80	36.10 $\pm$ 5.19	34.87 $\pm$ 4.89	11.79 $\pm$ 1.43	30.77 $\pm$ 0.51	16.92 $\pm$ 1.32	33.68 $\pm$ 3.44	11.33 $\pm$ 1.41	49.06 $\pm$ 3.21
CricketY	42.67 $\pm$ 4.49	43.08 $\pm$ 3.40	43.28 $\pm$ 6.01	30.41 $\pm$ 3.78	36.92 $\pm$ 6.92	37.86 $\pm$ 2.42	9.06 $\pm$ 1.21	36.58 $\pm$ 2.06	15.23 $\pm$ 1.85	37.26 $\pm$ 2.44	12.05 $\pm$ 2.18	49.66 $\pm$ 1.41
CricketZ	51.59 $\pm$ 3.02	49.64 $\pm$ 3.11	47.79 $\pm$ 6.26	26.31 $\pm$ 2.67	36.31 $\pm$ 4.35	37.18 $\pm$ 1.18	10.77 $\pm$ 0.51	34.87 $\pm$ 1.43	15.44 $\pm$ 1.54	37.18 $\pm$ 0.77	10.92 $\pm$ 1.59	53.68 $\pm$ 2.06
Crop	49.93 $\pm$ 1.45	46.93 $\pm$ 2.27	49.93 $\pm$ 1.45	31.91 $\pm$ 2.21	38.13 $\pm$ 2.76	46.92 $\pm$ 2.60	44.47 $\pm$ 0.97	47.82 $\pm$ 2.55	47.76 $\pm$ 1.51	48.69 $\pm$ 2.69	8.15 $\pm$ 3.22	52.18 $\pm$ 2.24
DiatomSizeReduction	85.75 $\pm$ 6.98	70.26 $\pm$ 16.84	95.95 $\pm$ 1.26	61.57 $\pm$ 2.74	65.49 $\pm$ 10.17	87.69 $\pm$ 0.59	68.63 $\pm$ 2.55	66.34 $\pm$ 8.52	73.53 $\pm$ 9.37	87.25 $\pm$ 0.00	41.83 $\pm$ 13.83	93.66 $\pm$ 1.97
DistalPhalanxOutlineAgeGroup	65.76 $\pm$ 4.54	68.63 $\pm$ 3.96	67.19 $\pm$ 5.75	58.99 $\pm$ 14.81	63.31 $\pm$ 4.46	64.75 $\pm$ 3.74	67.63 $\pm$ 2.16	69.30 $\pm$ 3.24	68.20 $\pm$ 2.35	68.59 $\pm$ 2.31	53.67 $\pm$ 12.57	69.35 $\pm$ 1.20
DistalPhalanxOutlineCorrect	66.81 $\pm$ 3.92	61.96 $\pm$ 7.85	68.62 $\pm$ 6.57	56.23 $\pm$ 10.81	60.07 $\pm$ 6.07	62.44 $\pm$ 3.65	62.68 $\pm$ 3.32	58.94 $\pm$ 3.37	63.41 $\pm$ 7.40	59.30 $\pm$ 3.08	55.87 $\pm$ 8.58	63.77 $\pm$ 11.63
DistalPhalanxTW	58.99 $\pm$ 2.69	57.12 $\pm$ 1.40	58.56 $\pm$ 3.40	49.64 $\pm$ 7.01	60.58 $\pm$ 2.61	57.31 $\pm$ 0.42	51.32 $\pm$ 3.62	59.47 $\pm$ 4.40	54.10 $\pm$ 4.24	56.59 $\pm$ 5.05	31.94 $\pm$ 15.91	56.40 $\pm$ 4.87
DodgerLoopDay	37.75 $\pm$ 3.35	40.50 $\pm$ 4.11	37.00 $\pm$ 3.26	46.75 $\pm$ 4.01	44.25 $\pm$ 2.09	52.50 $\pm$ 5.45	22.08 $\pm$ 0.72	45.42 $\pm$ 2.60	24.25 $\pm$ 3.49	50.83 $\pm$ 4.73	44.25 $\pm$ 2.09	45.75 $\pm$ 3.26
DodgerLoopGame	66.52 $\pm$ 4.02	82.32 $\pm$ 3.38	72.17 $\pm$ 3.64	76.09 $\pm$ 5.15	84.49 $\pm$ 1.89	75.36 $\pm$ 5.02	50.00 $\pm$ 2.61	78.99 $\pm$ 7.63	53.33 $\pm$ 2.09	77.29 $\pm$ 4.82	84.49 $\pm$ 1.89	72.61 $\pm$ 10.25
DodgerLoopWeekend	95.51 $\pm$ 1.65	92.46 $\pm$ 3.46	94.78 $\pm$ 1.88	97.39 $\pm$ 0.40	97.25 $\pm$ 0.61	96.62 $\pm$ 0.84	64.43 $\pm$ 4.25	97.83 $\pm$ 0.00	72.90 $\pm$ 1.82	96.86 $\pm$ 0.42	97.25 $\pm$ 0.61	93.62 $\pm$ 2.37
ECG200	76.40 $\pm$ 3.58	77.80 $\pm$ 6.26	76.20 $\pm$ 4.60	75.00 $\pm$ 1.58	62.80 $\pm$ 16.36	77.67 $\pm$ 3.06	76.67 $\pm$ 3.79	78.00 $\pm$ 2.00	77.60 $\pm$ 3.21	79.33 $\pm$ 4.93	69.60 $\pm$ 4.51	67.40 $\pm$ 8.91
ECG5000	78.85 $\pm$ 9.03	75.51 $\pm$ 9.29	75.56 $\pm$ 6.88	84.88 $\pm$ 3.51	71.69 $\pm$ 7.99	82.01 $\pm$ 5.12	76.33 $\pm$ 7.90	83.48 $\pm$ 5.73	74.58 $\pm$ 8.80	82.23 $\pm$ 5.84	18.64 $\pm$ 16.84	73.89 $\pm$ 8.21
ECGFiveDays	71.54 $\pm$ 13.30	70.27 $\pm$ 8.41	70.22 $\pm$ 9.52	69.38 $\pm$ 6.83	77.54 $\pm$ 3.65	72.94 $\pm$ 9.66	87.65 $\pm$ 8.42	75.22 $\pm$ 2.60	83.30 $\pm$ 11.51	75.61 $\pm$ 4.26	60.49 $\pm$ 6.59	70.96 $\pm$ 3.91
EOGHorizontalSignal	24.97 $\pm$ 1.88	32.71 $\pm$ 1.62	27.35 $\pm$ 3.43	29.56 $\pm$ 3.95	37.73 $\pm$ 6.23	36.92 $\pm$ 1.62	10.96 $\pm$ 1.15	38.40 $\pm$ 2.19	9.94 $\pm$ 1.90	35.17 $\pm$ 4.76	31.99 $\pm$ 1.89	46.35 $\pm$ 2.27
EOGVerticalSignal	18.18 $\pm$ 3.62	22.10 $\pm$ 3.22	18.78 $\pm$ 4.22	29.01 $\pm$ 2.45	38.23 $\pm$ 4.43	31.77 $\pm$ 5.00	8.47 $\pm$ 1.80	31.95 $\pm$ 3.43	10.39 $\pm$ 1.59	31.22 $\pm$ 4.29	22.82 $\pm$ 2.72	36.74 $\pm$ 2.24
Earthquakes	50.94 $\pm$ 12.41	50.22 $\pm$ 8.85	52.23 $\pm$ 11.61	56.83 $\pm$ 2.92	52.09 $\pm$ 5.05	56.35 $\pm$ 11.87	51.32 $\pm$ 9.11	56.12 $\pm$ 3.14	53.38 $\pm$ 4.56	54.68 $\pm$ 7.61	52.81 $\pm$ 9.99	54.68 $\pm$ 9.31
ElectricDevices	36.69 $\pm$ 5.52	36.63 $\pm$ 6.97	36.08 $\pm$ 5.74	23.65 $\pm$ 1.86	38.77 $\pm$ 8.33	38.73 $\pm$ 7.16	22.89 $\pm$ 3.05	25.09 $\pm$ 5.42	31.29 $\pm$ 6.08	38.61 $\pm$ 5.70	11.47 $\pm$ 4.02	47.06 $\pm$ 4.18
EthanolLevel	27.52 $\pm$ 1.92	26.36 $\pm$ 0.86	26.60 $\pm$ 1.66	27.88 $\pm$ 1.25	24.88 $\pm$ 0.18	28.40 $\pm$ 1.73	28.60 $\pm$ 2.78	26.53 $\pm$ 0.12	25.40 $\pm$ 0.75	29.53 $\pm$ 0.64	26.32 $\pm$ 1.07	29.52 $\pm$ 1.40
FaceAll	67.63 $\pm$ 4.66	67.75 $\pm$ 7.28	67.63 $\pm$ 4.66	53.96 $\pm$ 2.14	41.22 $\pm$ 3.29	55.76 $\pm$ 2.36	53.79 $\pm$ 5.19	55.44 $\pm$ 2.51	51.51 $\pm$ 5.08	54.30 $\pm$ 2.41	17.64 $\pm$ 1.73	54.54 $\pm$ 1.26
FaceFour	88.18 $\pm$ 2.85	61.82 $\pm$ 13.74	89.77 $\pm$ 4.62	78.86 $\pm$ 3.17	57.50 $\pm$ 8.82	86.36 $\pm$ 1.14	64.77 $\pm$ 2.27	87.88 $\pm$ 1.74	71.82 $\pm$ 1.24	85.61 $\pm$ 1.74	41.14 $\pm$ 6.88	75.68 $\pm$ 4.37
FacesUCR	78.67 $\pm$ 2.69	73.00 $\pm$ 4.70	76.25 $\pm$ 3.16	50.94 $\pm$ 2.11	43.00 $\pm$ 7.53	68.33 $\pm$ 3.31	58.46 $\pm$ 2.45	68.99 $\pm$ 1.59	60.62 $\pm$ 1.49	67.61 $\pm$ 2.78	27.20 $\pm$ 3.34	68.58 $\pm$ 2.57
FiftyWords	30.07 $\pm$ 3.14	27.91 $\pm$ 3.50	30.07 $\pm$ 3.14	47.03 $\pm$ 2.26	37.71 $\pm$ 3.50	52.23 $\pm$ 1.46	9.45 $\pm$ 3.91	56.19 $\pm$ 3.63	25.10 $\pm$ 2.19	49.38 $\pm$ 1.41	40.84 $\pm$ 2.35	61.58 $\pm$ 2.55
Fish	61.26 $\pm$ 15.63	48.69 $\pm$ 6.91	67.66 $\pm$ 6.55	56.80 $\pm$ 6.22	42.97 $\pm$ 5.32	68.76 $\pm$ 3.49	25.52 $\pm$ 2.93	56.57 $\pm$ 5.51	40.80 $\pm$ 3.78	67.05 $\pm$ 1.44	26.40 $\pm$ 8.24	64.57 $\pm$ 5.92
ForestA	71.41 $\pm$ 10.22	71.94 $\pm$ 9.09	70.55 $\pm$ 10.29	50.33 $\pm$ 1.09	63.62 $\pm$ 6.82	50.28 $\pm$ 2.11	51.67 $\pm$ 1.61	49.67 $\pm$ 1.72	51.11 $\pm$ 2.04	49.82 $\pm$ 1.42	50.91 $\pm$ 1.25	54.00 $\pm$ 2.22
FordB	61.26 $\pm$ 2.75	58.96 $\pm$ 1.47	59.04 $\pm$ 3.16	50.00 $\pm$ 1.41	55.88 $\pm$ 4.39	50.86 $\pm$ 0.77	49.51 $\pm$ 0.81	48.81 $\pm$ 1.26	49.56 $\pm$ 0.80	50.21 $\pm$ 1.36	50.27 $\pm$ 2.02	52.32 $\pm$ 2.78
FreezerRegularTrain	71.93 $\pm$ 7.16	86.24 $\pm$ 2.60	79.17 $\pm$ 4.44	76.42 $\pm$ 0.69	77.88 $\pm$ 9.36	76.43 $\pm$ 0.19	57.					

TABLE 8: Per-dataset 5-shot classification results on 128 UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
MixedShapesSmallTrain	66.24 ± 12.23	57.21 ± 4.26	72.37 ± 7.91	66.37 ± 2.62	66.13 ± 3.66	68.30 ± 1.42	22.61 ± 0.88	67.60 ± 3.67	25.23 ± 4.04	66.78 ± 1.26	60.00 ± 5.54	78.71 ± 1.70
MoteStrain	84.74 ± 4.04	83.85 ± 4.63	85.10 ± 2.03	80.19 ± 8.46	77.96 ± 3.58	85.73 ± 4.20	72.68 ± 2.25	80.09 ± 3.68	77.11 ± 6.04	85.36 ± 1.87	60.30 ± 15.64	80.37 ± 4.50
NonInvasiveFetalECGThorax1	40.80 ± 1.72	31.63 ± 3.55	40.80 ± 1.72	72.10 ± 1.96	51.79 ± 9.05	70.09 ± 3.02	22.04 ± 0.40	70.72 ± 1.13	32.89 ± 2.08	69.19 ± 0.16	28.56 ± 4.30	67.42 ± 0.56
NonInvasiveFetalECGThorax2	41.15 ± 5.46	43.13 ± 8.61	41.15 ± 5.46	78.55 ± 0.96	60.71 ± 3.55	75.84 ± 1.88	37.61 ± 1.37	75.40 ± 1.50	50.66 ± 2.46	74.93 ± 2.49	37.30 ± 2.25	75.31 ± 0.82
OSULeaf	57.85 ± 5.97	50.74 ± 10.99	72.23 ± 4.84	32.40 ± 1.86	35.37 ± 3.13	38.84 ± 1.09	16.80 ± 1.04	32.64 ± 2.71	24.05 ± 2.60	36.91 ± 1.04	23.06 ± 4.07	49.92 ± 6.46
OliveOil	59.33 ± 13.21	28.67 ± 13.66	74.00 ± 6.83	31.33 ± 9.60	26.67 ± 10.00	42.22 ± 25.02	40.00 ± 10.00	32.22 ± 13.47	40.00 ± 21.73	32.22 ± 22.19	26.67 ± 10.00	28.67 ± 11.69
PLAID	24.02 ± 4.96	24.51 ± 3.44	22.87 ± 2.95	30.06 ± 4.34	47.93 ± 3.93	36.56 ± 2.69	26.88 ± 2.01	40.16 ± 2.24	27.26 ± 3.48	37.93 ± 4.09	47.93 ± 3.93	52.96 ± 5.39
PhalangesOutlinesCorrect	56.22 ± 7.89	55.22 ± 9.82	56.85 ± 6.96	52.75 ± 11.57	53.15 ± 7.53	51.83 ± 11.55	50.35 ± 9.47	51.20 ± 13.21	53.05 ± 9.33	50.85 ± 11.66	53.61 ± 9.97	53.61 ± 8.69
Phoneme	19.05 ± 2.97	18.15 ± 2.47	19.05 ± 2.97	6.61 ± 0.25	9.00 ± 1.08	10.58 ± 0.95	6.87 ± 0.83	9.00 ± 0.29	9.61 ± 1.88	9.30 ± 1.34	8.21 ± 0.75	15.63 ± 1.56
PickupGestureWiimoteZ	56.00 ± 10.20	66.80 ± 5.93	73.60 ± 4.56	52.40 ± 1.67	56.80 ± 9.55	57.33 ± 3.06	27.33 ± 1.15	58.00 ± 3.46	42.00 ± 2.00	58.67 ± 1.15	56.80 ± 9.55	68.00 ± 2.45
PigAirwayPressure	25.00 ± 2.61	13.37 ± 2.05	48.17 ± 0.40	5.10 ± 0.26	4.62 ± 0.55	4.81 ± 0.00	4.17 ± 2.22	6.73 ± 0.48	4.04 ± 0.87	5.93 ± 0.56	4.42 ± 0.86	9.81 ± 0.73
PigArtPressure	50.96 ± 5.81	20.77 ± 3.29	68.17 ± 4.63	8.46 ± 0.65	5.10 ± 1.82	8.97 ± 1.21	6.57 ± 2.00	8.97 ± 0.73	6.92 ± 1.65	9.29 ± 1.11	18.17 ± 2.24	33.08 ± 1.33
PigCVP	29.33 ± 2.90	18.75 ± 2.40	62.98 ± 1.73	6.83 ± 0.22	3.37 ± 1.32	7.53 ± 0.56	4.65 ± 1.11	7.53 ± 0.28	3.75 ± 1.24	4.81 ± 0.48	10.38 ± 1.21	19.52 ± 0.55
Plane	100.00 ± 0.00	98.10 ± 2.43	99.62 ± 0.52	88.95 ± 7.93	95.24 ± 3.16	97.78 ± 0.55	96.51 ± 1.98	95.87 ± 1.45	97.14 ± 1.51	95.87 ± 0.55	23.05 ± 11.32	99.24 ± 1.24
PowerCons	80.56 ± 3.36	84.44 ± 5.02	84.00 ± 6.30	70.11 ± 5.26	81.11 ± 10.74	75.37 ± 5.45	54.26 ± 3.90	80.93 ± 10.72	52.22 ± 4.39	78.52 ± 10.32	68.89 ± 15.97	76.11 ± 6.55
ProximalPhalanxOutlineAgeGroup	73.37 ± 8.19	81.76 ± 3.12	79.90 ± 3.95	65.56 ± 9.78	75.41 ± 5.61	79.02 ± 7.19	79.02 ± 8.46	84.07 ± 1.71	74.34 ± 7.73	80.00 ± 3.41	43.61 ± 12.69	76.88 ± 5.79
ProximalPhalanxOutlineCorrect	69.35 ± 11.27	69.00 ± 5.91	69.28 ± 9.35	64.47 ± 0.52	63.71 ± 8.48	65.18 ± 3.44	64.60 ± 1.57	63.69 ± 1.73	63.16 ± 5.61	66.67 ± 4.47	55.26 ± 17.89	65.84 ± 4.70
ProximalPhalanxTW	62.34 ± 7.16	66.44 ± 4.62	69.37 ± 2.52	53.37 ± 11.45	65.27 ± 5.74	58.05 ± 7.67	58.21 ± 7.39	55.45 ± 4.15	60.00 ± 4.28	55.28 ± 3.66	27.32 ± 18.90	66.24 ± 3.19
RefrigerationDevices	42.83 ± 1.65	43.20 ± 6.13	44.00 ± 6.18	33.33 ± 2.73	37.92 ± 6.42	38.93 ± 3.24	31.20 ± 3.23	32.98 ± 2.22	32.32 ± 2.29	38.04 ± 3.74	34.61 ± 1.83	40.96 ± 5.59
Rock	42.00 ± 5.29	36.80 ± 7.16	38.40 ± 4.34	80.40 ± 2.97	60.80 ± 5.93	80.67 ± 1.15	81.33 ± 3.06	88.67 ± 2.31	59.20 ± 3.63	85.33 ± 3.06	33.60 ± 5.90	76.00 ± 3.16
ScreenType	40.64 ± 8.17	35.41 ± 2.78	36.16 ± 4.77	35.56 ± 2.39	32.18 ± 2.24	36.09 ± 2.30	32.53 ± 1.41	35.02 ± 0.41	34.08 ± 2.12	34.51 ± 4.14	34.51 ± 4.14	34.51 ± 4.14
SemgHandGenderCh2	58.13 ± 2.49	66.47 ± 7.67	56.07 ± 2.82	65.13 ± 3.61	67.13 ± 2.65	50.28 ± 5.31	55.83 ± 13.67	55.06 ± 2.04	54.63 ± 4.30	47.94 ± 16.07	59.97 ± 7.08	66.80 ± 7.25
SemgHandMovementCh2	30.89 ± 3.99	33.91 ± 3.59	34.40 ± 6.06	32.84 ± 5.27	26.49 ± 1.89	27.70 ± 2.36	20.59 ± 1.85	29.48 ± 2.44	21.64 ± 2.79	23.41 ± 3.03	33.16 ± 2.38	36.31 ± 5.92
SemgHandSubjectCh2	37.11 ± 4.58	44.09 ± 1.20	35.60 ± 3.13	50.76 ± 1.90	49.02 ± 0.83	46.67 ± 2.12	27.78 ± 0.59	51.19 ± 2.74	30.62 ± 1.93	41.48 ± 6.03	36.62 ± 4.30	54.31 ± 3.37
ShakeGestureWiimoteZ	80.40 ± 4.34	91.60 ± 2.61	89.20 ± 1.10	56.40 ± 1.67	72.80 ± 5.22	62.00 ± 0.00	35.33 ± 1.15	60.67 ± 3.06	44.00 ± 2.45	55.33 ± 2.31	72.80 ± 5.22	92.00 ± 0.00
ShapeletSim	82.78 ± 10.96	99.67 ± 0.75	92.56 ± 6.55	49.44 ± 2.69	88.22 ± 10.06	55.93 ± 2.31	49.81 ± 2.80	49.63 ± 4.21	50.67 ± 1.82	49.44 ± 3.33	49.67 ± 3.98	92.00 ± 3.88
ShapesAll	57.57 ± 3.68	48.33 ± 5.58	57.57 ± 3.68	58.13 ± 1.68	51.50 ± 3.69	64.67 ± 1.64	9.22 ± 0.79	65.83 ± 3.88	13.83 ± 3.47	59.22 ± 2.28	56.47 ± 1.46	76.20 ± 1.83
SmallKitchenAppliances	59.47 ± 8.51	57.39 ± 6.38	62.35 ± 6.85	39.04 ± 4.11	57.55 ± 2.52	57.51 ± 6.62	38.67 ± 2.54	39.56 ± 5.26	46.67 ± 3.53	56.62 ± 3.96	44.00 ± 7.15	51.63 ± 12.25
SmoothSubspace	86.80 ± 2.13	82.00 ± 2.83	88.67 ± 2.36	33.87 ± 1.19	57.07 ± 4.58	65.78 ± 2.04	69.33 ± 3.71	61.11 ± 0.77	77.07 ± 5.51	84.00 ± 4.81	37.07 ± 6.74	88.00 ± 2.87
SonyAIBORobotSurface1	93.94 ± 2.22	95.71 ± 1.41	92.85 ± 2.46	61.83 ± 10.72	84.73 ± 2.02	71.21 ± 3.16	67.39 ± 1.20	71.60 ± 6.11	64.96 ± 2.40	64.89 ± 3.03	54.54 ± 12.34	84.86 ± 2.67
SonyAIBORobotSurface2	92.51 ± 3.32	91.14 ± 0.89	91.54 ± 4.19	78.15 ± 2.59	85.83 ± 4.10	79.96 ± 1.73	83.11 ± 2.73	81.78 ± 2.73	80.76 ± 4.57	83.21 ± 2.88	61.15 ± 14.84	77.21 ± 3.67
StarLightCurves	81.52 ± 8.85	60.67 ± 16.70	76.44 ± 1.22	74.70 ± 4.25	76.72 ± 3.78	75.63 ± 4.10	28.48 ± 9.45	75.33 ± 3.39	27.99 ± 0.00	75.69 ± 5.23	75.95 ± 2.54	81.86 ± 7.93
Strawberry	63.51 ± 2.99	64.22 ± 4.90	60.22 ± 9.83	58.49 ± 2.90	63.03 ± 7.09	61.89 ± 4.92	67.21 ± 4.90	59.82 ± 1.80	61.03 ± 3.64	65.77 ± 7.40	52.92 ± 6.34	58.05 ± 7.04
SwedishLeaf	79.81 ± 1.70	76.00 ± 4.97	79.81 ± 1.70	47.97 ± 1.68	65.66 ± 4.54	62.61 ± 4.53	49.23 ± 4.26	67.15 ± 4.00	56.93 ± 3.43	62.29 ± 1.68	15.90 ± 6.80	79.78 ± 2.56
Symbols	88.78 ± 7.99	88.74 ± 7.74	91.24 ± 0.96	83.44 ± 1.29	84.92 ± 3.31	84.29 ± 1.01	36.11 ± 0.23	86.40 ± 1.63	58.19 ± 7.40	83.75 ± 0.90	27.56 ± 4.43	88.04 ± 1.30
SyntheticControl	87.80 ± 2.38	96.33 ± 0.33	92.60 ± 1.01	48.53 ± 3.68	92.73 ± 2.94	73.22 ± 3.10	52.11 ± 3.20	66.22 ± 3.66	56.27 ± 3.45	72.78 ± 3.66	20.93 ± 4.87	95.07 ± 1.62
ToeSegmentation1	85.96 ± 7.05	77.37 ± 5.14	81.58 ± 5.14	54.74 ± 3.58	68.16 ± 3.79	55.56 ± 2.16	53.95 ± 2.74	51.61 ± 0.91	53.25 ± 3.38	54.82 ± 1.32	48.77 ± 4.31	70.53 ± 10.39
ToeSegmentation2	74.77 ± 12.53	77.85 ± 7.33	74.77 ± 8.79	50.77 ± 2.88	68.46 ± 6.25	58.97 ± 1.60	50.26 ± 0.44	59.49 ± 6.72	37.54 ± 3.14	64.10 ± 2.70	52.31 ± 9.12	78.00 ± 6.36
Trace	88.60 ± 7.37	99.00 ± 1.00	93.00 ± 8.00	53.00 ± 4.42	76.40 ± 5.77	51.67 ± 4.93	53.33 ± 2.08	49.67 ± 3.79	59.80 ± 2.59	55.67 ± 8.50	48.80 ± 10.71	100.00 ± 0.00
TwoLeadECG	97.98 ± 4.17	95.47 ± 5.89	75.84 ± 5.58	58.75 ± 2.38	82.55 ± 14.28	62.16 ± 1.52	60.81 ± 0.10	56.34 ± 2.59	61.88 ± 1.65	68.51 ± 9.61	52.10 ± 2.73	96.68 ± 5.58
TwoPatterns	64.62 ± 3.64	74.98 ± 5.04	74.94 ± 1.43	38.09 ± 3.07	51.91 ± 7.48	40.83 ± 3.89	26.00 ± 0.92	39.83 ± 3.80	26.50 ± 1.83	37.54 ± 1.63	27.50 ± 4.13	74.22 ± 2.81
UMD	87.08 ± 10.10	91.53 ± 5.28	96.67 ± 2.05	56.53 ± 3.69	74.44 ± 17.00	61.57 ± 1.45	62.96 ± 5.82	68.52 ± 10.79	71.81 ± 2.17	70.14 ± 10.69	48.47 ± 7.32	95.78 ± 2.33
UWaveGestureLibraryAll	42.53 ± 1.15	39.33 ± 5.11	44.53 ± 1.84	77.02 ± 5.21	57.17 ± 2.24	78.18 ± 3.39	26.42 ± 1.62	77.47 ± 3.61	35.67 ± 1.53	76.96 ± 4.32	65.94 ± 6.48	48.36 ± 0.30
UWaveGestureLibraryX	39.64 ± 3.12	46.31 ± 4.46	46.23 ± 4.40	56.88 ± 2.64	50.95 ± 2.22	58.42 ± 2.37	19.27 ± 3.18	58.58 ± 2.34	31.22 ± 3.55	57.86 ± 1.47	26.48 ± 5.59	47.36 ± 0.30
UWaveGestureLibraryY	37.00 ± 1.20	38.44 ± 2.57	40.23 ± 3.63	52.69 ± 2.24	48.41 ± 3.12	51.42 ± 3.10	21.22 ± 1.11	51.72 ± 2.98	30.92 ± 5.43	50.81 ± 4.53	19.79 ± 3.07	51.36 ± 0.31
UWaveGestureLibraryZ	38.74 ± 2.41	44.95 ± 4.51	40.85 ± 4.13	48.35 ± 3.41	47.76 ± 4.10	50.34 ± 2.37	20.23 ± 1.68	49.48 ± 1.66	31.23 ± 4.16	48.05 ± 3.36	25.20 ± 6.78	55.54 ± 2.61
Wafer	76.83 ± 9.15	69.99 ± 14.88	71.79 ± 15.46	55.60 ± 16.77	73.34 ± 19.66	80.42 ± 16.85	76.25 ± 9.34	83.86 ± 16.37	75.80 ± 12.54	83.26 ± 17.85	64.01 ± 22.29	71.65 ± 9.58
Wine	57.41 ± 13.03	48.15 ± 3.93	61.11 ± 5.40	55.93 ± 3.56	53.33 ± 5.62	47.53 ± 8.35	54.94 ± 8.55	50.00 ± 0.00	46.67 ± 6.47	37.65 ± 19.80	50.00 ± 0.00	51.48 ± 2.03
WordSynonyms	25.08 ± 1.54	21.22 ± 1.56	25.08 ± 1.54	29.56 ± 2.06	30.38 ± 2.74	45.87 ± 4.63	7.99 ± 1.24	44.67 ± 4.83	18.81 ± 2.07	41.33 ± 4.67	22.26 ± 2.64	47.05 ± 3.87
Worms	57.14 ± 9.76	55.58 ± 5.91	62.86 ± 4.55	24.68 ± 4.68	40.26 ± 5.11	22.94 ± 0.75	19.91 ± 3.27	25.97 ± 0.00	16.62 ± 2.32	22.08 ± 1.30	22.08 ± 5.11	44.42 ± 6.26
WormsTwoClass	68.05 ± 7.21	59.74 ± 6.69	62.08 ± 7.97	48.83 ± 3.13	51.17 ± 8.09	52.38 ± 2.70	42.86 ± 3.44	51.08 ± 5.25	48.83 ± 3.26	52.81 ± 3.27	48.83 ± 5.92	54.81 ± 7.59
Yoga	55.71 ± 3.37	53.57 ± 4.72	55.19 ± 2.57	50.13 ± 2.22	53.89 ± 3.78	49.79 ± 3.21	52.94 ± 1.57	47.50 ± 3.55	51.67 ± 1.99	49.68 ± 4.28	48.87 ± 2.05	51.31 ± 4.34
Avg. Acc.	61.27	59.47	63.55	52.30	56.94	57.86	43.20	57.20	47.30	58.35	42.94	64.14
Avg. Rank	4.77	5.34	4.23	7.73	6.60	5.69	9.55	5.94	8.70	5.79	9.48	4.19
#Best	21	13	30	6	4	8	4	7	2	8	1	36

TABLE 9: Per-dataset 10-shot classification results on 128 UCR datasets. Each value is reported as mean $\pm$ standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
ACSF1	85.40 $\pm$ 2.19	76.80 $\pm$ 3.35	80.80 $\pm$ 0.84	38.60 $\pm$ 0.89	58.80 $\pm$ 2.59	64.67 $\pm$ 0.58	42.67 $\pm$ 1.53	67.00 $\pm$ 2.00	53.60 $\pm$ 2.51	67.00 $\pm$ 2.00	36.80 $\pm$ 3.27	81.33 $\pm$ 1.53
Adiac	62.46 $\pm$ 4.53	56.98 $\pm$ 3.84	62.46 $\pm$ 4.53	45.32 $\pm$ 1.06	47.72 $\pm$ 5.48	56.01 $\pm$ 2.77	48.68 $\pm$ 0.78	47.91 $\pm$ 6.68	71.71 $\pm$ 1.94	48.00 $\pm$ 3.58	12.84 $\pm$ 1.42	76.42 $\pm$ 1.68
AllGestureWiimoteX	49.37 $\pm$ 3.15	43.11 $\pm$ 2.57	39.26 $\pm$ 2.50	23.91 $\pm$ 1.60	43.54 $\pm$ 2.85	37.14 $\pm$ 0.52	15.00 $\pm$ 0.87	35.19 $\pm$ 1.33	25.80 $\pm$ 2.24	31.00 $\pm$ 1.57	43.54 $\pm$ 2.85	52.31 $\pm$ 2.65
AllGestureWiimoteY	54.89 $\pm$ 2.09	50.74 $\pm$ 4.62	40.17 $\pm$ 2.24	27.26 $\pm$ 1.11	48.03 $\pm$ 1.58	42.57 $\pm$ 3.19	16.95 $\pm$ 2.22	42.57 $\pm$ 3.81	30.37 $\pm$ 4.65	41.24 $\pm$ 2.84	48.03 $\pm$ 1.58	56.34 $\pm$ 3.54
AllGestureWiimoteZ	49.34 $\pm$ 1.38	46.03 $\pm$ 2.76	39.80 $\pm$ 2.97	22.43 $\pm$ 1.80	34.54 $\pm$ 4.20	31.29 $\pm$ 1.13	16.67 $\pm$ 1.15	29.67 $\pm$ 3.32	21.71 $\pm$ 1.69	32.19 $\pm$ 0.59	34.54 $\pm$ 4.20	44.43 $\pm$ 2.91
ArrowHead	72.23 $\pm$ 5.22	63.89 $\pm$ 7.06	65.49 $\pm$ 9.09	64.34 $\pm$ 0.65	62.86 $\pm$ 7.57	60.38 $\pm$ 1.19	48.19 $\pm$ 2.58	59.62 $\pm$ 2.88	57.83 $\pm$ 5.80	63.43 $\pm$ 3.48	45.83 $\pm$ 13.23	81.49 $\pm$ 3.32
BME	64.27 $\pm$ 9.29	69.33 $\pm$ 1.56	78.67 $\pm$ 4.64	64.53 $\pm$ 0.30	97.60 $\pm$ 3.29	71.11 $\pm$ 2.34	78.89 $\pm$ 1.02	88.44 $\pm$ 1.02	91.33 $\pm$ 2.31	86.67 $\pm$ 0.00	46.40 $\pm$ 7.46	99.20 $\pm$ 0.20
Beef	51.33 $\pm$ 6.91	44.00 $\pm$ 6.83	52.00 $\pm$ 2.98	61.33 $\pm$ 4.47	46.00 $\pm$ 6.41	70.00 $\pm$ 3.33	51.11 $\pm$ 5.09	54.44 $\pm$ 3.85	64.00 $\pm$ 7.23	80.00 $\pm$ 0.00	37.33 $\pm$ 4.35	68.00 $\pm$ 1.83
BeetleFly	71.00 $\pm$ 8.94	74.00 $\pm$ 6.52	75.00 $\pm$ 6.12	86.00 $\pm$ 4.18	86.00 $\pm$ 2.24	81.67 $\pm$ 2.89	48.33 $\pm$ 2.89	83.33 $\pm$ 2.89	49.00 $\pm$ 11.94	75.00 $\pm$ 5.00	70.00 $\pm$ 12.25	96.67 $\pm$ 2.89
BirdChicken	84.00 $\pm$ 12.45	92.00 $\pm$ 5.70	94.00 $\pm$ 4.18	68.00 $\pm$ 2.74	81.00 $\pm$ 9.62	71.67 $\pm$ 7.64	41.67 $\pm$ 7.64	68.33 $\pm$ 16.07	56.00 $\pm$ 5.48	56.67 $\pm$ 7.64	56.00 $\pm$ 4.18	89.00 $\pm$ 5.48
CBF	98.96 $\pm$ 0.23	98.27 $\pm$ 0.96	99.24 $\pm$ 0.32	67.49 $\pm$ 2.09	95.98 $\pm$ 4.64	90.07 $\pm$ 1.09	39.37 $\pm$ 1.51	91.44 $\pm$ 1.93	51.24 $\pm$ 7.49	90.11 $\pm$ 0.88	49.31 $\pm$ 9.05	97.71 $\pm$ 0.69
Car	57.67 $\pm$ 10.84	58.67 $\pm$ 3.21	65.33 $\pm$ 2.17	73.00 $\pm$ 2.98	62.67 $\pm$ 5.22	70.00 $\pm$ 2.89	27.22 $\pm$ 5.85	65.56 $\pm$ 0.96	37.33 $\pm$ 3.84	72.22 $\pm$ 1.92	50.33 $\pm$ 8.28	76.33 $\pm$ 5.45
Chinatown	98.19 $\pm$ 0.13	98.13 $\pm$ 0.16	98.54 $\pm$ 0.21	54.58 $\pm$ 24.80	97.73 $\pm$ 0.75	94.75 $\pm$ 2.10	95.63 $\pm$ 1.01	97.28 $\pm$ 0.61	96.62 $\pm$ 0.60	96.60 $\pm$ 1.10	63.56 $\pm$ 20.21	94.56 $\pm$ 3.09
ChlorineConcentration	41.18 $\pm$ 3.66	34.41 $\pm$ 8.63	37.86 $\pm$ 5.20	34.42 $\pm$ 4.40	35.51 $\pm$ 6.32	35.06 $\pm$ 1.39	38.85 $\pm$ 1.62	32.99 $\pm$ 4.01	39.05 $\pm$ 4.08	39.50 $\pm$ 4.07	33.01 $\pm$ 11.68	34.77 $\pm$ 6.93
CinCECGTorso	54.03 $\pm$ 3.03	65.94 $\pm$ 5.62	64.64 $\pm$ 5.11	45.36 $\pm$ 1.09	80.65 $\pm$ 7.24	81.59 $\pm$ 1.76	35.82 $\pm$ 2.41	84.42 $\pm$ 2.94	63.93 $\pm$ 1.27	85.19 $\pm$ 1.24	52.77 $\pm$ 4.47	94.71 $\pm$ 4.65
Coffee	97.14 $\pm$ 6.39	97.14 $\pm$ 1.60	100.00 $\pm$ 0.00	96.43 $\pm$ 4.37	97.86 $\pm$ 1.96	91.67 $\pm$ 2.06	86.90 $\pm$ 16.50	92.86 $\pm$ 0.00	98.57 $\pm$ 3.19	90.48 $\pm$ 2.06	56.43 $\pm$ 7.74	95.71 $\pm$ 2.99
Computers	66.48 $\pm$ 3.03	63.20 $\pm$ 6.81	68.08 $\pm$ 4.62	48.88 $\pm$ 2.52	56.40 $\pm$ 10.98	52.13 $\pm$ 9.60	52.13 $\pm$ 6.02	50.93 $\pm$ 12.73	49.68 $\pm$ 5.04	53.87 $\pm$ 12.94	51.52 $\pm$ 3.69	65.60 $\pm$ 6.66
CricketX	60.72 $\pm$ 2.35	55.18 $\pm$ 4.36	51.28 $\pm$ 3.40	30.15 $\pm$ 2.84	42.92 $\pm$ 1.99	46.50 $\pm$ 1.21	14.27 $\pm$ 1.71	45.38 $\pm$ 2.45	20.87 $\pm$ 2.94	45.38 $\pm$ 1.94	19.33 $\pm$ 2.95	66.92 $\pm$ 2.71
CricketY	56.92 $\pm$ 2.81	49.44 $\pm$ 1.94	47.69 $\pm$ 2.43	32.82 $\pm$ 4.21	42.15 $\pm$ 5.17	44.62 $\pm$ 0.44	9.32 $\pm$ 1.04	44.02 $\pm$ 2.09	17.90 $\pm$ 2.44	41.11 $\pm$ 0.53	23.95 $\pm$ 3.97	57.95 $\pm$ 1.33
CricketZ	63.18 $\pm$ 1.66	57.38 $\pm$ 2.58	48.82 $\pm$ 4.98	30.56 $\pm$ 3.52	43.18 $\pm$ 5.27	46.32 $\pm$ 1.71	12.91 $\pm$ 2.15	47.78 $\pm$ 1.41	18.26 $\pm$ 2.08	45.30 $\pm$ 1.50	21.95 $\pm$ 3.02	65.30 $\pm$ 1.16
Crop	57.92 $\pm$ 1.22	54.52 $\pm$ 1.67	57.92 $\pm$ 1.22	44.46 $\pm$ 2.10	44.96 $\pm$ 1.35	54.14 $\pm$ 2.07	52.84 $\pm$ 0.15	54.82 $\pm$ 1.79	54.74 $\pm$ 0.79	54.60 $\pm$ 1.45	25.02 $\pm$ 2.10	57.97 $\pm$ 1.25
DiatomSizeReduction	87.32 $\pm$ 3.50	71.05 $\pm$ 13.58	96.80 $\pm$ 0.91	44.77 $\pm$ 1.16	62.68 $\pm$ 4.64	87.04 $\pm$ 1.80	66.88 $\pm$ 2.86	51.96 $\pm$ 5.03	48.89 $\pm$ 18.63	87.47 $\pm$ 0.19	30.13 $\pm$ 0.15	92.61 $\pm$ 2.00
DistalPhalanxOutlineAgeGroup	68.20 $\pm$ 2.98	68.78 $\pm$ 4.24	70.50 $\pm$ 1.61	69.50 $\pm$ 1.66	69.50 $\pm$ 2.81	66.67 $\pm$ 1.10	65.71 $\pm$ 3.62	70.50 $\pm$ 1.25	68.78 $\pm$ 2.72	66.91 $\pm$ 1.25	48.06 $\pm$ 11.76	70.65 $\pm$ 2.18
DistalPhalanxOutlineCorrect	64.78 $\pm$ 12.49	65.07 $\pm$ 11.29	63.62 $\pm$ 12.87	50.00 $\pm$ 13.55	60.14 $\pm$ 9.92	58.94 $\pm$ 4.04	67.15 $\pm$ 2.41	55.43 $\pm$ 14.84	62.39 $\pm$ 9.46	61.71 $\pm$ 6.72	47.61 $\pm$ 8.80	60.51 $\pm$ 11.85
DistalPhalanxTW	55.83 $\pm$ 3.65	59.14 $\pm$ 2.12	60.58 $\pm$ 3.58	47.34 $\pm$ 6.42	57.55 $\pm$ 6.08	53.96 $\pm$ 3.14	53.00 $\pm$ 2.91	54.92 $\pm$ 0.83	52.52 $\pm$ 4.35	54.92 $\pm$ 2.91	32.37 $\pm$ 14.50	54.10 $\pm$ 3.11
DodgerLoopDay	49.25 $\pm$ 3.11	47.25 $\pm$ 4.28	39.75 $\pm$ 3.11	41.50 $\pm$ 2.24	47.00 $\pm$ 4.11	49.58 $\pm$ 5.64	28.75 $\pm$ 2.50	45.42 $\pm$ 1.91	29.00 $\pm$ 2.40	48.75 $\pm$ 1.25	47.00 $\pm$ 4.11	48.75 $\pm$ 1.77
DodgerLoopGame	68.70 $\pm$ 6.09	83.33 $\pm$ 5.28	75.07 $\pm$ 2.64	79.28 $\pm$ 0.97	88.84 $\pm$ 1.67	76.81 $\pm$ 1.26	48.79 $\pm$ 3.42	81.40 $\pm$ 1.51	53.48 $\pm$ 2.83	78.99 $\pm$ 1.45	88.84 $\pm$ 1.67	83.62 $\pm$ 0.65
DodgerLoopWeekend	94.06 $\pm$ 1.39	94.20 $\pm$ 2.11	96.81 $\pm$ 0.83	97.83 $\pm$ 0.00	96.81 $\pm$ 0.65	96.86 $\pm$ 0.42	72.71 $\pm$ 1.82	97.58 $\pm$ 0.42	76.38 $\pm$ 1.82	97.10 $\pm$ 0.00	96.81 $\pm$ 0.65	97.83 $\pm$ 0.00
ECG200	80.00 $\pm$ 4.00	83.00 $\pm$ 2.00	80.40 $\pm$ 4.28	77.60 $\pm$ 4.34	79.80 $\pm$ 6.53	79.33 $\pm$ 3.21	78.00 $\pm$ 3.61	81.00 $\pm$ 5.00	82.00 $\pm$ 2.92	82.00 $\pm$ 3.61	63.00 $\pm$ 15.26	77.40 $\pm$ 4.88
ECG5000	83.56 $\pm$ 5.09	78.23 $\pm$ 2.20	80.72 $\pm$ 4.37	88.37 $\pm$ 1.22	76.31 $\pm$ 5.83	79.59 $\pm$ 2.75	73.64 $\pm$ 2.21	84.75 $\pm$ 1.16	79.28 $\pm$ 5.11	82.19 $\pm$ 1.79	24.56 $\pm$ 15.15	73.78 $\pm$ 6.18
ECGFiveDays	95.40 $\pm$ 1.88	88.99 $\pm$ 4.44	97.56 $\pm$ 1.20	74.87 $\pm$ 4.61	94.61 $\pm$ 2.00	83.82 $\pm$ 7.05	96.44 $\pm$ 1.42	91.64 $\pm$ 1.90	97.51 $\pm$ 1.12	86.18 $\pm$ 8.29	57.47 $\pm$ 4.38	75.28 $\pm$ 1.71
EthanolHorizontalSignal	30.61 $\pm$ 3.99	43.15 $\pm$ 5.84	30.94 $\pm$ 3.48	29.89 $\pm$ 1.60	38.78 $\pm$ 6.14	39.59 $\pm$ 1.67	8.01 $\pm$ 1.27	41.62 $\pm$ 1.42	10.61 $\pm$ 3.44	40.06 $\pm$ 3.72	39.56 $\pm$ 4.17	49.17 $\pm$ 1.62
EOGVerticalSignal	21.71 $\pm$ 2.53	28.23 $\pm$ 2.47	19.83 $\pm$ 3.08	31.55 $\pm$ 2.19	40.06 $\pm$ 1.98	36.00 $\pm$ 1.12	8.47 $\pm$ 0.84	37.75 $\pm$ 1.67	11.99 $\pm$ 1.33	37.02 $\pm$ 1.99	30.55 $\pm$ 3.17	42.21 $\pm$ 2.26
Earthquakes	53.81 $\pm$ 8.62	53.96 $\pm$ 9.50	51.37 $\pm$ 3.47	52.52 $\pm$ 6.49	54.82 $\pm$ 2.85	60.91 $\pm$ 4.63	49.16 $\pm$ 8.34	59.71 $\pm$ 6.86	54.68 $\pm$ 5.26	61.87 $\pm$ 3.60	47.91 $\pm$ 6.66	54.68 $\pm$ 7.63
ElectricDevices	40.12 $\pm$ 4.42	38.65 $\pm$ 3.96	36.00 $\pm$ 4.64	28.04 $\pm$ 1.63	39.08 $\pm$ 8.26	48.59 $\pm$ 6.21	30.07 $\pm$ 3.26	33.74 $\pm$ 3.43	37.44 $\pm$ 1.49	48.14 $\pm$ 2.51	16.08 $\pm$ 5.57	49.42 $\pm$ 5.02
EthanolLevel	26.56 $\pm$ 1.96	25.76 $\pm$ 1.05	26.16 $\pm$ 0.71	29.28 $\pm$ 1.78	25.92 $\pm$ 1.70	28.07 $\pm$ 2.55	25.20 $\pm$ 3.34	26.80 $\pm$ 0.69	25.60 $\pm$ 1.13	29.87 $\pm$ 2.47	25.56 $\pm$ 1.05	28.24 $\pm$ 5.05
FaceAll	77.73 $\pm$ 3.48	80.30 $\pm$ 1.67	77.73 $\pm$ 3.48	67.09 $\pm$ 2.40	54.49 $\pm$ 5.41	67.46 $\pm$ 1.82	59.68 $\pm$ 1.83	67.32 $\pm$ 2.82	60.69 $\pm$ 2.63	64.06 $\pm$ 2.32	32.89 $\pm$ 2.71	63.01 $\pm$ 1.34
FaceFour	87.95 $\pm$ 3.07	59.32 $\pm$ 7.51	92.95 $\pm$ 2.03	74.77 $\pm$ 0.95	66.36 $\pm$ 10.77	85.98 $\pm$ 1.31	65.91 $\pm$ 2.27	88.26 $\pm$ 0.66	69.55 $\pm$ 1.69	83.71 $\pm$ 1.31	34.55 $\pm$ 3.82	79.09 $\pm$ 1.90
FacesUCR	87.80 $\pm$ 1.60	84.26 $\pm$ 2.06	82.41 $\pm$ 2.48	63.80 $\pm$ 1.79	54.40 $\pm$ 3.05	80.54 $\pm$ 1.15	70.11 $\pm$ 1.05	83.69 $\pm$ 1.13	72.08 $\pm$ 1.16	79.51 $\pm$ 1.53	34.67 $\pm$ 3.28	79.08 $\pm$ 1.41
FiftyWords	45.36 $\pm$ 2.39	36.70 $\pm$ 1.89	45.36 $\pm$ 2.39	53.67 $\pm$ 1.47	50.46 $\pm$ 2.83	64.32 $\pm$ 1.87	16.85 $\pm$ 0.67	67.99 $\pm$ 1.84	37.05 $\pm$ 3.62	57.36 $\pm$ 1.54	56.40 $\pm$ 1.98	71.34 $\pm$ 2.10
Fish	83.31 $\pm$ 3.78	56.00 $\pm$ 4.48	71.54 $\pm$ 6.28	65.60 $\pm$ 3.49	62.74 $\pm$ 2.15	68.95 $\pm$ 4.29	24.38 $\pm$ 3.80	71.05 $\pm$ 1.44	57.14 $\pm$ 3.45	75.05 $\pm$ 2.31	38.17 $\pm$ 5.05	78.29 $\pm$ 3.92
FordA	78.26 $\pm$ 5.14	75.61 $\pm$ 0.71	78.70 $\pm$ 3.26	49.09 $\pm$ 0.93	67.88 $\pm$ 3.90	51.99 $\pm$ 2.52	52.22 $\pm$ 0.95	48.81 $\pm$ 1.31	53.05 $\pm$ 1.09	48.21 $\pm$ 1.67	50.06 $\pm$ 2.65	57.15 $\pm$ 1.85
FordB	63.83 $\pm$ 5.74	62.17 $\pm$ 3.33	61.70 $\pm$ 3.19	48.54 $\pm$ 0.72	59.06 $\pm$ 1.85	46.75 $\pm$ 0.61	49.79 $\pm$ 1.45	47.24 $\pm$ 1.24	50.47 $\pm$ 2.25	47.33 $\pm$ 0.14	49.33 $\pm$ 1.67	

TABLE 9: Per-dataset 10-shot classification results on 128 UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
MixedShapesSmallTrain	70.61 $\pm$ 10.30	69.46 $\pm$ 11.49	75.93 $\pm$ 3.55	73.56 $\pm$ 2.63	73.16 $\pm$ 8.70	78.08 $\pm$ 2.18	23.04 $\pm$ 1.80	77.03 $\pm$ 1.02	27.06 $\pm$ 6.78	77.29 $\pm$ 1.92	65.95 $\pm$ 4.04	81.61 $\pm$ 2.47
MoteStrain	89.38 $\pm$ 0.88	89.14 $\pm$ 3.04	90.18 $\pm$ 0.68	86.26 $\pm$ 0.20	86.58 $\pm$ 1.30	87.19 $\pm$ 1.69	77.40 $\pm$ 1.04	87.43 $\pm$ 0.48	85.32 $\pm$ 0.75	90.65 $\pm$ 0.48	67.41 $\pm$ 11.72	89.23 $\pm$ 1.53
NonInvasiveFetalECGThorax1	55.91 $\pm$ 3.81	50.28 $\pm$ 3.45	55.91 $\pm$ 3.81	81.90 $\pm$ 0.59	63.56 $\pm$ 4.40	77.13 $\pm$ 1.08	36.83 $\pm$ 0.46	78.18 $\pm$ 0.43	46.96 $\pm$ 3.44	75.71 $\pm$ 1.08	58.08 $\pm$ 1.54	76.84 $\pm$ 1.40
NonInvasiveFetalECGThorax2	62.50 $\pm$ 1.19	67.13 $\pm$ 2.71	62.50 $\pm$ 1.19	86.79 $\pm$ 1.25	75.21 $\pm$ 2.50	83.55 $\pm$ 0.48	57.79 $\pm$ 2.45	83.27 $\pm$ 1.00	69.23 $\pm$ 2.65	80.68 $\pm$ 0.66	70.22 $\pm$ 2.79	82.53 $\pm$ 1.33
OSULeaf	72.64 $\pm$ 5.62	69.26 $\pm$ 4.17	69.01 $\pm$ 3.75	34.88 $\pm$ 3.54	39.34 $\pm$ 3.85	46.42 $\pm$ 2.52	19.83 $\pm$ 4.60	40.50 $\pm$ 2.19	29.83 $\pm$ 2.82	37.19 $\pm$ 2.07	28.10 $\pm$ 5.12	59.42 $\pm$ 3.58
OliverOil	56.00 $\pm$ 14.79	40.00 $\pm$ 0.00	66.67 $\pm$ 9.72	40.00 $\pm$ 0.00	40.00 $\pm$ 0.00	40.00 $\pm$ 0.00	45.56 $\pm$ 9.62	40.00 $\pm$ 0.00	40.00 $\pm$ 0.00	40.00 $\pm$ 0.00	40.00 $\pm$ 0.00	40.00 $\pm$ 0.00
PLAID	28.04 $\pm$ 4.01	25.14 $\pm$ 3.42	22.46 $\pm$ 6.07	34.45 $\pm$ 1.13	54.53 $\pm$ 5.93	50.16 $\pm$ 2.34	32.59 $\pm$ 0.99	44.82 $\pm$ 3.36	37.77 $\pm$ 2.12	46.80 $\pm$ 0.84	54.53 $\pm$ 5.93	59.89 $\pm$ 3.28
PhalangesOutlinesCorrect	56.88 $\pm$ 6.98	56.74 $\pm$ 8.68	57.48 $\pm$ 7.09	54.27 $\pm$ 11.23	49.91 $\pm$ 6.62	58.78 $\pm$ 4.01	55.79 $\pm$ 3.74	55.36 $\pm$ 17.14	52.42 $\pm$ 4.71	54.23 $\pm$ 12.22	53.99 $\pm$ 12.03	55.73 $\pm$ 8.07
Phoneme	28.10 $\pm$ 1.26	25.96 $\pm$ 1.60	28.10 $\pm$ 1.26	8.58 $\pm$ 0.50	11.88 $\pm$ 0.54	13.82 $\pm$ 0.86	9.55 $\pm$ 1.16	11.71 $\pm$ 0.45	10.91 $\pm$ 1.85	10.88 $\pm$ 0.47	10.94 $\pm$ 0.56	21.95 $\pm$ 1.27
PickupGestureWiimoteZ	50.80 $\pm$ 15.01	68.40 $\pm$ 2.61	73.60 $\pm$ 2.19	51.60 $\pm$ 1.67	54.40 $\pm$ 6.84	56.67 $\pm$ 3.06	26.67 $\pm$ 1.15	54.67 $\pm$ 1.15	42.00 $\pm$ 5.10	60.67 $\pm$ 1.15	54.40 $\pm$ 6.84	70.40 $\pm$ 2.61
PigAirwayPressure	26.35 $\pm$ 2.05	12.88 $\pm$ 0.53	50.58 $\pm$ 0.92	4.81 $\pm$ 0.00	4.13 $\pm$ 0.43	4.81 $\pm$ 0.48	2.08 $\pm$ 1.00	6.25 $\pm$ 1.27	4.62 $\pm$ 0.87	6.73 $\pm$ 0.48	5.00 $\pm$ 0.65	9.62 $\pm$ 0.48
PigArtPressure	54.71 $\pm$ 1.57	25.19 $\pm$ 3.85	69.81 $\pm$ 3.29	8.27 $\pm$ 0.40	6.06 $\pm$ 2.62	8.97 $\pm$ 0.28	6.41 $\pm$ 1.00	9.78 $\pm$ 0.73	6.35 $\pm$ 0.92	8.65 $\pm$ 1.27	18.08 $\pm$ 2.06	31.73 $\pm$ 1.92
PigCVP	28.85 $\pm$ 2.68	19.62 $\pm$ 1.84	61.35 $\pm$ 2.42	7.12 $\pm$ 0.40	3.27 $\pm$ 1.61	7.37 $\pm$ 0.73	7.05 $\pm$ 1.21	8.33 $\pm$ 1.69	4.13 $\pm$ 0.80	5.93 $\pm$ 1.11	11.63 $\pm$ 1.24	19.52 $\pm$ 1.34
Plane	100.00 $\pm$ 0.00	99.43 $\pm$ 1.28	100.00 $\pm$ 0.00	96.57 $\pm$ 1.28	96.95 $\pm$ 1.24	98.10 $\pm$ 0.88	98.73 $\pm$ 0.55	97.14 $\pm$ 0.95	98.48 $\pm$ 0.52	97.14 $\pm$ 0.95	49.71 $\pm$ 5.36	99.24 $\pm$ 1.24
PowerCons	84.00 $\pm$ 3.74	85.67 $\pm$ 2.68	88.11 $\pm$ 1.91	73.22 $\pm$ 7.34	83.67 $\pm$ 3.78	85.93 $\pm$ 2.85	53.33 $\pm$ 0.56	86.30 $\pm$ 2.25	59.67 $\pm$ 5.39	84.63 $\pm$ 0.85	74.33 $\pm$ 14.06	82.33 $\pm$ 5.46
ProximalPhalanxOutlineAgeGroup	69.07 $\pm$ 5.10	80.49 $\pm$ 4.77	80.98 $\pm$ 2.76	80.98 $\pm$ 2.95	75.90 $\pm$ 8.78	81.30 $\pm$ 5.55	81.14 $\pm$ 3.66	81.63 $\pm$ 4.89	76.88 $\pm$ 9.31	77.56 $\pm$ 6.40	68.78 $\pm$ 16.36	79.80 $\pm$ 3.30
ProximalPhalanxOutlineCorrect	63.71 $\pm$ 11.74	64.60 $\pm$ 2.91	63.85 $\pm$ 3.21	56.01 $\pm$ 10.53	61.86 $\pm$ 4.47	64.60 $\pm$ 0.69	66.55 $\pm$ 3.51	64.26 $\pm$ 0.69	61.86 $\pm$ 6.47	65.98 $\pm$ 3.31	51.34 $\pm$ 15.85	60.07 $\pm$ 4.20
ProximalPhalanxTW	60.78 $\pm$ 11.91	70.05 $\pm$ 3.85	72.39 $\pm$ 1.88	51.12 $\pm$ 7.39	70.73 $\pm$ 1.42	64.07 $\pm$ 2.20	60.16 $\pm$ 2.69	65.53 $\pm$ 2.25	57.95 $\pm$ 5.18	59.84 $\pm$ 3.91	25.07 $\pm$ 15.12	69.66 $\pm$ 3.47
RefrigerationDevices	42.67 $\pm$ 7.00	44.27 $\pm$ 6.42	43.41 $\pm$ 5.31	34.72 $\pm$ 1.59	37.71 $\pm$ 6.85	44.00 $\pm$ 2.97	34.40 $\pm$ 1.33	34.76 $\pm$ 2.56	35.31 $\pm$ 4.05	44.62 $\pm$ 2.22	32.37 $\pm$ 1.67	41.23 $\pm$ 4.90
Rock	41.60 $\pm$ 6.84	34.40 $\pm$ 4.56	36.00 $\pm$ 1.41	81.60 $\pm$ 0.89	57.20 $\pm$ 9.01	82.67 $\pm$ 1.15	84.67 $\pm$ 6.11	88.00 $\pm$ 0.00	62.00 $\pm$ 6.16	86.67 $\pm$ 3.06	30.80 $\pm$ 4.82	76.40 $\pm$ 2.97
ScreenType	35.84 $\pm$ 5.22	36.48 $\pm$ 5.07	37.23 $\pm$ 5.64	36.37 $\pm$ 2.19	39.15 $\pm$ 5.32	40.53 $\pm$ 1.87	35.73 $\pm$ 2.12	36.09 $\pm$ 2.74	33.12 $\pm$ 2.51	39.82 $\pm$ 2.27	36.32 $\pm$ 3.91	37.12 $\pm$ 3.93
SemgHandGenderCh2	63.47 $\pm$ 5.58	74.93 $\pm$ 6.24	65.43 $\pm$ 6.18	67.83 $\pm$ 4.45	66.37 $\pm$ 2.41	62.89 $\pm$ 2.12	51.67 $\pm$ 5.43	65.78 $\pm$ 6.45	55.43 $\pm$ 2.94	60.72 $\pm$ 4.24	58.77 $\pm$ 7.13	63.80 $\pm$ 10.73
SemgHandMovementCh2	37.69 $\pm$ 5.41	36.27 $\pm$ 1.15	37.82 $\pm$ 1.58	37.42 $\pm$ 2.10	32.22 $\pm$ 2.74	30.67 $\pm$ 2.78	25.04 $\pm$ 0.56	34.52 $\pm$ 3.16	25.69 $\pm$ 1.85	33.04 $\pm$ 3.33	33.64 $\pm$ 4.19	45.56 $\pm$ 2.93
SemgHandSubjectCh2	43.56 $\pm$ 2.78	45.20 $\pm$ 5.06	40.22 $\pm$ 5.31	62.80 $\pm$ 4.54	54.13 $\pm$ 4.47	60.37 $\pm$ 0.71	34.22 $\pm$ 3.27	68.07 $\pm$ 2.65	39.96 $\pm$ 2.48	60.96 $\pm$ 1.10	38.18 $\pm$ 2.42	66.71 $\pm$ 4.45
ShakeGestureWiimoteZ	82.00 $\pm$ 6.16	91.20 $\pm$ 2.28	88.80 $\pm$ 1.10	55.60 $\pm$ 2.97	68.40 $\pm$ 7.67	60.00 $\pm$ 4.00	33.33 $\pm$ 3.06	62.67 $\pm$ 3.06	45.20 $\pm$ 7.29	56.67 $\pm$ 1.15	68.40 $\pm$ 7.67	91.60 $\pm$ 2.97
ShapeletSim	86.33 $\pm$ 6.80	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	48.78 $\pm$ 0.82	95.22 $\pm$ 4.26	59.07 $\pm$ 5.56	48.70 $\pm$ 1.40	50.19 $\pm$ 2.80	51.00 $\pm$ 1.98	49.81 $\pm$ 1.95	52.33 $\pm$ 4.64	96.67 $\pm$ 0.96
ShapesAll	68.60 $\pm$ 2.31	66.63 $\pm$ 1.82	68.60 $\pm$ 2.31	62.47 $\pm$ 0.34	65.47 $\pm$ 2.49	72.11 $\pm$ 0.69	12.89 $\pm$ 4.16	73.28 $\pm$ 0.38	31.00 $\pm$ 5.80	65.56 $\pm$ 1.08	70.00 $\pm$ 1.20	83.13 $\pm$ 1.32
SmallKitchenAppliances	67.20 $\pm$ 8.07	61.87 $\pm$ 4.21	64.69 $\pm$ 7.08	36.59 $\pm$ 1.57	60.00 $\pm$ 3.85	55.11 $\pm$ 4.59	41.51 $\pm$ 2.40	38.76 $\pm$ 2.56	47.25 $\pm$ 3.08	51.02 $\pm$ 1.54	41.33 $\pm$ 5.03	57.60 $\pm$ 5.56
SmoothSubspace	92.00 $\pm$ 1.56	87.07 $\pm$ 3.67	92.00 $\pm$ 3.02	38.13 $\pm$ 6.59	67.73 $\pm$ 10.31	79.56 $\pm$ 2.78	80.67 $\pm$ 0.67	60.00 $\pm$ 11.02	86.13 $\pm$ 3.38	90.67 $\pm$ 4.00	43.60 $\pm$ 9.96	89.47 $\pm$ 2.60
SonyAIBORobotSurface1	95.11 $\pm$ 2.42	96.01 $\pm$ 0.87	92.45 $\pm$ 1.40	42.96 $\pm$ 0.07	87.95 $\pm$ 1.92	64.28 $\pm$ 1.83	67.05 $\pm$ 1.01	76.43 $\pm$ 0.69	64.86 $\pm$ 0.43	67.83 $\pm$ 1.06	45.09 $\pm$ 4.84	86.09 $\pm$ 3.97
SonyAIBORobotSurface2	96.73 $\pm$ 3.11	92.82 $\pm$ 3.91	94.67 $\pm$ 2.40	77.50 $\pm$ 0.71	90.95 $\pm$ 2.30	81.25 $\pm$ 2.70	83.07 $\pm$ 2.21	86.08 $\pm$ 2.22	82.71 $\pm$ 1.50	85.73 $\pm$ 2.18	61.15 $\pm$ 12.88	84.11 $\pm$ 2.29
StarLightCurves	71.47 $\pm$ 14.56	82.56 $\pm$ 3.87	75.10 $\pm$ 5.06	74.16 $\pm$ 3.22	77.51 $\pm$ 4.09	75.29 $\pm$ 4.53	29.60 $\pm$ 9.11	76.31 $\pm$ 3.24	30.20 $\pm$ 4.95	74.55 $\pm$ 5.89	70.71 $\pm$ 14.11	85.26 $\pm$ 3.54
Strawberry	79.73 $\pm$ 8.80	76.70 $\pm$ 3.26	76.70 $\pm$ 4.55	60.81 $\pm$ 1.94	79.89 $\pm$ 6.30	61.98 $\pm$ 1.58	80.63 $\pm$ 1.13	59.10 $\pm$ 1.09	77.51 $\pm$ 4.36	68.11 $\pm$ 2.15	50.54 $\pm$ 10.66	75.84 $\pm$ 0.99
SwedishLeaf	88.03 $\pm$ 1.39	83.68 $\pm$ 1.26	88.03 $\pm$ 1.39	66.11 $\pm$ 1.36	73.47 $\pm$ 2.54	75.95 $\pm$ 1.36	60.69 $\pm$ 4.66	81.39 $\pm$ 2.49	68.80 $\pm$ 1.87	74.72 $\pm$ 1.82	28.96 $\pm$ 3.20	85.89 $\pm$ 1.27
Symbols	93.83 $\pm$ 2.55	89.89 $\pm$ 7.08	92.52 $\pm$ 1.13	75.94 $\pm$ 0.93	85.87 $\pm$ 1.87	83.32 $\pm$ 0.52	34.77 $\pm$ 0.30	85.96 $\pm$ 1.03	61.21 $\pm$ 7.54	82.51 $\pm$ 0.70	21.11 $\pm$ 2.93	89.33 $\pm$ 0.82
SyntheticControl	95.60 $\pm$ 1.14	97.87 $\pm$ 0.61	97.60 $\pm$ 0.43	59.07 $\pm$ 3.98	94.20 $\pm$ 1.64	90.78 $\pm$ 1.64	59.44 $\pm$ 1.95	79.44 $\pm$ 4.86	68.40 $\pm$ 3.43	90.11 $\pm$ 1.35	22.60 $\pm$ 7.62	97.33 $\pm$ 0.85
ToeSegmentation1	93.25 $\pm$ 1.60	89.56 $\pm$ 3.33	91.58 $\pm$ 1.68	55.53 $\pm$ 3.14	68.95 $\pm$ 9.85	56.73 $\pm$ 5.97	50.29 $\pm$ 6.71	55.85 $\pm$ 3.55	49.39 $\pm$ 2.18	55.70 $\pm$ 4.98	50.53 $\pm$ 1.40	80.53 $\pm$ 2.75
ToeSegmentation2	81.38 $\pm$ 10.25	87.08 $\pm$ 4.53	85.85 $\pm$ 2.64	50.15 $\pm$ 3.62	73.38 $\pm$ 4.70	68.21 $\pm$ 10.44	56.67 $\pm$ 7.86	58.46 $\pm$ 6.30	46.77 $\pm$ 3.41	61.54 $\pm$ 3.35	43.54 $\pm$ 14.75	80.31 $\pm$ 4.70
Trace	92.20 $\pm$ 4.60	100.00 $\pm$ 0.00	95.20 $\pm$ 5.07	55.80 $\pm$ 5.72	86.20 $\pm$ 10.35	72.67 $\pm$ 1.53	55.67 $\pm$ 4.51	61.33 $\pm$ 4.62	74.80 $\pm$ 3.27	78.00 $\pm$ 4.36	46.80 $\pm$ 8.17	100.00 $\pm$ 0.00
TwoLeadECG	99.86 $\pm$ 0.08	98.79 $\pm$ 1.25	97.56 $\pm$ 2.37	58.05 $\pm$ 1.68	91.89 $\pm$ 5.00	70.85 $\pm$ 3.05	75.62 $\pm$ 1.63	65.26 $\pm$ 1.91	78.86 $\pm$ 1.56	78.99 $\pm$ 2.23	54.01 $\pm$ 3.73	99.68 $\pm$ 0.24
TwoPatterns	70.89 $\pm$ 4.77	77.56 $\pm$ 2.96	78.86 $\pm$ 3.59	42.28 $\pm$ 1.08	74.73 $\pm$ 6.48	42.55 $\pm$ 1.37	26.22 $\pm$ 0.85	42.35 $\pm$ 1.82	29.82 $\pm$ 3.35	37.95 $\pm$ 1.72	25.60 $\pm$ 4.32	70.06 $\pm$ 0.70
UMD	92.78 $\pm$ 3.35	98.33 $\pm$ 1.05	98.75 $\pm$ 0.58	64.17 $\pm$ 3.05	91.53 $\pm$ 4.77	76.62 $\pm$ 2.44	75.93 $\pm$ 7.23	89.58 $\pm$ 4.22	93.06 $\pm$ 1.47	87.27 $\pm$ 3.21	47.50 $\pm$ 8.96	98.67 $\pm$ 7.42
UWaveGestureLibraryAll	48.45 $\pm$ 3.31	48.15 $\pm$ 2.07	49.06 $\pm$ 3.81	80.57 $\pm$ 3.71	68.35 $\pm$ 4.39	84.19 $\pm$ 2.90	27.54 $\pm$ 1.68</					

TABLE 10
Win/Tie/Loss of TimeMorph against selected baselines on 128 UCR datasets.

Baseline	1-shot	2-shot	5-shot	10-shot
COSCO-ResNet	68 / 0 / 60	68 / 1 / 59	58 / 0 / 70	64 / 0 / 64
PatchTST	97 / 0 / 31	97 / 0 / 31	99 / 0 / 29	94 / 2 / 32
TimesNet	88 / 0 / 40	91 / 1 / 36	86 / 0 / 42	93 / 1 / 34
GPT4TS	106 / 2 / 20	110 / 0 / 18	114 / 1 / 13	121 / 1 / 6
ResNet	64 / 1 / 63	66 / 1 / 61	67 / 0 / 61	69 / 0 / 59
TapNet	75 / 0 / 53	75 / 3 / 50	76 / 1 / 51	72 / 3 / 53

A.1 Win–Tie–Loss Counts

Table 10 reports the complete win/tie/loss counts visualized in Figure 4.

TABLE 11
Key target-adaptation settings for TimeMorph.

Component	Setting
Base LLM	meta-llama/Llama-3.2-1B
Vision backbone	facebook/dinov2-base
Classification prompt suffix / answer	Class: / one class token, e.g., $\langle \text{cls}_y \rangle$ followed by EOS
Morphology-map construction	Overlapping-window grid transform with per-sequence median/IQR normalization
Temporal patch length / stride	16 / 8
Temporal hidden size / heads	128 / 8
Branch projectors	LayerNorm-Linear-GELU-Dropout MLPs when dimensions differ
Language projector	MLP projector into the LLM hidden dimension followed by output LayerNorm
Default UCR checkpoint	Stage II (TSQA transfer)
Warm-up / joint epochs	5 / 55
Batch size / eval batch size	8 / 8
Gradient accumulation	4
LR (encoder / projector / LoRA)	2×10^{-4} / 1×10^{-4} / 1×10^{-4}
Language-side adaptation	LoRA on <code>q/k/v/o_proj</code> , <code>gate_proj</code> , <code>up_proj</code> , and <code>down_proj</code> ($r = 16$, $\alpha = 32$)
Tokenizer training mode	<code>class_rows</code>
Anonymous class-token initialization	pretrained vocabulary-row mean plus small random perturbation
Prediction rule	anonymous class-token scoring; semantic-prior class-token scoring when label text is meaningful
Freeze vision backbone	true
Downstream data augmentation	none

TABLE 12
Stage data sources and roles for TimeMorph.

Stage	Training data	Validation / monitoring data	Role in reported experiments
Stage I: Representation Pretraining	Raw TSQA-train series, raw M4-train series, and official TRAIN-split raw series from a 98-dataset UCR training pool.	Raw TSQA validation/test series and raw M4 validation/test series. No UCR official TEST split is used.	Learns dual-view representations before the LLM interface; corresponds to <code>stage0_encoder_ssl</code> in the implementation.
Stage II: Embedding-Space Alignment Pretraining	TSQA train split only, with answer-token language-model training on time-series question answering.	TSQA validation/test splits.	Provides the default checkpoint used by all main UCR few-shot results in Section 4; corresponds to <code>stage1_tsqa_transfer</code> .
Stage III: Optional Label-Semantic Calibration	Mixture of TSQA retention examples, M4 captioning examples, and synthetic attribute supervision derived from TSQA/M4 raw series.	Corresponding validation/test splits of the same mixture.	Provides the calibrated checkpoint used by the external semantic-label evaluations; not used for the default UCR benchmark. This corresponds to <code>stage2_semantic_bridge</code> and later caption specialization in code.

APPENDIX B ADDITIONAL EXPERIMENTAL DETAILS

We summarize the key implementation choices of TimeMorph for the few-shot benchmark in Table 11. These settings are fixed across the main benchmark reported in Section 4.

B.0.0.1 Morphology-map transform.: The operator Γ_{img} in Eq. 5 applies per-sample min-max normalization over the two-dimensional grid and a fixed contrast adjustment $u \mapsto u^{0.8}$. In the default morphology mode, the effective window length is $\max(1, \lfloor \sqrt{T} \rfloor)$, the stride length is $\max(1, \lfloor 0.1q \rfloor)$, incomplete windows are completed by replicate padding on the left, and the resulting grid is resized to the DINOv2 image resolution with nearest-neighbor interpolation before RGB replication. We use DINOv2 patch tokens from the configured intermediate layer and exclude the class token.

B.0.0.2 Stage I pretraining hyperparameters.: Stage I uses mask ratio 0.4, branch-dropout probability 0.1, jitter standard deviation 0.03, scaling range $[0.8, 1.2]$, and time-mask ratio 0.1 for the two augmented views. The Stage I loss weights are $\lambda_{\text{rec}} = 1.0$, $\lambda_t = 0.25$, $\lambda_v = 0.25$, and $\lambda_f = 0.5$. These values are fixed before the target few-shot runs.

B.1 Pretraining Stages and Data Sources

Table 12 summarizes the three training stages discussed in the main paper and clarifies which checkpoint is used by the UCR and external semantic-label evaluations.

After staged pretraining, target adaptation uses only the sampled support labels of the target dataset. For UCR targets contained in the Stage I pool, unlabeled TRAIN sequences may already have contributed to the encoder pretraining.

B.2 Code Release and Assets

We will release an anonymized code package with the submission. The release includes the TimeMorph implementation, target-adaptation and evaluation scripts, baseline launchers, reporting scripts for the tables in this paper, environment files, and README instructions for preparing the public datasets used in our experiments. The code is released under the MIT license. The package does not redistribute raw third-party datasets or gated pretrained model weights; users should obtain those assets from the original providers and comply with their licenses and terms of use. The released scripts document which experiments are directly reproducible from the package and which require separately downloaded datasets or pretrained checkpoints.

B.3 Asset and License Table

Table 13 summarizes the main existing assets used in this paper. We list the original source, role, version when available, license or stated terms, and whether the asset is redistributed in our release. For assets whose official source does not state a separate dataset license, we cite the official repository or archive page and do not redistribute the raw data.

TABLE 13: Existing assets used in this paper and their licenses or stated terms. URLs were checked during manuscript preparation.

Asset	Role	Source / version	License or stated terms	Redistribution in our release
UCR 2018	Archive Main few-shot benchmark and unlabeled Stage I source pool	TSC website; Zenodo v1	Zenodo record lists Creative Commons Attribution 4.0 International (CC BY 4.0); archive should be cited via Dataset Card et al.	Not redistributed; scripts expect users to download from the original archive or cited mirror.
TSQA / Chat-Time	Stage I raw series and card Stage II time-series question answering	Hugging Face dataset	Apache-2.0 according to the Hugging Face dataset metadata	Not redistributed; scripts load from the original dataset source.
M4 Competition dataset	Stage I raw series and Stage III captioning / label-semantic calibration data	Official M4 repository	The official repository publicly provides the dataset and asks users to cite the M4 competition resources; no separate dataset LICENSE file was found in the repository.	Not redistributed; users download from the official repository.
MIT-BIH Arrhythmia Database	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet citation required	Not redistributed; preprocessing scripts require a local PhysioNet download.
PhysioNet/CinC 2017 AF dataset	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet and challenge citations required	Not redistributed; preprocessing scripts require a local PhysioNet download.
Llama 3.2 1B	Pretrained causal LLM backbone	Hugging Face model card	Llama 3.2 Community License and Acceptable Use Policy	Not redistributed; users must obtain access from Meta / Hugging Face.
DINOv2-base	Frozen visual backbone	Hugging Face model card	Apache-2.0	Not redistributed; loaded from the original model repository.
OpenTSLM base code	Starting codebase and time-series language-model components	Project repository and release package	MIT license in the repository license file	Modified code is released in anonymized form under MIT, preserving applicable notices.

TABLE 14
Compute-resource budget for the reported experiment groups. Downstream jobs use single RTX 3090 workers unless noted otherwise.

Experiment group	Scope	Compute workers	Mean per run	Total GPU-hours	Notes
Stage I-II pretraining	Staged pretraining checkpoint used to initialize target adaptation	1 RTX 3090 worker	–	19.81	Stage I encoder SSL used 9.57 GPU-hours; Stage I-II together used 19.81 GPU-hours.
Optional Stage III calibration	Label-semantic calibration and caption-specialization training used outside the default UCR results	1 RTX 3090 worker	–	29.03	Not used for the default UCR benchmark.
TimeMorph main UCR benchmark	128 UCR datasets, 1/2/5/10-shot, five runs per setting; 2,507 completed downstream runs	6 single-GPU workers	0.67 GPU-h/run	1,677	Batch size 8, gradient accumulation 4, 60 downstream epochs.
UCR baseline runs	ResNet, TapNet, COSCO-ResNet, PatchTST, GPT4TS, and TSLib-family baselines under the same few-shot protocol	6 single-GPU workers	0.40–0.49 GPU-h/run	10,937	Includes the retained main UCR baseline logs; GPT4TS coverage is reported for the completed 60-dataset run.
External-dataset TimeMorph runs	MIT-BIH and CinC2017 under 10/20/30-shot protocols	single-GPU workers	1.69 GPU-h/run	102	Same target-adaptation recipe as the external few-shot experiments.
External-dataset baseline runs	Reported baseline families on MIT-BIH and CinC2017	single-GPU workers	0.68 GPU-h/job	15	Launcher jobs cover both external datasets and baseline families.
Top-level ablations	No-pretraining, temporal-only, visual-only, class-token decoding, and no-LLM backbone ablations	6 single-GPU workers	1.67 GPU-h/run	3,895	Includes the two retained no-LLM linear runs added together.
Semantic-prior ablation	External semantic-label interface runs	single-GPU workers	1.69 GPU-h/run	20	Uses the same 10/20/30-shot external evaluation protocol.
Efficiency probe	Representative UCR runtime subset with parameter, memory, latency, and throughput measurements	one RTX 3090	–	–	Detailed runtime and memory values are reported in Tables 3, 17, and 19.

TABLE 13: Existing assets used in this paper and their licenses or stated terms (continued).

Asset	Role	Source / version	License or stated terms	Redistribution in our release
Python libraries, including PyTorch, Transformers, PEFT, NumPy, pandas, scikit-learn, and related dependencies	ML Training, adaptation, evaluation, and reporting	Version constraints are listed in the released environment files	Governed by their respective open-source licenses	Not vendored; installed by the environment files.

B.4 Computational Resource Reporting

All reported downstream experiments were run on a local server with six NVIDIA GeForce RTX 3090 GPUs (24GB each), two Intel Xeon Silver 4314 CPUs, and 251GiB system memory. Unless otherwise stated, each training job used one GPU, and the batch launchers executed up to six single-GPU workers in parallel. Table 14 summarizes the compute used by the main experiment groups. GPU-hours are summed over single-GPU jobs; wall-clock time can be lower when multiple workers run concurrently. The efficiency measurements in Appendix B.7 additionally report peak memory, train-step latency, inference latency, throughput, total parameters, and downstream-updated parameters on the representative runtime subset.

Retained launcher summaries also include failed or exploratory jobs. These runs account for less than 2 additional GPU-hours in the preserved logs, and dry-run checks add a negligible amount of compute.

B.5 External Dataset Results

Table 15 reports the full 10-, 20-, and 30-shot results on MIT-BIH and CinC2017. All methods use the same external few-shot settings and reported run count; we use these datasets as transfer checks beyond the primary UCR benchmark rather than as a claim of uniform dominance across external domains.

B.6 Semantic-Prior Decision Ablation

Table 16 expands the semantic-prior comparison summarized in Table 5. Both variants use one-pass class-token scoring; the semantic-prior variant additionally initializes and regularizes class-token rows from label-verbalizer prototypes. The semantic prior improves the averaged performance on both external datasets, while individual shot-level deltas remain dataset-dependent.

TABLE 15

Few-shot classification accuracy (%) on two external semantic-label datasets. Each cell reports mean $\pm$ standard deviation over 10 runs under the same few-shot protocol. Best is bold and second-best is underlined within each dataset-shot column.

Model	MIT-BIH			CinC2017		
	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot
TimeMorph	55.03 $\pm$ 9.40	63.95 $\pm$ 13.80	64.15 $\pm$ 5.32	32.80 $\pm$ 9.08	36.98 $\pm$ 4.27	34.13 $\pm$ 9.73
ResNet	37.97 $\pm$ 14.13	55.66 $\pm$ 8.58	57.07 $\pm$ 9.05	30.95 $\pm$ 12.41	33.73 $\pm$ 7.43	36.20 $\pm$ 4.84
TapNet	31.94 $\pm$ 7.18	40.15 $\pm$ 10.28	44.11 $\pm$ 11.14	30.05 $\pm$ 6.73	<u>32.80 $\pm$ 8.81</u>	31.01 $\pm$ 11.54
COSCO-ResNet	32.33 $\pm$ 14.25	35.08 $\pm$ 13.75	41.17 $\pm$ 8.94	<u>31.41 $\pm$ 4.48</u>	31.04 $\pm$ 6.97	31.98 $\pm$ 8.37
PatchTST	38.94 $\pm$ 14.35	55.36 $\pm$ 8.84	57.08 $\pm$ 9.40	26.07 $\pm$ 6.24	30.35 $\pm$ 7.51	31.21 $\pm$ 3.36
OneFitsAll	32.92 $\pm$ 16.63	42.48 $\pm$ 10.82	54.87 $\pm$ 10.04	30.80 $\pm$ 6.07	27.56 $\pm$ 4.33	29.23 $\pm$ 3.03
TimesNet	51.01 $\pm$ 5.84	54.21 $\pm$ 7.81	56.15 $\pm$ 8.77	27.25 $\pm$ 2.81	28.41 $\pm$ 4.24	29.43 $\pm$ 2.44
DLinear	55.57 $\pm$ 9.85	57.45 $\pm$ 7.95	58.84 $\pm$ 8.69	24.92 $\pm$ 1.16	25.17 $\pm$ 1.22	24.53 $\pm$ 1.70
Autoformer	<u>38.05 $\pm$ 5.65</u>	42.49 $\pm$ 8.58	<u>44.39 $\pm$ 7.25</u>	28.37 $\pm$ 5.44	27.67 $\pm$ 2.76	26.30 $\pm$ 4.36
Crossformer	53.48 $\pm$ 9.36	<u>58.55 $\pm$ 10.43</u>	58.24 $\pm$ 9.71	24.02 $\pm$ 6.10	25.50 $\pm$ 2.86	27.35 $\pm$ 2.35
FEDformer	58.88 $\pm$ 5.77	56.80 $\pm$ 9.98	55.12 $\pm$ 10.09	20.35 $\pm$ 8.32	26.90 $\pm$ 3.54	27.91 $\pm$ 4.87
Informer	53.19 $\pm$ 6.03	54.20 $\pm$ 7.61	52.35 $\pm$ 10.80	26.93 $\pm$ 9.30	28.14 $\pm$ 4.01	30.12 $\pm$ 5.95

TABLE 16

Semantic-prior class-token ablation on two external semantic-label datasets (accuracy, %). Each cell reports mean $\pm$ standard deviation over 10 runs under the same few-shot protocol. The better variant within each dataset-shot column is bold. Δ denotes Semantic prior minus Anonymous labels.

Variant	MIT-BIH			CinC2017		
	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot
Anonymous labels	55.03 $\pm$ 9.40	63.95 $\pm$ 13.80	64.15 $\pm$ 5.32	32.80 $\pm$ 9.08	36.98 $\pm$ 4.27	34.13 $\pm$ 9.73
Semantic prior	57.05 $\pm$ 10.96	66.62 $\pm$ 6.54	68.87 $\pm$ 11.23	34.72 $\pm$ 3.15	35.24 $\pm$ 1.24	38.86 $\pm$ 7.00
Semantic prior – Anonymous	+2.02	+2.67	+4.72	+1.92	-1.74	+4.73

B.7 Efficiency Summary

Table 17 reports computational efficiency and adaptation cost under the 10-shot UCR protocol. Runtime and memory are measured on the predefined representative UCR subset with batch size 8 on a single RTX 3090, while parameter counts are reported separately for total deployed parameters and downstream-updated parameters.

B.8 Baseline Training Recipes and Hyperparameters

Table 18 summarizes the training rules used by the reported baseline families. The goal of this table is to make the few-shot comparison protocol explicit: all methods share the same sampled support/query split, but they do not all use the same checkpoint-selection rule or the same epoch schedule.

B.9 Computational Efficiency Probe Details

The efficiency probe is implemented by the repository’s `efficiency_probe.py` script. It uses the same model families as the main benchmark and omits TimeMorph-only ablations. The default runtime subset contains COFFEE, GUNPOINT, ECG5000, ELECTRICDEVICES, CROP, ACSF1, PHONEME, and NONINVASIVEFETALECGTHORAX1, covering short, medium, and long UCR series as well as different class counts. The script writes the LaTeX tables, CSV summary, and accuracy-versus-updated-parameters plot to the `efficiency-report` directory.

APPENDIX C

ADDITIONAL PLOTS

Figure 7 visualizes the shot-scaling trend over the 128 UCR datasets. TimeMorph remains the strongest method on the averaged benchmark curve across the entire few-shot range, with the largest absolute advantage appearing at 10-shot.

TABLE 17

Computational efficiency and adaptation cost under the 10-shot UCR protocol. Total parameters count the full deployed model, while updated parameters count only the parameters optimized during downstream adaptation. Adapter size estimates the per-dataset trainable-state footprint. Runtime is measured on representative UCR datasets with batch size 8 on a single RTX 3090.

Model	Total params	Updated params	Updated %	Adapter MB	Peak GB	Train ms/step	Infer ms/ex.	Infer ex./s
TimeMorph	1.28B	12.3M	0.96	46.7	13.75	1207.8	156.85	6.4
ResNet	629.1K	629.1K	100.00	2.4	0.33	13.2	0.26	4273.3
TapNet	912.1K	912.1K	100.00	3.5	0.22	15.0	0.47	3345.0
COSCO-ResNet	662.1K	662.1K	100.00	2.5	0.58	50.8	0.25	4315.1
PatchTST	665.4K	598.3K	90.80	2.3	1.54	32.6	0.75	2149.8
DLinear	1.2M	1.2M	100.00	4.6	0.04	2.7	0.07	14389.5
TimesNet	2.5M	2.5M	100.00	9.7	0.14	66.7	2.46	614.5
Autoformer	1.2M	1.2M	100.00	4.5	0.40	23.1	11.10	91.3
Crossformer	2.8M	2.8M	100.00	10.8	0.06	24.9	0.59	1709.6
FEDformer	1.8M	1.8M	100.00	6.7	0.26	244.1	5.17	201.9
Informer	1.4M	1.4M	100.00	5.3	0.22	18.0	0.62	1690.1
GPT4TS	82.2M	1.1M	1.34	4.2	0.56	22.6	0.86	1181.2

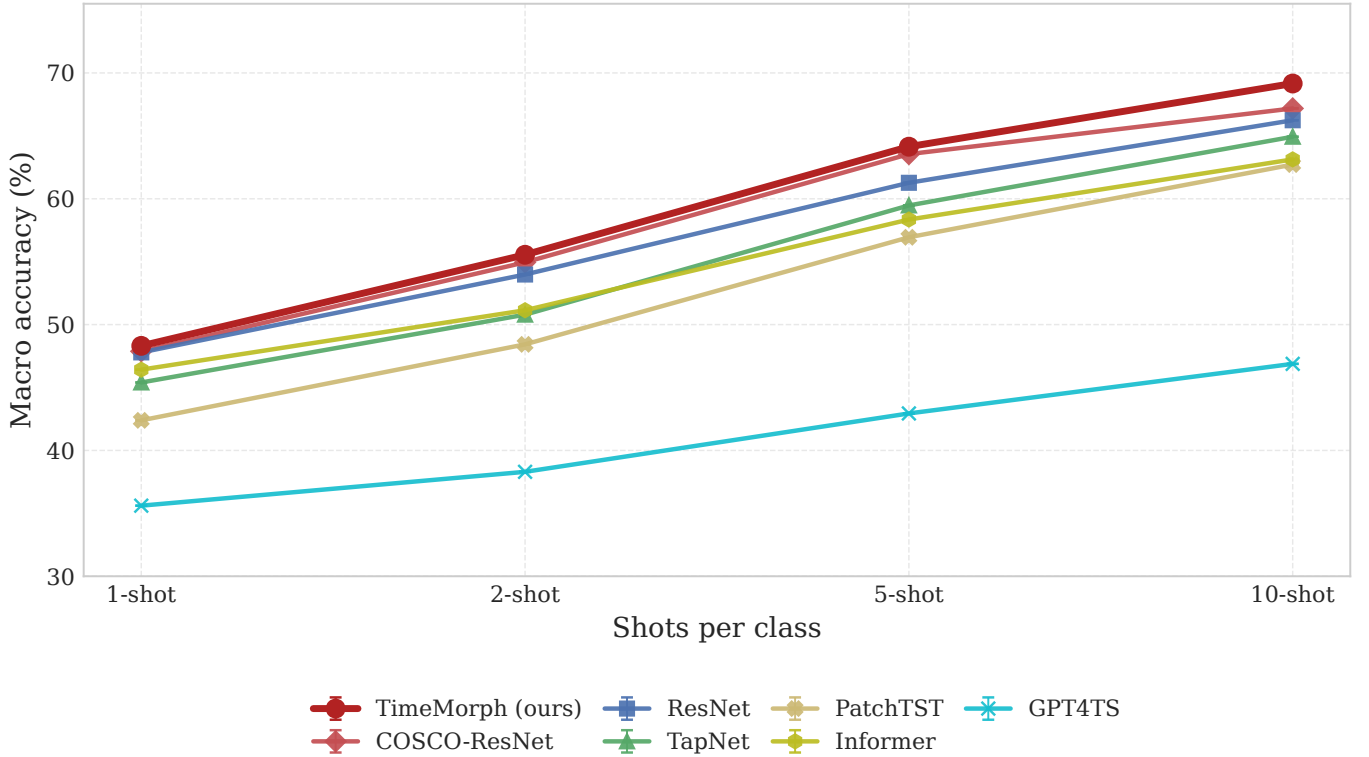


Fig. 7. Shot-scaling trend on the 128 UCR datasets. Each point reports the mean accuracy over five runs. TimeMorph remains the strongest method across all four shot settings.

TABLE 18

Few-shot training recipes for TimeMorph and the reported baseline families. All methods use the same sampled support sets within each run and evaluate on all official TEST examples from the target dataset.

Method / family	Optimizer	Epochs	Batch / LR	Model selection rule	Val	ES	Few-shot notes
TimeMorph	AdamW	5 warm-up + 55 joint	$8 / 2 \times 10^{-4}$ (encoder), 1×10^{-4} (projector, LoRA)	Report the last joint-adaptation checkpoint	No	No	Class-token rows remain trainable; language-side adaptation uses LoRA in the joint adaptation phase.
ResNet, TapNet	AdamW	Fixed schedule in the few-shot runner (60 in the benchmark configs)	model-specific / benchmark configs use 10^{-4}	Best support-set accuracy, tie-broken by support loss	No	No	Trained only on the sampled support set; the support set doubles as the model-selection set.
COSCO-ResNet	SAM with SGD base optimizer + prototypical loss	Fixed schedule (100 in the benchmark configs)	$128 / 10^{-2}$ by default	Final trained model; support centroids computed after training	No	No	Query predictions are made by distances to support centroids rather than a learned classifier head.
PatchTST	AdamW + linear warmup	Two-step fixed schedule (head warm-up + joint fine-tuning; 60 total epochs in the benchmark configs)	16 / separate backbone-head LR	Report the last joint fine-tuning checkpoint	No	No	Uses the same support/query split and full dataset class set as the main protocol.
TSLib family (DLINEAR, AUTOFORMER, CROSSFORMER, FEDFORMER, INFORMER, TIMESNET)	RAdam + cosine schedule	Fixed model-specific schedule from the TSLib profile or benchmark override	usually $16 / 10^{-3}$	Report the last checkpoint	No	No	Single-stage training; evaluation masks logits to the target dataset's class set. TimesNet uses a shorter profile than the other TSLib baselines.
GPT4TS	RAdam (default runner setting)	Fixed schedule (60 in the benchmark configs)	64 / 10^{-4} in the benchmark configs	Report the last checkpoint	No	No	Implemented with the One-Fits-All classification runner under the same sampled support/query protocol.

TABLE 19

Available per-dataset runtime details for the computational-efficiency probe. The full machine-readable runtime file is produced by the efficiency script; rows with missing measurements are omitted from the paper appendix.

Dataset	Model	Peak GB	Train ms/step	Infer ms/ex.	Infer ex./s
ACSF1	TimeMorph	13.75	1262.4	173.87	5.8
ACSF1	ResNet	1.01	26.0	0.32	3167.0
ACSF1	TapNet	0.59	33.7	1.45	688.2
ACSF1	COSCO-ResNet	1.64	119.2	0.43	2303.7
ACSF1	PatchTST	6.50	107.6	2.23	447.7
ACSF1	DLinear	0.10	4.5	0.07	13898.9
ACSF1	TimesNet	0.27	121.8	4.54	220.5
ACSF1	Autoformer	1.19	40.5	12.53	79.8
ACSF1	Crossformer	0.12	25.0	0.58	1723.7
ACSF1	FEDformer	0.61	277.0	7.03	142.3
ACSF1	Informer	0.56	23.4	1.09	917.6
ACSF1	GPT4TS	1.00	29.1	0.89	1121.4

TABLE 20

Ablation of aggregate curriculum pretraining on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full TimeMorph model.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
w/o Curriculum Pretraining	44.70	51.81	60.65	66.60	55.94	3.35
TimeMorph	48.31	55.55	64.14	69.16	59.29	–

APPENDIX D

ABLATION TABLES

Table 4 in the main paper reports the compact ablation summary. Tables 20–22 expand that summary into three focused views under the same few-shot protocol, and Table 23 summarizes the LLM-backbone ablation. Because UCR class identifiers are anonymous, the label-space ablation should be interpreted as testing closed-set class-token scoring; the semantic-prior class-token ablation is reported on external datasets with meaningful class names in Appendix B.6.

TABLE 21

Ablation of dual-view time-series-to-language modeling on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. These comparisons assess the effect of removing one top-level view at a time. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full TimeMorph model.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
Temporal only	43.51	50.55	58.40	64.39	54.21	5.08
Visual only	45.88	50.02	60.86	63.58	55.09	4.20
TimeMorph	48.31	55.55	64.14	69.16	59.29	–

TABLE 22

Ablation of closed-set class-token label-space scoring on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full TimeMorph model. Since UCR labels are anonymous, this ablation does not evaluate semantic label priors.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
w/o Class-Token Label-Space Scoring	44.46	50.44	60.08	65.59	55.14	4.15
TimeMorph	48.31	55.55	64.14	69.16	59.29	–

TABLE 23

Ablation of the LLM backbone on all 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs. Drop denotes the average accuracy decrease relative to full TimeMorph.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
TimeMorph	48.31	55.55	64.14	69.16	59.29	–
w/o LLM Backbone	44.89	48.29	54.22	58.43	51.46	7.83

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